Paying to Sharpen Identification: Optimal Probing of Coarse Data

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Abstract

Observations that are rounded, mismeasured, missing, or otherwise coarsened can often be probed through follow-up questions, remeasurement, or linkage across data sources. This paper develops a framework for selecting which observations to probe under a budget constraint. The value of probing is measured by the reduction in identified-set diameter. For scalar expectations, and for univariate distributions under Wasserstein-1 distance, identified-set diameter decomposes into additive observation-level contributions. This decomposition yields a simple optimal probing rule: probe observations in decreasing order of expected diameter reduction per unit cost until the budget is exhausted. In a counterfactual survey-design application based on Manski and Straub's (2000) analysis of worker expectations, targeting follow-up questions to the coarsest 25% of initially ambiguous responses captures 71% of the estimated identified-set diameter reduction attainable by questioning all respondents.

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1 Introduction

Empirical data are frequently rounded, heaped, censored, mismeasured, misclassified, or missing. Following Heitjan and Rubin (1991), I group these processes together under the common term *coarsening* and call the resulting data *coarse*. Data coarsening can leave statistical parameters only partially identified: knowing the population distribution from which the observed data are drawn is only sufficient to exclude certain parameter values, not to point identify a single value. For example, observing exact household incomes point identifies the mean, whereas observing only the number of households in each of several income brackets restricts the mean to an interval of possible values.

In certain contexts, data coarsening can be reduced or resolved: ambiguous responses on surveys can trigger clarifying follow-up questions (e.g. Juster and Smith, 1997); initial survey nonrespondents can be tracked and an interview reattempted (e.g. Kling, Liebman, and Katz, 2007); administrative record and survey data can be linked across sources (e.g. Abowd and Stinson, 2013); audits can identify misreporting (e.g. Kleven et al., 2011); and self-reports can be compared to objective measures (e.g. Carletto, Savastano, and Zezza, 2013). In this paper, these and similar methods of addressing data coarsening are called *data probing* (cf. Manski and Molinari, 2010). Where coarse data give rise to partial identification, data probing has the potential to sharpen identification. That is, the identified set for the parameter can be replaced with a proper subset.

Since data probing is costly, in many cases it is infeasible to probe all coarse observations. The researcher must therefore select which observations to probe. I solve this decision problem for a class of expectation-based and distributional targets whose identified-set diameters admit additive pointwise decompositions. To this end, I develop a formal model of how data coarsening gives rise to partial identification, and how probing can sharpen identification.

The researcher observes coarse data $\mathbf{z}$, which restrict a latent outcome $\mathbf{y}$ to an *outcome set* $Y(\mathbf{z})$. These restrictions may come from an explicit reporting or measurement process, such as interval reporting; or from maintained assumptions, such as bounded measurement error or a rounding model. Importantly, data ambiguity in the model is strictly observation-level, in that feasibility constraints are imposed observation-by-observation. This contrasts with population-level constraints, such as some fraction of the data being drawn from a contaminating distribution, and with cross-observation constraints such as $\mathbf{y}$ being monotone in $\mathbf{z}$.

The outcome sets $Y(\mathbf{z})$ determine an identified set for the parameter of interest, which is the range of parameter values consistent with each latent value being contained within its respective outcome set. The body of the paper focuses on the case where the parameter depends on $\mathbf{y}$ only through some real-valued transformation $g : \mathcal{Z} \times \mathcal{Y} \to \mathbb{R}$. If the parameter is the expectation of $g(\mathbf{z}, \mathbf{y})$, then the identified set will be defined by the extreme values of g on each outcome set. Figure 1 illustrates this for five observations, the function $g(z, y) := y$, and outcome sets that are intervals. The extreme values of the expectation are attained by the distributions that place probability one on the corresponding extreme values of the

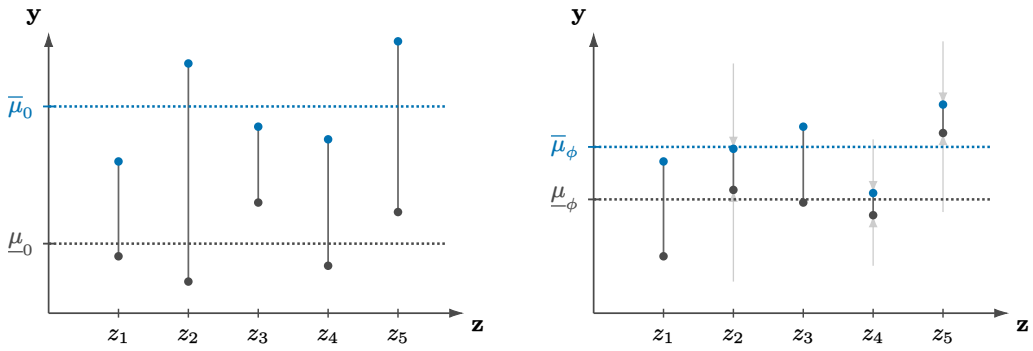


Figure 1: The vertical line segments are the outcome sets $Y(z_j)$. The dashed lines give the lower and upper endpoints of the identified set, in this case an interval, for $\mathsf{E}[\mathbf{y}]$. In the second graph, observations z_2, z_4, and z_5 have been probed, shrinking the corresponding outcome sets $Y(z_j)$, thereby sharpening the identified set for $\mathsf{E}[\mathbf{y}]$.

intervals. In this context, probing a particular data point replaces its outcome set with a subset. The loss of a probing rule is the expected diameter of the post-probing identified set. For a scalar expectation and the standard absolute distance on $\mathbb{R}$, the identified-set diameter decomposes additively into the expected image diameter. The same decomposition extends to univariate distributions under Wasserstein-1 distance. The cost of probing is assumed to be additively separable across observations, so that pointwise probing cost can be integrated to obtain total expenditure. This expenditure is then constrained not to exceed a budget, in expectation. Under these assumptions, the optimal decision rule is a pointwise threshold rule: probe at z if the ratio of the expected reduction in $\operatorname{diam} g(\{z\} \times Y(z))$ to the cost of probing at z exceeds some threshold, possibly randomising at the boundary to satisfy the expected-budget constraint. Appendices C, D, and E extend the allocation problem to ambiguity about probing effectiveness, multivariate targets, and multiple alternative probe types. Appendix B relates identified-set diameter to downstream decision regret.

Two survey-design applications illustrate gains from targeting initial coarseness and predicted probing effectiveness, respectively. In the first, probing 25% of initially coarse responses captures 71% of the identified-set diameter reduction attainable under full probing. In the second, imposing equal predicted sharpening across schooling groups substantially reduces disparities in the identified-set diameters for their conditional means, while only slightly increasing the diameter for the overall mean.

Related Literature. This paper is most closely related to work in the partial identification literature that treats data collection as a formal decision problem. Manski and Molinari (2008) study whether a survey item should be asked of all respondents, asked only following a specified response to an opening question, or omitted. They evaluate these designs using a loss that trades data collection cost against the diameter of the resulting identified set. In their main analysis, the target is the mean of a bounded transformation of the item

response, cost is proportional to the fraction of respondents queried, and the trade-off is operationalised by a loss function that sums this cost and the identified-set diameter. This skip-sequencing problem is a special case of the probing problem: I consider the general problem of probing in the presence of coarse data, allowing the degree of coarsening, and the cost and expected effectiveness of probing, to vary over the observed data. I obtain a pointwise threshold rule that specifies the optimal probing decision at each observation, generalising the collection of inequalities that define the optimal rule in Manski and Molinari. Additionally, I focus on a budget-constrained formulation of the probing problem; in this respect, their approach is more closely related to the penalised form of the problem (see Appendix C).

Dominitz and Manski (2017) allocate a budget between sampling processes that differ in cost and data quality, focusing on nonresponse and interval-valued measurements. They choose a sampling design before observing sample draws, and minimise maximum regret for a downstream prediction problem. Thus, sampling uncertainty and identification uncertainty are jointly relevant. My analysis differs in three respects. First, coarse data are observed before a probing decision is made, and probing is conditioned on the observed values. In particular, this timing allows probing to target realised coarseness at each observation rather than requiring it to be allocated ex ante across sampling designs based on the expected coarseness of the data they produce. Second, I abstract from sampling uncertainty to isolate identification uncertainty. My analysis therefore compares alternative strategies for reducing identification uncertainty, rather than focusing on the trade-off between identification and sampling uncertainty. Third, I use the diameter of the identified set as the objective. Appendix B discusses connections between identified-set diameter and minimax-regret evaluation of downstream decision problems.

Dominitz and Manski (2022) extend their previous analysis to a two-period panel data setting with attrition. Tetenov (2012) also studies the trade-off between sampling uncertainty and identification uncertainty, in a Gaussian setting. He shows that the relative value of reducing each depends on the criterion used to evaluate precision. Dominitz and Sherman (2004) study how verifying a subset of records sharpens identification under data contamination, building on the model of Horowitz and Manski (1995). Verification is analogous to probing, but contamination models link observations through population-level mixture shares. The outcome-set restriction $y \in Y(z)$ is instead observation-level, and the pointwise separability of this restriction is what permits the pointwise threshold rule.

My identification framework combines Heitjan and Rubin's (1991) model of coarse data with the partial-identification approach developed by Manski (2003). A large literature studies partial identification induced by particular forms of data coarsening. Representative examples include interval-valued covariates or outcomes (Manski and Tamer, 2002); rounded reports of probabilistic expectations (Manski and Molinari, 2010); survey nonresponse (Horowitz and Manski, 1998); censored outcomes (Khan, Ponomareva, and Tamer, 2011); and misclassified discrete variables (Molinari, 2008). The outcome-set correspondence $z \rightrightarrows Y(z)$ is a random set, as are several derived objects such as the outcome-image correspondence $z \rightrightarrows g(\{z\} \times Y(z))$, and the post-probing identified set. Random-set meth-

ods for partial identification analysis are developed by Beresteanu, Molchanov, and Molinari (2012) and treated in detail by Molchanov and Molinari (2018).

2 Data Coarsening and Partial Identification

Data coarsening is represented by a correspondence that maps each observed value to a set of latent outcomes that are consistent with the observation. These *outcome sets* are defined at the observation level, in the sense that a separate set is associated with each observed value. Together, the outcome sets determine an *identified set* for the parameter: for example, income brackets determine an interval of possible values for mean income. The identified set is thus defined at the population level, by aggregating observation-level restrictions over the distribution of observed data.

2.1 Setup

Data. Let $\mathbf{y} \in \mathcal{Y} \subseteq \mathbb{R}^{d_\mathcal{Y}}$ denote a latent outcome of interest.[1] The researcher observes $\mathbf{z} \in \mathcal{Z} \subseteq \mathbb{R}^{d_\mathcal{Z}}$, with distribution P. The observable z may include a reported outcome, covariates, observation-process information, and variables that determine probing cost. Examples of observation-process information include indicators for missingness or other coarsening, information about the data source or measurement device, and flags for imputation, editing, or record linkage. Let R denote the unobserved joint distribution of $(\mathbf{z}, \mathbf{y})$, with $\mathbf{z}$-marginal $\mathsf{R}_\mathbf{z} = \mathsf{P}$. Individual observational units, which may share a common $\mathbf{z}$, are referred to as units. The observable $\mathbf{z}$ encodes all information available to distinguish units. A low-resolution $\mathbf{z}$ sorts the units into a small number of bins; a sufficiently detailed $\mathbf{z}$ could make all units observationally distinguishable. The analysis abstracts from sampling uncertainty in order to isolate identification uncertainty, taking the population observed-data distribution P as directly observed, rather than merely estimated.

Outcome sets. Each z determines a nonempty outcome set, $Y(z) \subseteq \mathcal{Y}$, consisting of those y-values consistent with the observed z and maintained assumptions about the form of data coarsening.[2] The outcome set restriction is

$$\mathsf{R}(\mathbf{y} \in Y(\mathbf{z})) = 1.$$

The outcome-set correspondence $Y : \mathcal{Z} \rightrightarrows \mathcal{Y}$ maps observables z to outcome sets for y. Since $\mathbf{z}$ is a random variable, $Y(\mathbf{z})$ is a random set.

Example 1 (Interval-valued data). Suppose that $\mathbf{z} := (\mathbf{y}^L, \mathbf{y}^U) \in \mathbb{R}^2$ and $\mathbf{y}^L \leqslant \mathbf{y} \leqslant \mathbf{y}^U$ R-a.s. The outcome set is

$$Y(y^L, y^U) := [y^L, y^U].$$

[1] Throughout the paper, random variables are denoted by upright bold characters, and their realisations by italic, non-bold characters; for example, $\mathbf{z}$ has realisation z.

[2] Regularity conditions ensuring measurability and existence of measurable selections are discussed in Appendix A but omitted from the main text.

Remark 1 (Outcome sets from reporting models). I treat outcome sets as primitives, but they can be derived from an explicit measurement or reporting model. Suppose $\text{Rep} : \mathcal{Y} \times \mathcal{X} \times \mathcal{E} \to \mathcal{Y}^{\text{obs}}$ maps the latent outcome $\mathbf{y}$, a covariate vector $\mathbf{x}$, and an unobservable disturbance $\boldsymbol{\varepsilon}$ to a reported outcome $\mathbf{y}^{\text{obs}}$. Then the outcome sets are

$$Y(y^{\text{obs}}, x) = \{y \in \mathcal{Y} \mid \text{Rep}(y, x, \varepsilon) = y^{\text{obs}} \text{ for some } \varepsilon \in \mathcal{E}\}.$$

Identified set. Let $\mathcal{R}(Y, \mathsf{P})$ denote the set of distributions on $\mathcal{Z} \times \mathcal{Y}$ that are consistent with the observed distribution P and the outcome-set correspondence Y,

$$\mathcal{R}(Y, \mathsf{P}) := \{\mathsf{R} \in \mathcal{P}(\mathcal{Z} \times \mathcal{Y}) \mid \mathsf{R}_{\mathbf{z}} = \mathsf{P},\ \mathsf{R}(\mathbf{y} \in Y(\mathbf{z})) = 1\},$$

where $\mathcal{P}(\mathcal{Z} \times \mathcal{Y})$ is the set of Borel probability distributions on $\mathcal{Z} \times \mathcal{Y}$. Let $(\mathcal{T}, d)$ be a metric space. For a parameter $\tau : \mathcal{R}(Y, \mathsf{P}) \to \mathcal{T}$, the identified set is

$$\Theta_\tau(Y, \mathsf{P}) \ := \ \{\tau(\mathsf{R}) \mid \mathsf{R} \in \mathcal{R}(Y, \mathsf{P})\}. \tag{1}$$

That is, it is the image of $\mathcal{R}(Y, \mathsf{P})$ under the parameter mapping. When τ is understood, write $\Theta(Y, \mathsf{P})$.

Identified-set diameter. The imprecision of the identified set (1) is measured by its diameter,

$$\operatorname{diam}_d \Theta(Y, \mathsf{P}) := \sup_{\theta, \theta' \in \Theta(Y, \mathsf{P})} d(\theta, \theta').$$

Why diameter? Diameter is a natural measure of identification uncertainty for three reasons. First, it is a standard and intuitive measure of set size. Second, for scalar expectations, and for the larger collection of parameter-metric pairs given in Section 4.1, identified-set diameter decomposes into the expected image diameter, which permits the simple threshold form of the optimal probing rule. Third, diameter has a decision-theoretic interpretation, developed in detail in Appendix B, and briefly summarised here. If the identified set is used in a downstream minimax-regret point-estimation problem, the value of that problem is bounded above and below by monotone functions of the identified-set diameter (Proposition B.3). A related robustness result holds when the downstream decision problem is unknown but belongs to a sufficiently rich uniformly equicontinuous class (Proposition B.1).

Diameter is not always the appropriate objective. In certain settings the researcher may also care about the location of the identified set—for example, whether the identified set for a treatment effect lies entirely above or below a particular threshold. However, when this threshold is itself uncertain, minimising diameter again becomes defensible. Proposition B.2 formalises this point for downstream binary treatment-choice problems.

2.2 Identified-Set Diameter Decomposition

Proposition 1 (Identified-set diameter decomposition)**.** *Let $g : \mathcal{Z} \times \mathcal{Y} \to \mathbb{R}$ be a bounded measurable function, and let the target parameter be $\tau(\mathsf{R}) := \mathsf{E}_{\mathsf{R}}[g(\mathbf{z}, \mathbf{y})]$. Then*

$$\operatorname{diam} \Theta(Y, \mathsf{P}) = \mathsf{E}_{\mathsf{P}}[\Delta(\mathbf{z})], \qquad \Delta(z) := \operatorname{diam} g(\{z\} \times Y(z)).$$

Proof. See appendix. □

Remark 2. Proposition 1 is closely related to the support function characterisation of a selection expectation; see Molchanov (2017, Theorem 2.1.35) and Molchanov and Molinari (2018, Theorem 3.11 and Example 3.14).

Remark 3 (CDF values and conditional expectations)*.* Proposition 1 applies to many familiar scalar targets. For example, if y is real-valued, taking $g(z, y) := \mathbf{1}\{y \leqslant t\}$ gives the CDF value $\mathsf{R}(\mathbf{y} \leqslant t)$; and if $\mathsf{P}(C) > 0$, taking $g(z, y) := \mathbf{1}\{z \in C\}y/\mathsf{P}(C)$ gives the conditional expectation $\mathsf{E}_{\mathsf{R}}[\mathbf{y} \mid \mathbf{z} \in C]$.

In general, we work with the diameter of $g(\{z\} \times Y(z))$, not with the diameter of $Y(z)$ itself. However, when y is real-valued, $g(z, y) = y$ recovers the diameter of $Y(z)$ as a special case.

Example 2 (Image diameters of equal-width intervals)*.* Suppose that z includes component $y^U > 1$. Let $Y(z) := [y^U - 1, y^U]$, so that every outcome set has diameter one. The corresponding image diameters need not be constant. For example, if $g(z, y) = \log y$ then

$$\Delta(z) = \log\left(\frac{y^U}{y^U - 1}\right)$$

is strictly decreasing in y^U. Alternatively, if $g(z, y) = w(z)y$, where $w(z) > 0$ is a sample weight, then

$$\Delta(z) = w(z)$$

varies with the sample weight.

3 Probing

3.1 The Probing Decision Problem

Probing. Probing a data point with initial observed value z_0 replaces the observed value with a refined observation $(z_0, z_1) \in \mathcal{Z}_0 \times \mathcal{Z}_1 \subseteq \mathbb{R}^{d_{\mathcal{Z}_0} + d_{\mathcal{Z}_1}}$, and replaces the outcome set $Y_0(z_0)$ with a refined outcome set $Y_1(z_0, z_1) \subseteq Y_0(z_0)$. If probing exactly recovers the true latent outcome, the post-probing set will be a singleton.

The refinement z_1 could be the latent outcome y, a sharper approximation of y, or additional information about the reporting process. The post-probing outcome-set correspondence $Y_1 : \mathcal{Z}_0 \times \mathcal{Z}_1 \rightrightarrows \mathcal{Y}$ is known at the time the probing decision is made. The realised

set $Y_1(z_0, z_1)$ is determined once the refinement $\mathbf{z}_1$ is observed. Denote the pre-probing image diameter by

$$\Delta_0(z_0) := \operatorname{diam} g(\{z_0\} \times Y_0(z_0)),$$

and its post-probing counterpart by

$$\Delta_1(z_0, z_1) := \operatorname{diam} g(\{z_0\} \times Y_1(z_0, z_1)).$$

Let P_{01} denote the joint distribution of $(\mathbf{z}_0, \mathbf{z}_1)$, with $\mathbf{z}_0$-marginal P_0. Write E_0 and E_{01} for the corresponding expectations. The researcher is not assumed to know the full P_{01}: such knowledge would eliminate the need for probing. However, the researcher is initially assumed to know the induced conditional expected post-probing diameters,

$$\overline{\Delta}_1(z_0) := \mathsf{E}_{01}\left[\Delta_1(\mathbf{z}_0, \mathbf{z}_1) \mid \mathbf{z}_0 = z_0\right].$$

Section 4.2 considers the case in which $\overline{\Delta}_1$ is unknown and must be estimated.[3]

A *nonrandomised probing rule* is a measurable function $\phi : \mathcal{Z}_0 \to \{0, 1\}$ specifying which observations to probe. It can be identified with a measurable probing region, $C \subseteq \mathcal{Z}_0$, the set on which the rule takes value one. Randomised rules $\phi : \mathcal{Z}_0 \to [0, 1]$ are then defined by[4]

$$\phi(z_0) = \sum_{j=1}^{\infty} q_j \mathbf{1}\{z_0 \in C_j\}, \tag{2}$$

where $q_1, q_2, \ldots$ are nonnegative and sum to 1. Denote the space of all such rules by $\Phi \subseteq [0, 1]^{\mathcal{Z}_0}$. Lemma A.5 in the appendix shows that Φ is exactly the set of all measurable functions from $\mathcal{Z}_0$ into $[0, 1]$.

Example 3 (Skip-sequencing). Manski and Molinari (2008) study skip-sequencing in surveys. A response $z_0 \in \{0, 1\}$ to an opening question determines whether a follow-up question is administered. Let $\phi(z_0) = 1$ indicate that all respondents with opening response z_0 receive the follow-up, and let $\phi(z_0) = 0$ indicate that none do. The rule

$$\phi(z_0) = \mathbf{1}\{z_0 = 1\}$$

implements skip-sequencing: only respondents with $z_0 = 1$ are asked the follow-up. The rule $\phi \equiv 1$ administers the follow-up to every respondent; $\phi \equiv 0$ administers it to none; and $\phi \equiv 1/2$ administers it to each respondent with probability 1/2.

[3] In some settings, knowledge or learnability of $\overline{\Delta}_1$ implies knowledge or learnability of conditional expected probing outcomes before probing. In such cases, the probing rule provided in Theorem 1 minimises the diameter of an outer set for the parameter which may not necessarily be the sharp identified set.

[4] Strictly speaking, a randomised rule should also specify joint probing probabilities across observations. The discussion preceding Lemma A.5 defines randomised rules in this more general way; Lemma A.8 then shows that only the marginal probing probabilities $\phi(z_0)$ are decision-relevant.

For any probing region $C \subseteq \mathcal{Z}_0$, define the induced outcome-set correspondence

$$Y_C(z_0, z_1) := \begin{cases} Y_0(z_0), & z_0 \notin C, \\ Y_1(z_0, z_1), & z_0 \in C, \end{cases}$$

and image diameter

$$\Delta_C(z_0, z_1) := \begin{cases} \Delta_0(z_0), & z_0 \notin C, \\ \Delta_1(z_0, z_1), & z_0 \in C. \end{cases}$$

Assumption 1 (Additive diameter decomposition)**.** The parameter-metric pair (τ, d) admits the additive diameter decompositions

$$\operatorname{diam}_d \Theta_\tau(Y_0, \mathsf{P}_0) = \mathsf{E}_0 \left[\Delta_0(\mathbf{z}_0)\right],$$

and for every probing region $C \subseteq \mathcal{Z}_0$,

$$\operatorname{diam}_d \Theta_\tau(Y_C, \mathsf{P}_{01}) = \mathsf{E}_{01} \left[\Delta_C(\mathbf{z}_0, \mathbf{z}_1)\right].$$

By Proposition 1, this assumption is satisfied by the absolute distance on $\mathbb{R}$ and the scalar expectation target

$$\tau(\mathsf{R}) := \mathsf{E}_{\mathsf{R}} \left[g(\mathbf{z}_0, \mathbf{y})\right],$$

for any bounded measurable real-valued g. Section 4.1 gives a larger collection of parameter-metric pairs for which the assumption holds.

Loss function. Define the loss of nonrandomised probing rules by

$$L(\mathbf{1}_C) := \operatorname{diam}_d \Theta_\tau(Y_C, \mathsf{P}_{01}).$$

This loss can be extended to randomised rules (2) by writing

$$L(\phi) := \sum_{j=1}^{\infty} q_j \operatorname{diam}_d \Theta_\tau(Y_{C_j}, \mathsf{P}_{01}).$$

Budget constraint. The researcher faces the expected budget constraint

$$\mathsf{E}_0[c(\mathbf{z}_0)\phi(\mathbf{z}_0)] \leqslant b$$

for some $b > 0$ and some pointwise cost function $c : \mathcal{Z}_0 \to \mathbb{R}_{>0}$ that is positive, bounded, and measurable. Denote the set of $\phi \in \Phi$ that satisfy this budget constraint by Φ_b. When ϕ is nonrandomised and pointwise costs are deterministic, this is also the ex-post realised cost.

3.2 Optimal Probing

Marginal value. The marginal value of probing at $z_0 \in \mathcal{Z}_0$ is

$$v(z_0) := \Delta_0(z_0) - \overline{\Delta}_1(z_0).$$

It is the expected reduction in image diameter from probing at z_0. Lemma A.8 gives the decomposition

$$\begin{aligned} L(\phi) &= \mathsf{E}_{01}\left[\Delta_1(\mathbf{z}_0, \mathbf{z}_1)\phi(\mathbf{z}_0) + \Delta_0(\mathbf{z}_0)(1 - \phi(\mathbf{z}_0))\right] \\ &= \mathsf{E}_0\left[\Delta_0(\mathbf{z}_0)\right] - \mathsf{E}_0\left[v(\mathbf{z}_0)\phi(\mathbf{z}_0)\right]. \end{aligned} \tag{3}$$

Theorem 1 (Optimal probing rule)**.** *Under Assumption 1, the problem*

$$\inf_{\phi \in \Phi_b} L(\phi)$$

is solved by $\phi^* \in \Phi_b$ if and only if, for some $\lambda^* \geqslant 0$ and some measurable $\gamma^* : \mathcal{Z}_0 \to [0,1]$,

$$\phi^*(z_0) = \mathbf{1}\left\{\frac{v(z_0)}{c(z_0)} > \lambda^*\right\} + \gamma^*(z_0)\mathbf{1}\left\{\frac{v(z_0)}{c(z_0)} = \lambda^*\right\} \quad \mathsf{P}_0\text{-}a.e. \tag{4}$$

and

$$\lambda^*\left(b - \mathsf{E}_0\left[c(\mathbf{z}_0)\phi^*(\mathbf{z}_0)\right]\right) = 0.$$

Proof. See appendix. The proof uses the loss decomposition (3) to show that minimising $L(\phi)$ is equivalent to maximising $\mathsf{E}_0\left[v\phi\right]$. When $\mathsf{E}_0[v] > 0$, it defines probability measures

$$\mathsf{P}_v(A) := \frac{\mathsf{E}_0[v\mathbf{1}_A]}{\mathsf{E}_0[v]}, \qquad \mathsf{P}_c(A) := \frac{\mathsf{E}_0[c\mathbf{1}_A]}{\mathsf{E}_0[c]},$$

under which this maximisation problem becomes a most-powerful-test problem that is solved using the Neyman-Pearson Lemma (Lehmann and Romano, 2022, Theorem 3.2.1). □

Remark 4 (Neyman-Pearson interpretation). Under the corresponding measures, v/c is the analogue of a likelihood ratio, and the budget is the analogue of a size constraint. That the Neyman-Pearson lemma can be used to solve general economic problems has previously been demonstrated by Berliant (1984), in the context of a consumer's decision to purchase land.

Remark 5. Appendix A.4 includes several corollaries to Theorem 1. Corollary A.1 says that the optimal probing rule can always be implemented by randomising over at most two nonrandomised rules. Corollary A.2 says that if the restriction of P_0 to the boundary set $\{v/c = \lambda^*\}$ is nonatomic, then randomisation can be replaced by deterministically probing a subset of the boundary. More generally, Corollary A.3 bounds the additional loss from restricting to nonrandomised probing rules. Corollary A.4 generalises Theorem 1 to settings in which probing decisions cannot be chosen observation-wise, but instead must be conditioned on some coarsening of the observed data. For example, a researcher faced with individual-level survey data may have to choose which of a small number of geographical regions to target for follow-up.

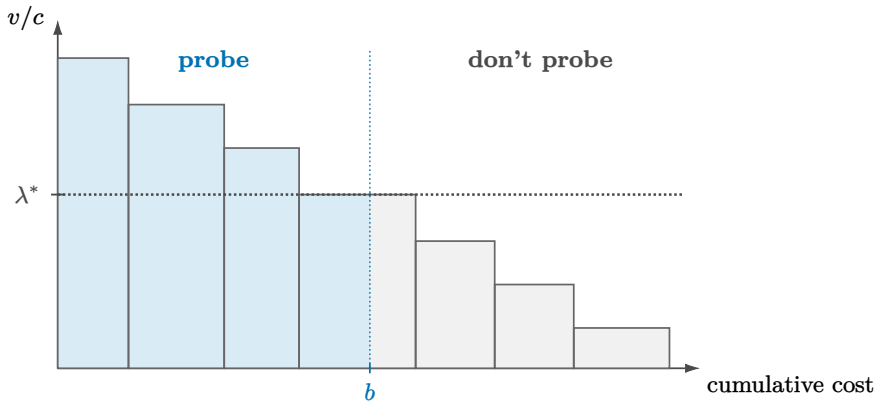


Figure 2: Visualisation of Theorem 1. Each rectangle represents a bin of observational units with a common marginal diameter reduction per unit cost, v/c. The height of the rectangle is this common ratio; the width is the expected cost of probing the entire bin. Bins are ordered from left to right by decreasing v/c, so a point on the horizontal axis records the cumulative expected cost of fully probing all bins to its left and, when it intersects a bin, the indicated fraction of that bin's cost. The optimal rule probes all observations satisfying $v/c > \lambda^*$, and randomises as necessary in the boundary bin $v/c = \lambda^*$ to exhaust the budget b.

The following proposition says that selectively targeted probing improves on untargeted probing when the marginal value-cost ratio v/c varies across observations. Denote the expected cost of probing all observations by

$$b^U := \mathsf{E}_0[c(\mathbf{z}_0)].$$

For $b \in [0, b^U]$, define the value function

$$L^*(b) := \inf \{L(\phi) \mid \mathsf{E}_0[c(\mathbf{z}_0)\phi(\mathbf{z}_0)] \leqslant b\}.$$

Write $L(0)$ and $L(1)$ for the losses under no probing and full probing, respectively, and suppose $L(0) > L(1)$.

Proposition 2 (Value of targeting)**.** *L^* is weakly decreasing and convex, and*

$$L(0) - L^*(b) \geqslant \frac{b}{b^U}(L(0) - L(1))$$

for every $b \in [0, b^U]$. For any interior $b \in (0, b^U)$, this holds with equality if and only if there exists $t > 0$ such that

$$\frac{v(z_0)}{c(z_0)} = t \qquad \mathsf{P}_0\text{-}a.e.$$

Proof. See appendix. □

Remark 6 (Inequality and targeting gains). Proposition 2 establishes that targeting has value at an interior budget whenever v/c varies across responses. Corollary A.5 in the appendix shows that greater inequality in the cost-weighted distribution of v/c, in the sense of Lorenz dominance, permits a greater share of full-probing gain to be attained at each budget share.

Remark 7 (The scope of the probing model). The pointwise threshold form of the optimal probing rule results from an additive decomposition of both loss and expenditure across observations. Several features of the model permit this decomposition.

(i) Identification restrictions are imposed at the level of the observation: each z_0 determines an outcome set $Y_0(z_0)$, with no restrictions linking latent outcomes across observational units.

(ii) The parameter-metric pair admits the additive decomposition of identified-set diameter in Assumption 1. Remarks A.2 and D.1 in the appendix illustrate settings where this decomposition fails.

(iii) Probing is pointwise and one-shot: a probe refines only the outcome sets of the probed units, and all probing decisions are made before any probe outcomes are observed.

(iv) Total cost is additive across observations, and the budget is imposed in expectation.

Without these features, the threshold rule in Theorem 1 need not be optimal, but it may still be possible to compute an alternative optimal probing rule. For example, in Remarks A.2 and D.1, the number of nonrandomised probing rules is sufficiently small that their costs and post-probing diameters can be enumerated and compared.

4 Extensions

4.1 Distributional Targets

Let $g : \mathcal{Z} \times \mathcal{Y} \to \mathbb{R}$ be a bounded measurable function, and recall that $\Delta(z) := \operatorname{diam} g(\{z\} \times Y(z))$. Denote the closed convex hull of $g(\mathcal{Z} \times \mathcal{Y})$ by $\mathcal{G} \subseteq \mathbb{R}$. A researcher may be interested in learning about the full distribution of $g(\mathbf{z}, \mathbf{y})$ rather than only its mean. Write $g_{\#}\mathsf{R}$ for the distribution of $g(\mathbf{z}, \mathbf{y})$ induced by $(\mathbf{z}, \mathbf{y}) \sim \mathsf{R}$, and denote the corresponding CDF and quantile function by

$$F_{\mathsf{R}}(t) := \mathsf{R}(g(\mathbf{z}, \mathbf{y}) \leqslant t), \qquad Q_{\mathsf{R}}(\alpha) := \inf\{t \in \mathcal{G} \mid \alpha \leqslant F_{\mathsf{R}}(t)\}.$$

Equip the space of Borel probability measures $\mathcal{P}(\mathcal{G})$ with the Wasserstein-1 distance,

$$W_1(\mu, \nu) := \inf_{\eta \in \Pi(\mu,\nu)} \int_{\mathcal{G}\times\mathcal{G}} |t - s| \, \mathrm{d}\eta(t, s),$$

where $\Pi(\mu, \nu)$ is the set of all couplings between μ and ν. The Wasserstein-1 distance between two distributions is the minimum average distance through which probability mass

must be transported in order to convert one distribution into the other (see Villani, 2009, Definition 6.1). For any interval $\mathcal{I} \subseteq \mathbb{R}$, define the $L^1(\mathcal{I})$ distance by

$$\|f_1 - f_2\|_{L^1(\mathcal{I})} := \int_{\mathcal{I}} |f_1(t) - f_2(t)| \, \mathrm{d}t.$$

Proposition 3 (Distributional targets)**.** *Let the target parameter be* $\tau(\mathsf{R}) := g_{\#}\mathsf{R}$ *and equip* $\mathcal{P}(\mathcal{G})$ *with the Wasserstein-1 metric. Then*

$$\operatorname{diam}_{W_1} \Theta_{\mathrm{dis}}(Y, \mathsf{P}) = \mathsf{E}_{\mathsf{P}}[\Delta(\mathbf{z})].$$

Equivalently, if the target parameter is the CDF $F_{\mathsf{R}} \in L^1(\mathcal{G})$ *or the quantile function* $Q_{\mathsf{R}} \in L^1([0,1])$*, then*

$$\operatorname{diam}_{L^1(\mathcal{G})} \Theta_{\mathrm{cdf}}(Y, \mathsf{P}) = \operatorname{diam}_{L^1([0,1])} \Theta_{\mathrm{qnt}}(Y, \mathsf{P}) = \mathsf{E}_{\mathsf{P}}[\Delta(\mathbf{z})].$$

Proof. See appendix. To summarise: Kantorovich-Rubinstein duality and Proposition 1 immediately imply that

$$\operatorname{diam}_{W_1} \Theta_{\mathrm{dis}}(Y, \mathsf{P}) \geqslant \mathsf{E}_{\mathsf{P}}[\Delta(\mathbf{z})].$$

The reverse inequality also holds, because a 1-Lipschitz transformation cannot increase diameter. And for univariate distributions, Wasserstein-1 distance equals L^1-distance between quantile functions and between CDFs. □

Because Proposition 3 applies to every outcome-set correspondence, these parameter-metric pairs satisfy Assumption 1. The corresponding probing problems are therefore solved by the optimal rule obtained in Theorem 1.

Remark 8 (CDF and quantile function versus CDF values and quantiles). Targeting the CDF F_{R} or quantile function Q_{R} as an object in L^1-space is distinct from targeting a particular CDF value $F_{\mathsf{R}}(t)$ or quantile $Q_{\mathsf{R}}(\alpha)$ as an object in $\mathbb{R}$. A CDF value can be targeted by taking the expectation of $\mathbf{1}\{g(z, y) \leqslant t\}$. This transformation changes the image diameter $\Delta(z)$, so targeting $F_{\mathsf{R}}(t) = \mathsf{E}_{\mathsf{R}}[\mathbf{1}\{g(\mathbf{z}, \mathbf{y}) \leqslant t\}]$ is not, in general, equivalent to targeting $\mathsf{E}_{\mathsf{R}}[g(\mathbf{z}, \mathbf{y})]$, even though targeting the entire CDF under the L^1 metric is. An individual quantile does not generally admit a representation of the form $\mathsf{E}_{\mathsf{R}}[g(\mathbf{z}, \mathbf{y})]$ and therefore cannot generally be targeted using the scalar expectation framework.

Remark 9 (Other probability metrics). Let d_{TV} and d_K be the total variation and Kolmogorov distances, respectively. Corollaries A.6 and A.7 in the appendix show that under strong additional geometric restrictions,

$$\operatorname{diam}_{d_{\mathrm{TV}}} \Theta_{\mathrm{dis}}(Y, \mathsf{P}) = \operatorname{diam}_{d_K} \Theta_{\mathrm{dis}}(Y, \mathsf{P}) = \mathsf{P}(\Delta(\mathbf{z}) > 0).$$

Remark A.2 gives a simple case in which additive decomposition fails under several metrics, including Kolmogorov distance and Wasserstein-p distances with $p > 1$.

4.2 Estimated Probing Values

In Section 3 we assumed that the expected post-probing diameter $\overline{\Delta}_1$ is known. If these values must be estimated, or more generally, if probing value v must be estimated, the resulting plug-in rule need not coincide with the oracle rule of Section 3. The next proposition bounds the excess loss of the plug-in rule in terms of the estimation error in v at observations where the plug-in differs from the oracle rule.

Proposition 4 (Plug-in performance)**.** *Let $\widehat{v} : \mathcal{Z}_0 \to \mathbb{R}_{\geqslant 0}$ be a bounded measurable estimate of v. Suppose that*

$$\widehat{\phi} \in \operatorname*{arg\,max}_{\phi \in \Phi_b} \mathsf{E}_0 \left[\widehat{v}\phi\right].$$

Then

(i)

$$L(\widehat{\phi}) - L(\phi^*) \leqslant \mathsf{E}_0 \left[|\widehat{v} - v| \cdot \left| \phi^* - \widehat{\phi} \right| \right].$$

(ii) Moreover, if $\widehat{\phi}$ costs at least as much as ϕ^*, then

$$L(\widehat{\phi}) - L(\phi^*) \leqslant \inf_{\substack{h:\mathbb{R}_{\geqslant 0}\to\mathbb{R}_{\geqslant 0} \\ h \text{ weakly increasing}}} \mathsf{E}_0 \left[c \left| \frac{v}{c} - h\left(\frac{\widehat{v}}{c}\right) \right| \cdot \left| \phi^* - \widehat{\phi} \right| \right].$$

(iii) In particular, if $\widehat{\phi}$ costs at least as much as ϕ^* and there exists a weakly increasing $h : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$ such that

$$\frac{v}{c} = h\left(\frac{\widehat{v}}{c}\right)$$

for P_0-a.e. z_0 with $\phi^*(z_0) \neq \widehat{\phi}(z_0)$, then

$$L(\widehat{\phi}) = L(\phi^*).$$

Proof. See appendix. □

Under the expenditure condition in part (iii), correctly ranked but miscalibrated value-cost predictions can still yield an optimal probing rule.

4.3 Multiple Linear Constraints

In some settings there may be multiple constraints on probing. For example, a survey may have separate constraints on money, interviewer time, and respondent burden. Let $c_1, \ldots, c_K : \mathcal{Z}_0 \to \mathbb{R}$ be bounded measurable functions, and let $b_1, \ldots, b_K \in \mathbb{R}$. Denote

$$\Phi_{b_1,\ldots,b_K} := \{\phi \in \Phi \mid \mathsf{E}_0 \left[c_k \phi\right] \leqslant b_k,\ k = 1, \ldots, K\}.$$

The budget-constrained problem of Section 3 is the special case in which $K = 1$, $c_1 = c > 0$ and $b_1 = b \geqslant 0$. Equality constraints can be represented by two inequalities.

Proposition 5 (Multiple linear constraints)**.** *Under Assumption 1, the problem*

$$\inf_{\phi \in \Phi_{b_1,\ldots,b_K}} L(\phi) \tag{5}$$

is solved by $\phi^* \in \Phi_{b_1,\ldots,b_K}$ *if for some* $\lambda_1^*, \ldots, \lambda_K^* \geqslant 0$ *and some measurable* $\gamma^* : \mathcal{Z}_0 \to [0,1]$*,*

$$\phi^*(z_0) = \mathbf{1}\left\{v(z_0) > \sum_{k=1}^K \lambda_k^* c_k(z_0)\right\} + \gamma^*(z_0)\mathbf{1}\left\{v(z_0) = \sum_{k=1}^K \lambda_k^* c_k(z_0)\right\} \qquad \mathsf{P}_0\text{-}a.e. \tag{6}$$

and

$$\lambda_k^* \left(b_k - \mathsf{E}_0\left[c_k \phi^*\right]\right) = 0, \qquad k = 1, \ldots, K. \tag{7}$$

Proof. See Appendix. □

The following are examples of constraints that could be imposed alongside the original budget constraint.

Example 4 (Fairness and capacity constraints). Let $C_1, \ldots, C_K \subseteq \mathcal{Z}_0$ be observable groups with $p_k := \mathsf{P}_0(C_k) > 0$. Then setting

$$c_k(z_0) = -\frac{v(z_0)\mathbf{1}\{z_0 \in C_k\}}{p_k}, \qquad b_k := -a_k \tag{8}$$

imposes minimum probing (by value) within each group:

$$\mathsf{E}_0\left[v\phi \mid C_k\right] \geqslant a_k, \qquad k = 1, \ldots, K.$$

Replacing v with c in (8) would instead impose a minimum probing expenditure within each group; replacing v with a positive constant would impose a minimum probing volume; replacing it with a negative constant would turn this into a capacity constraint.

Example 5 (Protecting secondary target parameters). Let $g_1, \ldots, g_K : \mathcal{Z}_0 \times \mathcal{Y} \to \mathbb{R}$ be bounded measurable functions and let $v_1, \ldots, v_K : \mathcal{Z}_0 \to \mathbb{R}_{\geqslant 0}$ be the corresponding marginal-value-of-probing functions. Setting

$$c_k(z_0) = -v_k(z_0), \qquad b_k := -v_k^L$$

imposes minimum expected identified-set diameter reductions for each of $\mathsf{E}_\mathsf{R}[g_k(\mathbf{z}_0, \mathbf{y})]$, $k = 1, \ldots, K$.

5 Survey-Design Applications

I consider two retrospective counterfactual survey-design problems using data from the Survey of Economic Expectations (SEE), a cross-sectional telephone survey conducted in 16 waves between 1994 and 2002 (Dominitz and Manski, 2004). The first problem targets perceived job insecurity and follows Section III of Manski and Straub (2000); the second

targets stock-return expectations and relates to Section 2.2 of Dominitz and Manski (2011). The two problems illustrate distinct sources of probing value: heterogeneity in initial data coarseness versus heterogeneity in probing effectiveness.

In both settings, the probing rule is estimated using a pilot sample consisting of earlier survey waves and then evaluated on an independent sample consisting of later waves. For each realised probing region C, let $g_i^L(C)$ and $g_i^U(C)$ be the lower and upper endpoints, respectively, of the probing-induced outcome set for evaluation respondent i. The estimated identified set for the mean is

$$\widehat{\Theta}_n(C) := \left[\frac{1}{n} \sum_{i=1}^{n} g_i^L(C),\ \frac{1}{n} \sum_{i=1}^{n} g_i^U(C) \right].$$

I evaluate each probing rule by the diameter of this set, averaging across realisations when the rule randomises. This is the empirical counterpart of the paper's diameter criterion. For a realised allocation, the interval also gives the range of evaluation-sample means consistent with the remaining observation-level coarseness. Appendix F reports confidence intervals for the identified set.

5.1 Perceived Job Insecurity

Employed respondents to the SEE reported the percent chance that they would lose their job over the following 12 months and the percent chance that, conditional on such job loss, the job they subsequently find and accept would be at least as good as their current one, in terms of wages and benefits. Denote the reported job-loss chance by loss_i and the reported search-outcome chance by find_i. Using responses to the first eight waves of the survey, Manski and Straub define a composite measure of perceived job insecurity,

$$\mathsf{ins}_i := \frac{\mathsf{loss}_i(100 - \mathsf{find}_i)}{100}.$$

This is the implied percent chance that "a worker will lose his job in the year ahead and subsequently not obtain a position of comparable economic value."

Suppose instead that the search-outcome question find_i was asked only of a subset of respondents to the prior job-loss question loss_i. Absent find_i, loss_i defines an outcome set for ins_i,

$$Y_{0i} = [0, \mathsf{loss}_i],$$

with diameter

$$\Delta_{0i} = \mathsf{loss}_i.$$

I treat find_i as a probe which replaces Y_{0i} with a singleton,[5]

$$Y_{1i} = \{\mathsf{ins}_i\}, \qquad \Delta_{1i} = 0.$$

[5]About 1% of follow-up questions do not receive a valid response. I do not attempt to predict nonresponse in the pilot sample when constructing the probing rule, but retain nonrespondents in the evaluation. Nonresponse to a probe returns the initial outcome set.

Observe that respondents with $\mathsf{loss}_i = 0$ have no probing value. The initial target is mean job insecurity. Suppose find_i is asked of a subset C of respondents and let $C^S \subseteq C$ consist of those individuals that provided a valid response. Then

$$\widehat{\Theta}(C) := \left[\frac{1}{n} \sum_{i \in C^S} \mathsf{ins}_i,\ \frac{1}{n} \left(\sum_{i \in C^S} \mathsf{ins}_i + \sum_{i \notin C^S} \mathsf{loss}_i \right) \right]$$

and

$$\operatorname{diam} \widehat{\Theta}(C) = \frac{1}{n} \sum_{i \notin C^S} \mathsf{loss}_i.$$

With uniform probing cost, the targeted rule probes in decreasing order of loss_i:

$$\phi_i^* = \mathbf{1}\{\mathsf{loss}_i > \lambda^*\} + \gamma_i^* \mathbf{1}\{\mathsf{loss}_i = \lambda^*\}, \tag{9}$$

where $\gamma_i^* \in [0,1]$ is chosen to satisfy the expected-budget constraint. In particular, I impose uniform randomisation over boundary units.

To mimic a real survey-design setting, I split the SEE sample in two. For the first eight waves, those evaluated by Manski and Straub (2000), I suppose that only loss_i was elicited. I pool these waves and use them as a pilot sample to calibrate the threshold λ^* in (9). This is done for each budget in a 2,001-point uniform grid between 0 and the cost of probing all positive-loss_i respondents. I then apply this rule to the pooled evaluation sample, consisting of waves 9 through 16. After excluding nonrespondents to loss_i, the pilot sample consists of 3,708 individuals and the evaluation sample consists of 2,618 individuals. The share of respondents reporting $\mathsf{loss}_i > 0$ is similar in both samples: 65.4% and 64.3% respectively.

The top panel of Figure 3 plots the estimated identified set for mean job insecurity against the realised share of positive-loss_i respondents probed. Because probing costs are uniform, this share is proportional to the realised expenditure. The lower panel plots the corresponding diameters. The vertical separation between the two curves measures the additional reduction in identified-set diameter from targeted rather than random probing. Table 1 gives the identified-set diameter and the share of full-probing gain attained at selected probe shares. At a 25% probe share, for example, the targeted rule captures 70.6% of the identified-set diameter reduction attainable under full probing.[6]

Manski and Straub report several qualitative findings about how worker perceptions of job insecurity vary with demographic factors. I focus here on their observation that mean job insecurity among Black workers exceeds that among White workers. We can write this difference as

$$\frac{1}{n_B} \sum_{i \in B} \mathsf{ins}_i - \frac{1}{n_W} \sum_{i \in W} \mathsf{ins}_i = \frac{1}{n} \sum_{i=1}^{n} w_i^{\text{eval}} \cdot \mathsf{ins}_i,$$

where

$$w_i^{\text{eval}} := \frac{\mathbf{1}\{i \in B\}}{n_B/n} - \frac{\mathbf{1}\{i \in W\}}{n_W/n}, \tag{10}$$

[6]Endpoints and diameters for random probing are averaged analytically across realisations. The same is true for targeted probing when randomising over boundary units.

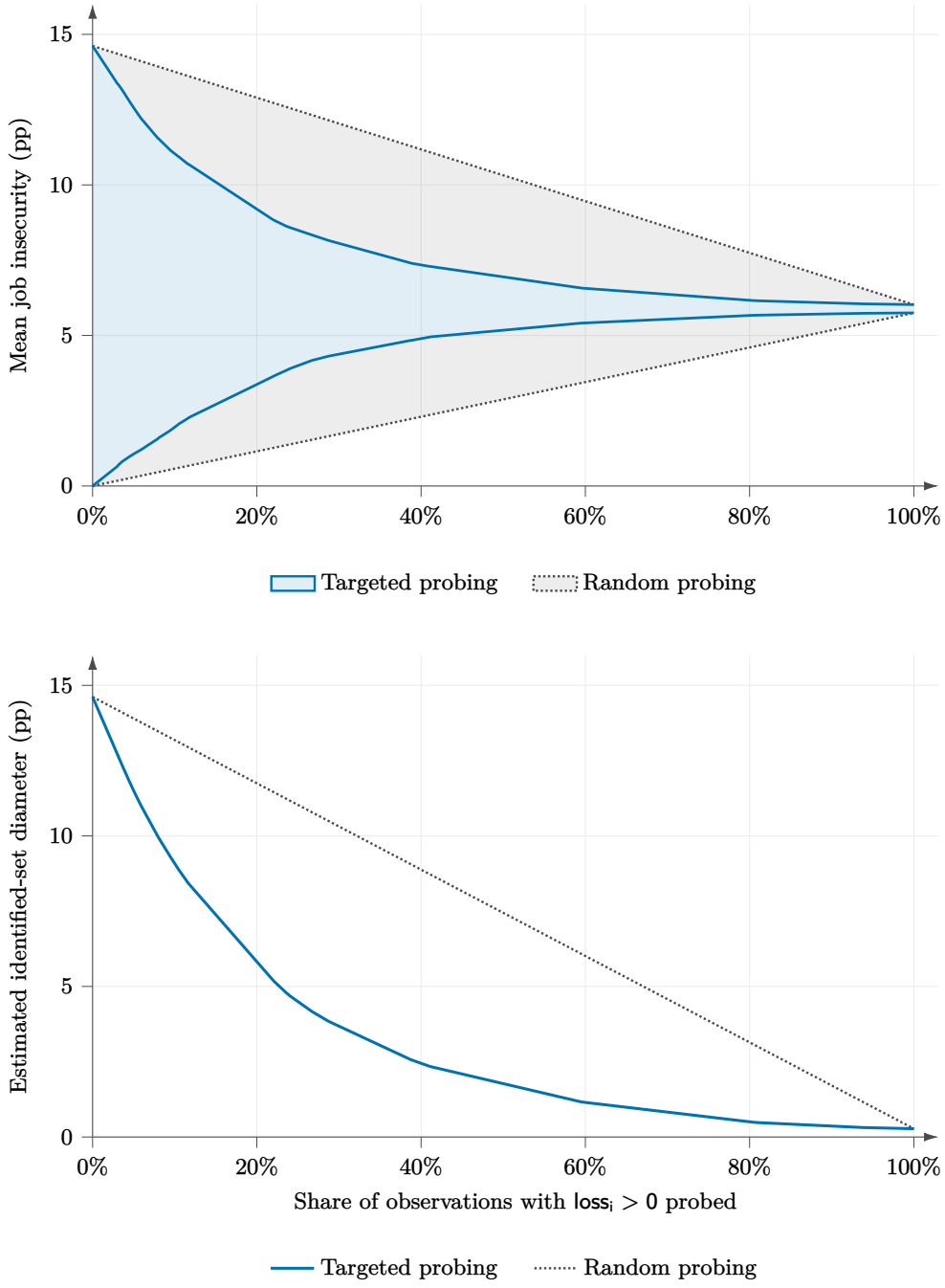


Figure 3: Estimated identified set for mean job insecurity (top) and its diameter (bottom) under targeted and random probing. Where probing is randomised, endpoints and diameters are averaged over realisations.

Probe share (%)	diam $\widehat{\Theta}_n$ (pp)	Gain share (%)
0	14.62	0.0
10	9.08	38.6
25	4.50	70.6
50	1.77	89.6
75	0.67	97.3
100	0.28	100.0

Table 1: Estimated identified-set diameter and gain share for mean job insecurity under targeted probing. The gain share is the fraction of the full-probing diam $\widehat{\Theta}_n$ reduction attained.

and B and W are the sets of Black and White respondents in the evaluation sample, respectively, and n_B and n_W are their cardinalities. The targeted rule probes in decreasing order of

$$\left| w_i^{\text{pilot}} \right| \cdot \mathsf{loss}_i,$$

where w_i^{pilot} is the analogue of (10) with n_B/n and n_W/n replaced by the corresponding pilot-sample proportions. Figure 4 shows the corresponding estimated identified set and diameter curves for the Black-White mean contrast, comparing the rule that directly targets the contrast with the earlier overall-mean targeted rule and random probing. The averaged estimated identified set excludes zero after 81.8% of positive-loss_i respondents are probed under untargeted probing, versus 33.8% and 22.8% under the rules targeting the overall mean and Black-White contrast, respectively.

5.2 Stock-Return Expectations

In the job-insecurity application, probing value arose from initial coarseness. This second application considers a setting in which probing frequently fails and heterogeneity in failure probability generates heterogeneity in probing value. Following Dominitz and Manski (2011), I use waves 12 through 14 of the SEE, consisting of 1,651 respondents in total. These waves asked respondents to report the percent chance that a \$1,000 investment in a diversified stock mutual fund would be worth more than \$1,000 one year later. Call this the subjective probability of positive nominal return and denote it pnr_i. The target parameter is the mean of pnr_i. Suppose, counterfactually, that the researcher selectively elicited pnr_i as a probe. Assume uniform probing cost across respondents. Before probing, each respondent's pnr_i is unobserved and so $Y_{0i} \equiv [0, 100]$. If probing elicits a valid response, this set is replaced with a point value; if probing fails, the original uninformative outcome set is returned. That is,

$$Y_{1i} = \begin{cases} \{\mathsf{pnr}_i\}, & \text{probing succeeds,} \\ [0, 100], & \text{probing fails.} \end{cases}$$

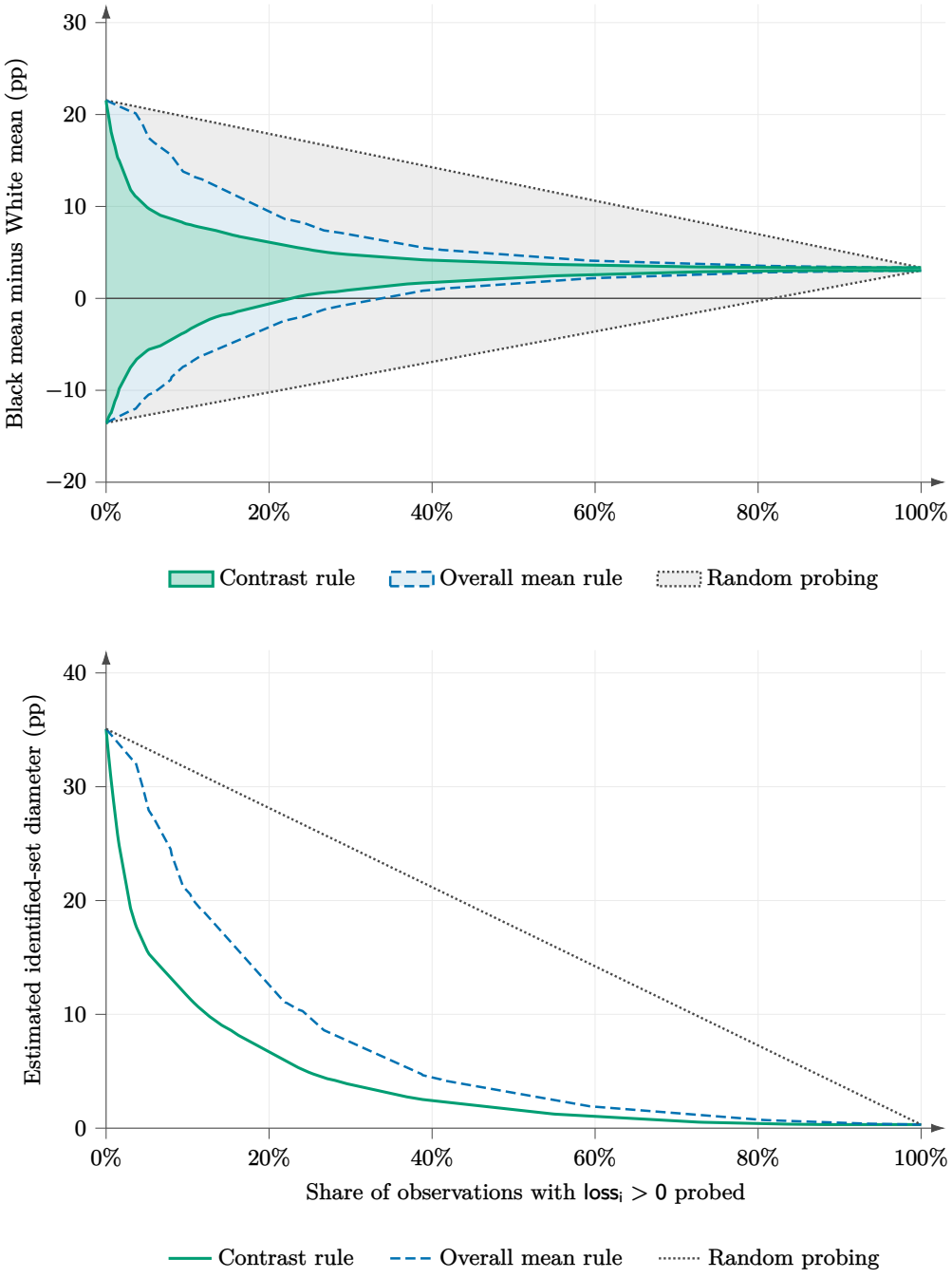


Figure 4: Estimated identified set for the Black-White mean job-insecurity contrast (top) and identified-set diameter (bottom) under alternative probing rules. Where probing is randomised, endpoints and diameters are averaged over realisations.

Because initial coarseness is common across respondents, variation in probing value arises entirely from variation in the probability of success. This probability is unknown when the probing decision is made. I assume, however, that the researcher can observe, and condition probing on, responses to earlier parts of the survey. I estimate probing success probability conditional on earlier responses using regularised logit. Table 2 compares the performance of the plug-in probing rule that uses these estimates with untargeted probing and an infeasible oracle rule that conditions probing on realised probing success. Compared to the job insecurity application, targeted-probing gains are modest in relative terms: 30.8% of full-probing gain at a 25% probing share, for example. This is largely because probing gains are more evenly distributed across observations here (see Corollary A.5). In absolute terms, the gains over untargeted probing are similar across both applications: 4.3 additional percentage points of identified-set diameter reduction here as against 6.5 percentage points in the job-insecurity application. The ex-post oracle provides a benchmark for the gains from perfectly predicting probing success. The plug-in captures a larger fraction of oracle gains over untargeted probing at small budgets than at larger budgets—71% at 10% probing share versus 33% at 75% probing share.

Dominitz and Manski (2011) note that nonresponse rates are substantially higher for those with less education. The targeted rule presented here exploits this difference to preferentially probe individuals with more education. Indeed, at 25% of respondents probed, the identified-set diameters for the conditional means by schooling levels are: 91.4pp for no postsecondary education; 82.1pp for some postsecondary education; and 63.8pp for bachelor's degree or higher. Following Section 4.3, I add two additional linear constraints,

$$\frac{1}{n_L}\sum_{i\in L}\widehat{v}_i\phi_i = \frac{1}{n_M}\sum_{i\in M}\widehat{v}_i\phi_i = \frac{1}{n_H}\sum_{i\in H}\widehat{v}_i\phi_i,$$

where n_L, n_M, and n_H are the numbers of low, middle, and high education respondents in the pilot sample, and $\widehat{v}_i$ is individual i's predicted probability of responding to the survey item. These constraints impose equal predicted identified-set diameter reduction for the three conditional means in the pilot sample. At a 25% realised probing share, adding these constraints increases the identified-set diameter for the overall mean from 77.4pp to 78.1pp. However, the disparity across educational groups is substantially reduced: the three diameters above become 81.1pp, 78.1pp, and 76.1pp, respectively.

Probe share (%)	Full-probing gain share (%)		
	Random probing	Plug-in rule	Ex-post oracle
10	10.0	12.6	13.6
25	25.0	30.8	34.0
50	50.0	59.8	68.1
75	75.0	83.3	100.0

Table 2: Full-probing gain shares for mean subjective probability of a positive nominal stock return under alternative probing rules. Gain shares measure the fraction of the full-probing identified-set diameter reduction attained. Probe shares are relative to the 1,104 evaluation respondents in waves 13 and 14.

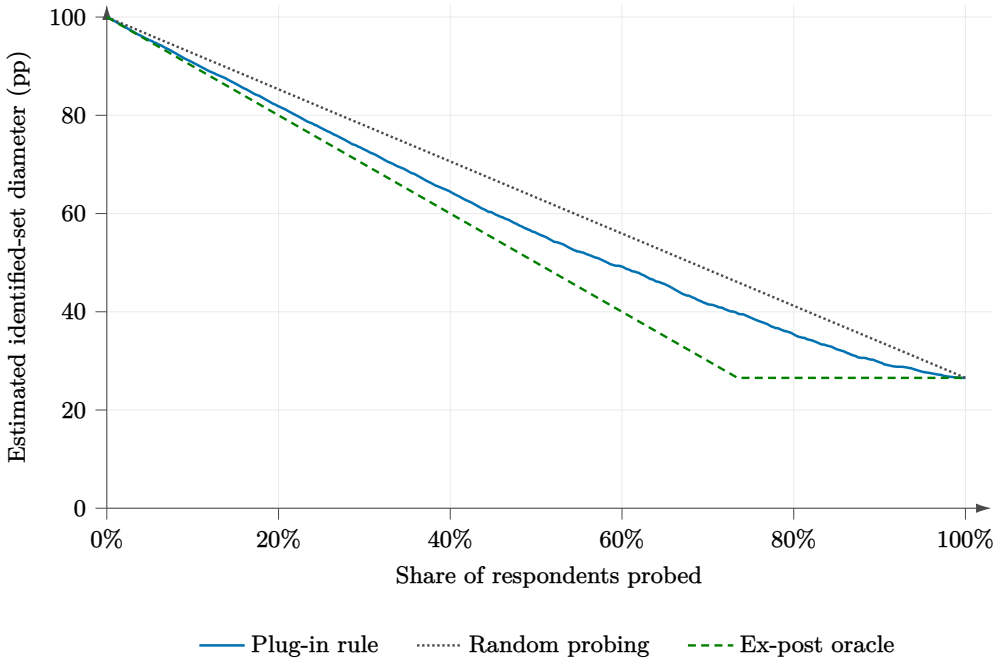


Figure 5: Estimated identified-set diameter for mean subjective probability of a positive nominal stock return under alternative probing rules.

6 Conclusion

I study the allocation of a limited budget to optimally probe coarse observations. The goal is to minimise identification uncertainty, measured by the diameter of the identified set. I show that for scalar expectations and univariate distributions under Wasserstein-1 distance, identified-set diameter decomposes into the expectation of observation-level coarseness. When probing cost also decomposes additively over observations, the optimal rule takes a simple form: probe observations in decreasing order of expected diameter reduction per unit cost, randomising when needed to exhaust the budget.

I apply this rule to two retrospective counterfactual survey-design problems. Both applications demonstrate that selectively targeted probing can significantly reduce identification uncertainty relative to untargeted probing. The job-insecurity application shows that the choice of target parameter can be important, changing the value of probing particular observations and thereby changing the optimal rule. The stock-return application shows that a plug-in rule using predicted probe effectiveness can improve on untargeted probing even when initial coarseness is identical across respondents. The stock-return application also illustrates the incorporation of additional linear constraints: requiring equal predicted diameter reductions across schooling groups substantially reduces disparities in uncertainty at a small cost in precision for the overall mean.

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A Proofs and Related Results

A.1 Notation

Notation	Meaning
$\mathcal{B}(\mathcal{X})$	Borel σ-algebra on a topological space $\mathcal{X}$
$\mathcal{P}(\mathcal{X})$	Family of all Borel probability measures on $\mathcal{X}$
E_μ	Expectation under probability measure μ
$\operatorname{diam}_d A$	Diameter of set A under metric d
$\operatorname{cl} A,\ \operatorname{conv} A$	Closure and convex hull of A
$\mathbf{z} \in \mathcal{Z},\ \mathbf{y} \in \mathcal{Y}$	Observed data and latent outcome
$\mathbf{z}_0 \in \mathcal{Z}_0$	Pre-probing observed data
$\mathbf{z}_1 \in \mathcal{Z}_1$	Probing outcome or refinement
P	Generic observed-data distribution
P_0	Observed distribution of the pre-probing data $\mathbf{z}_0$
P_{01}	Joint distribution of $(\mathbf{z}_0, \mathbf{z}_1)$, with $\mathbf{z}_0$-marginal P_0
$\mathsf{E}_0, \mathsf{E}_{01}$	Expectations under $\mathsf{P}_0, \mathsf{P}_{01}$, respectively
$Y(z)$	Generic outcome set
$Y_0(z_0),\ Y_1(z_0, z_1)$	Pre- and post-probing outcome sets
$Y_C(z_0, z_1)$	Outcome set induced by probing region C
$g^U(z),\ g^L(z)$	Supremum and infimum of $g(\{z\} \times Y(z))$
$\Delta(z)$	Image diameter, $\operatorname{diam} g(\{z\} \times Y(z))$, up to P-a.e. equality
$\Delta_0(z_0),\ \Delta_1(z_0, z_1)$	Pre-probing and realised post-probing image diameters
$\overline{\Delta}_1(z_0)$	Conditional mean post-probing image diameter
$\Delta_C(z_0, z_1)$	Image diameter induced by probing region C
$v(z_0)$	Marginal value of probing, $\Delta_0(z_0) - \overline{\Delta}_1(z_0)$
$\mathsf{R} \in \mathcal{R}(Y, \mathsf{P})$	Admissible joint distribution of observed and latent data
$\Theta_\tau(Y, \mathsf{P})$	Identified set for target parameter $\tau(\mathsf{R})$
$C \in \mathcal{B}(\mathcal{Z}_0)$	Probing region, inducing the nonrandomised rule $\mathbf{1}_C$
$(q_j, C_j)_{j \in \mathbb{N}}$	Countably supported randomised probing rule, with $\sum_{j=1}^{\infty} q_j = 1$
$\phi \in \Phi,\ \Phi_b$	Measurable behavioural probing rules $\phi : \mathcal{Z}_0 \to [0, 1]$ and the budget-feasible subfamily
$c(z_0),\ b$	Pointwise probing cost and probing budget
$L(\phi)$	Expected post-probing identified-set diameter, $\mathsf{E}_0[\Delta_0(\mathbf{z}_0) - v(\mathbf{z}_0)\phi(\mathbf{z}_0)]$
$\lambda^*,\ \gamma^*(z_0)$	Optimal threshold and boundary-randomisation probability
b^U	Expected cost of probing all observations
$L(0),\ L^*(b)$	Loss under zero probing and optimal loss at budget b

Table 3: Key notation. Bold lower-case symbols denote random variables or vectors; non-bold counterparts denote realisations.

A.2 Measurability Preliminaries

Definition A.1 (Completion)**.** Let $(\Omega, \mathcal{F}, \mu)$ be a probability space and define

$$\mathcal{F}_\mu := \sigma\left(\mathcal{F} \cup \{N \subseteq E \mid E \in \mathcal{F}, \mu(E) = 0\}\right).$$

That is, $\mathcal{F}_\mu$ is the smallest σ-algebra containing both $\mathcal{F}$ and the subsets of μ-null sets of $\mathcal{F}$. Call $(\Omega, \mathcal{F}_\mu, \mu)$ the *completion* of $(\Omega, \mathcal{F}, \mu)$.

In the following lemmas, let $(\Omega, \mathcal{F}, \mu)$ be a probability space, and $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ a Polish space[7] equipped with its Borel σ-algebra.

Lemma A.1 (Measurable versions)**.** *A function $f : \Omega \to \mathcal{X}$ is $\mathcal{F}_\mu$-measurable if and only if there exists an $\mathcal{F}$-measurable function $h : \Omega \to \mathcal{X}$ such that $h = f$ μ-a.s. In particular, for real-valued functions, if f is bounded or nonnegative then h can be chosen to have the same property.*

Proof. The first equivalence is immediate from Kallenberg (2021, Lemma 1.27). We prove the additional claim about real-valued f. Suppose that $\ell \leqslant f \leqslant u$ for some $\ell, u \in \mathbb{R}$. Then $\ell \leqslant h \leqslant u$ μ-a.s., and so the function $\min\{\max\{h, \ell\}, u\}$ is $\mathcal{F}$-measurable, almost surely equal to f, and satisfies the given bounds. If f is nonnegative, replace h with $\max\{h, 0\}$. □

Lemma A.2 (Measurable envelopes, diameters, and approximate selections)**.** *Let $\Gamma : \Omega \rightrightarrows \mathcal{X}$ be a nonempty-valued correspondence with an $(\mathcal{F} \otimes \mathcal{B}(\mathcal{X}))$-measurable graph, and $f : \Omega \times \mathcal{X} \to \mathbb{R}^J$ a bounded $(\mathcal{F} \otimes \mathcal{B}(\mathcal{X}))$-measurable function.*

(i) Let $J = 1$. The envelope function $f^U(\omega) := \sup_{x \in \Gamma(\omega)} f(\omega, x)$ is $\mathcal{F}_\mu$-measurable and bounded. Hence there exists an $\mathcal{F}$-measurable and bounded function $h : \Omega \to \mathbb{R}$ satisfying

$$h(\omega) = f^U(\omega) \qquad \mu\text{-a.e.}$$

(ii) Let $J = 1$. Fix $\varepsilon > 0$ and let the correspondence $\Gamma^\varepsilon : \Omega \rightrightarrows \mathcal{X}$ be defined by

$$\Gamma^\varepsilon(\omega) := \left\{ x \in \Gamma(\omega) \;\middle|\; f(\omega, x) \geqslant \sup_{x' \in \Gamma(\omega)} f(\omega, x') - \varepsilon \right\}.$$

Then there exists an $\mathcal{F}$-measurable $s^\varepsilon : \Omega \to \mathcal{X}$ such that

$$s^\varepsilon(\omega) \in \Gamma^\varepsilon(\omega) \qquad \mu\text{-a.e.} \tag{11}$$

[7] A separable, completely metrisable topological space.

(iii) Equip $\mathbb{R}^J$ with a norm $\|\cdot\|$. The diameter function

$$\omega \mapsto \operatorname{diam}_{\|\cdot\|} f(\{\omega\} \times \Gamma(\omega))$$

is $\mathcal{F}_\mu$-measurable and bounded. Hence there exists a bounded nonnegative $\mathcal{F}$-measurable function $h : \Omega \to \mathbb{R}$ satisfying

$$h(\omega) = \operatorname{diam}_{\|\cdot\|} f(\{\omega\} \times \Gamma(\omega)) \qquad \mu\text{-a.e.}$$

Proof.

(i) For all $t \in \mathbb{R}$,

$$\begin{aligned}\{\omega \in \Omega \mid f^U(\omega) > t\} &= \{\omega \in \Omega \mid \text{there exists } x \in \Gamma(\omega) \text{ with } f(\omega, x) > t\} \\ &= \operatorname{proj}_\Omega(\operatorname{Gr}\Gamma \cap f^{-1}(t, \infty)),\end{aligned}$$

where $\operatorname{proj}_\Omega : \Omega \times \mathcal{X} \to \Omega$ is the projection mapping of $\Omega \times \mathcal{X}$ onto Ω. $\operatorname{Gr}\Gamma$ is measurable by assumption: $\operatorname{Gr}\Gamma \in \mathcal{F} \otimes \mathcal{B}(\mathcal{X})$. The preimage of an open interval under the measurable function f is $\mathcal{F} \otimes \mathcal{B}(\mathcal{X})$-measurable. It follows that $\operatorname{Gr}\Gamma \cap f^{-1}(t, \infty) \in \mathcal{F} \otimes \mathcal{B}(\mathcal{X})$. The projection of a set in $\mathcal{F} \otimes \mathcal{B}(\mathcal{X})$ onto Ω is measurable in the μ-completion of $\mathcal{F}$ by the Projection Theorem (Aliprantis and Border, 2006, Theorem 18.25). Therefore, $\{\omega \in \Omega \mid f^U(\omega) > t\} \in \mathcal{F}_\mu$, for all $t \in \mathbb{R}$, and so f^U is an $\mathcal{F}_\mu$-measurable function. Since f is bounded, so too is f^U. A function $h : \Omega \to \mathbb{R}$ with the required properties exists by Lemma A.1.

(ii) Denote $f^U(\omega) := \sup_{x \in \Gamma(\omega)} f(\omega, x)$. By Part (i), f^U is $\mathcal{F}_\mu$-measurable. Then the mapping

$$(\omega, x) \mapsto f^U(\omega) - f(\omega, x)$$

is $\mathcal{F}_\mu \otimes \mathcal{B}(\mathcal{X})$-measurable. Thus,

$$\operatorname{Gr}\Gamma^\varepsilon = \operatorname{Gr}\Gamma \cap \{(\omega, x) \mid f^U(\omega) - f(\omega, x) \leqslant \varepsilon\} \in \mathcal{F}_\mu \otimes \mathcal{B}(\mathcal{X}).$$

It follows that Γ^ε admits an $\mathcal{F}_\mu$-measurable selection $s_0^\varepsilon : \Omega \to \mathcal{X}$, by the Aumann Selection Theorem (Aliprantis and Border, 2006, Theorem 18.26). By Lemma A.1, there exists an $\mathcal{F}$-measurable $s^\varepsilon : \Omega \to \mathcal{X}$ that is μ-a.e. equal to s_0^ε, and thus satisfies the selection condition (11).

(iii) Graph-measurability of Γ implies graph-measurability of the correspondence $\omega \rightrightarrows \Gamma(\omega) \times \Gamma(\omega)$, since

$$\operatorname{Gr}(\Gamma \times \Gamma) = \{(\omega, x, x') \mid (\omega, x), (\omega, x') \in \operatorname{Gr}\Gamma\} = \operatorname{proj}_{12}^{-1}(\operatorname{Gr}\Gamma) \cap \operatorname{proj}_{13}^{-1}(\operatorname{Gr}\Gamma),$$

where proj_{12} and proj_{13} denote the projections onto coordinates $(1,2)$ and $(1,3)$, respectively. Since f is measurable and norms are continuous, the mapping $(\omega, x, x') \mapsto \|f(\omega,x) - f(\omega,x')\|$ is also measurable. Then, by part (i),

$$\Delta_f(\omega) := \sup_{x,x' \in \Gamma(\omega)} \|f(\omega,x) - f(\omega,x')\| = \text{diam}_{\|\cdot\|} f(\{\omega\} \times \Gamma(\omega))$$

is $\mathcal{F}_\mu$-measurable. Since f is bounded, so too is Δ_f. The existence of a function $h : \Omega \to \mathbb{R}_{\geqslant 0}$ with the required properties is implied by Lemma A.1. □

A.3 Data Coarsening and Partial Identification

Remark A.1 (Assumption conventions). Standing assumptions such as the following are maintained from the point at which they are introduced, unless explicitly relaxed. This is distinct from assumptions that are only maintained locally, which are explicitly cited whenever they are assumed.

Standing Assumption A.1 (Borel measurability)**.** $\mathcal{Y}$ and $\mathcal{Z}$ are Polish spaces. Unless otherwise stated: Every Polish space $\mathcal{X}$ is equipped with its Borel σ-algebra $\mathcal{B}(\mathcal{X})$. If a probability measure is defined on $\mathcal{X}$, it is understood to be a measure on the measurable space $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$. The collection of all such Borel measures on $\mathcal{X}$ is denoted $\mathcal{P}(\mathcal{X})$, and is equipped with the topology of weak convergence.[8]

Standing Assumption A.2 (Generic outcome-set correspondence regularity)**.** The generic outcome-set correspondence $Y : \mathcal{Z} \rightrightarrows \mathcal{Y}$ is nonempty- and compact-valued, and has a measurable graph.

Lemma A.3. *$\mathcal{R}(Y, \mathsf{P})$ is nonempty.*

Proof. By Standing Assumption A.2, Y is nonempty- and compact-valued, has a measurable graph, and maps into a Polish space. It follows that Y admits a P-a.e. measurable selection $s : \mathcal{Z} \to \mathcal{Y}$ by the Aumann Selection Theorem (Aliprantis and Border, 2006, Theorem 18.26) and Lemma A.1. Define the pushforward measure $\mathsf{R}_s := (\text{id}, s)_\# \mathsf{P}$. For all measurable sets $A \subseteq \mathcal{Z}$, $\mathsf{R}_s(A \times \mathcal{Y}) = \mathsf{P}(A)$, so R_s has $\mathbf{z}$-marginal P. And R_s places probability 1 on $\{(z,y) \mid y = s(z)\}$ and thus on $\{(z,y) \mid y \in Y(z)\}$. Therefore, $\mathsf{R}_s \in \mathcal{R}(Y, \mathsf{P})$. □

Lemma A.4. *$\mathcal{R}(Y, \mathsf{P})$ is a convex set. Thus, if $\tau : \mathcal{R}(Y, \mathsf{P}) \to \mathcal{T}$ is affine, then $\Theta(Y, \mathsf{P})$ is also a convex set.*

Proof. Fix $\mathsf{R}, \mathsf{R}' \in \mathcal{R}(Y, \mathsf{P})$ and $\alpha \in [0,1]$, and define $\mathsf{R}'' = \alpha \mathsf{R} + (1-\alpha)\mathsf{R}'$. Then

$$\mathsf{R}''_{\mathbf{z}} = \alpha \mathsf{R}_{\mathbf{z}} + (1-\alpha)\mathsf{R}'_{\mathbf{z}} = \alpha \mathsf{P} + (1-\alpha)\mathsf{P} = \mathsf{P},$$

[8] The topology of weak convergence is the coarsest topology such that $\pi \mapsto \int_{\mathcal{X}} f \, \mathrm{d}\pi$ is continuous for every bounded continuous function $f : \mathcal{X} \to \mathbb{R}$.

and

$$\mathsf{R}''(\mathbf{y} \in Y(\mathbf{z})) = \alpha \mathsf{R}(\mathbf{y} \in Y(\mathbf{z})) + (1-\alpha)\mathsf{R}'(\mathbf{y} \in Y(\mathbf{z})) = \alpha + 1 - \alpha = 1.$$

Therefore, $\mathsf{R}'' \in \mathcal{R}(Y, \mathsf{P})$. The image of a convex set under an affine transformation is convex. □

Standing Assumption A.3. The function $g : \mathcal{Z} \times \mathcal{Y} \to \mathbb{R}$ is bounded and measurable.

Definition A.2. Let $\Delta : \mathcal{Z} \to \mathbb{R}_{\geqslant 0}$ be a bounded nonnegative measurable function such that

$$\Delta(z) = \operatorname{diam} g(\{z\} \times Y(z)) \quad \mathsf{P}\text{-a.e.}$$

Such a function exists, by Lemma A.2(iii).

Proof of Proposition 1. Let $\mathsf{R} \in \mathcal{R}(Y, \mathsf{P})$. By Lemma A.2(i) applied to g and $-g$, there exist measurable $g^L, g^U : \mathcal{Z} \to \mathbb{R}$ satisfying

$$g^L(z) = \inf_{y \in Y(z)} g(z,y), \quad g^U(z) = \sup_{y \in Y(z)} g(z,y) \qquad \mathsf{P}\text{-a.e.}$$

Since $\mathsf{R}(\mathbf{y} \in Y(\mathbf{z})) = 1$ and $\mathsf{R}_{\mathbf{z}} = \mathsf{P}$,

$$g^L(z) = \inf_{y \in Y(z)} g(z,y) \leqslant g(z,y) \leqslant \sup_{y \in Y(z)} g(z,y) = g^U(z) \qquad \mathsf{R}\text{-a.e.}$$

Monotonicity of expectations then implies

$$\mathsf{E}_{\mathsf{P}}\left[g^L(\mathbf{z})\right] \leqslant \mathsf{E}_{\mathsf{R}}\left[g(\mathbf{z},\mathbf{y})\right] \leqslant \mathsf{E}_{\mathsf{P}}\left[g^U(\mathbf{z})\right].$$

Since this holds for every $\mathsf{R} \in \mathcal{R}(Y, \mathsf{P})$,

$$\sup_{\mathsf{R} \in \mathcal{R}(Y,\mathsf{P})} \mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] \leqslant \mathsf{E}_{\mathsf{P}}\left[g^U(\mathbf{z})\right], \tag{12}$$

and

$$\inf_{\mathsf{R} \in \mathcal{R}(Y,\mathsf{P})} \mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] \geqslant \mathsf{E}_{\mathsf{P}}\left[g^L(\mathbf{z})\right]. \tag{13}$$

Therefore,

$$\operatorname{cl}\Theta(Y, \mathsf{P}) \subseteq \left[\mathsf{E}_{\mathsf{P}}\left[g^L(\mathbf{z})\right], \mathsf{E}_{\mathsf{P}}\left[g^U(\mathbf{z})\right]\right].$$

The next step is to show that these bounds are sharp. For each $k \in \mathbb{N}$, Lemma A.2(ii), applied to g and $-g$, gives measurable selections $s_k^+, s_k^- : \mathcal{Z} \to \mathcal{Y}$ from Y such that

$$g(z, s_k^+(z)) \geqslant g^U(z) - \frac{1}{k}, \qquad g(z, s_k^-(z)) \leqslant g^L(z) + \frac{1}{k} \qquad \mathsf{P}\text{-a.e.}$$

Define pushforward measures

$$\mathsf{R}_k^+ := (\mathrm{id}, s_k^+)_{\#}\mathsf{P}, \qquad \mathsf{R}_k^- := (\mathrm{id}, s_k^-)_{\#}\mathsf{P}.$$

Both have $\mathbf{z}$-marginal P and place probability one on $\operatorname{Gr} Y$, and so belong to $\mathcal{R}(Y, \mathsf{P})$. Furthermore,

$$\mathsf{E}_{\mathsf{R}_k^+}\left[g(\mathbf{z}, \mathbf{y})\right] = \mathsf{E}_{\mathsf{P}}\left[g(\mathbf{z}, s_k^+(\mathbf{z}))\right] \geqslant \mathsf{E}_{\mathsf{P}}\left[g^U(\mathbf{z})\right] - \frac{1}{k},$$

and

$$\mathsf{E}_{\mathsf{R}_k^-}\left[g(\mathbf{z}, \mathbf{y})\right] = \mathsf{E}_{\mathsf{P}}\left[g(\mathbf{z}, s_k^-(\mathbf{z}))\right] \leqslant \mathsf{E}_{\mathsf{P}}\left[g^L(\mathbf{z})\right] + \frac{1}{k}.$$

Taking $k \to \infty$ and combining with inequalities (12) and (13),

$$\sup_{\mathsf{R} \in \mathcal{R}(Y,\mathsf{P})} \mathsf{E}_{\mathsf{R}}\left[g(\mathbf{z}, \mathbf{y})\right] = \mathsf{E}_{\mathsf{P}}\left[g^U(\mathbf{z})\right], \qquad \inf_{\mathsf{R} \in \mathcal{R}(Y,\mathsf{P})} \mathsf{E}_{\mathsf{R}}\left[g(\mathbf{z}, \mathbf{y})\right] = \mathsf{E}_{\mathsf{P}}\left[g^L(\mathbf{z})\right].$$

Consequently,

$$\operatorname{diam} \Theta(Y, \mathsf{P}) = \mathsf{E}_{\mathsf{P}}\left[g^U(\mathbf{z})\right] - \mathsf{E}_{\mathsf{P}}\left[g^L(\mathbf{z})\right] = \mathsf{E}_{\mathsf{P}}\left[\Delta(\mathbf{z})\right]. \qquad \square$$

A.4 Probing

A *randomised probing rule* is a countably supported probability distribution over nonrandomised rules. Write its support as $C_1, C_2, \ldots$, with respective probabilities $q_1, q_2, \ldots$. Each randomised rule induces a *behavioural probing rule* $\phi : \mathcal{Z}_0 \to [0, 1]$, defined by[9]

$$\phi(z_0) = \sum_{j=1}^{\infty} q_j \mathbf{1}\{z_0 \in C_j\}.$$

A behavioural probing rule specifies only the marginal probability of probing at each observation; it does not determine the joint distribution of probing decisions across distinct values of z_0. Under this paper's assumptions, however, expected loss and expected cost do not depend on this cross-observation dependence (see Lemma A.8).

Lemma A.5 (Countable representation of behavioural probing rules)**.** *For any fixed measurable function $\phi : \mathcal{Z}_0 \to [0, 1]$, there exists a countable collection of measurable regions $(C_j)_{j=1}^{\infty}$ such that*

$$\phi(z_0) = \sum_{j=1}^{\infty} 2^{-j} \mathbf{1}\{z_0 \in C_j\} \tag{14}$$

for every $z_0 \in \mathcal{Z}_0$.

Proof. Every $x \in [0, 1]$ admits a binary representation

$$x = \sum_{j=1}^{\infty} 2^{-j} b_j(x),$$

[9]The distinction between randomised and behavioural probing rules parallels that between randomised and behavioural decision rules in Ferguson (1967, Section 1.5, Definitions 1 and 2).

where each $b_j : [0,1] \to \{0,1\}$ is measurable (Kallenberg, 2021, proof of Lemma 4.21). Applied pointwise to the values of ϕ,

$$\phi(z_0) = \sum_{j=1}^{\infty} 2^{-j} b_j(\phi(z_0)).$$

For $j \in \mathbb{N}$, define

$$C_j := \{z_0 \in \mathcal{Z}_0 \mid b_j(\phi(z_0)) = 1\}.$$

Since both b_j and ϕ are measurable functions, C_j is a measurable set. The representation (14) now follows. □

Standing Assumption A.4 (Outcome-set correspondence regularity)**.** $\mathcal{Y}$, $\mathcal{Z}_0$, and $\mathcal{Z}_1$ are Polish spaces. The pre- and post-probing outcome-set correspondences, $Y_0 : \mathcal{Z}_0 \rightrightarrows \mathcal{Y}$ and $Y_1 : \mathcal{Z}_0 \times \mathcal{Z}_1 \rightrightarrows \mathcal{Y}$ are nonempty- and compact-valued, and have measurable graphs

$$\operatorname{Gr} Y_0 := \{(z_0, y) \mid y \in Y_0(z_0)\} \in \mathcal{B}(\mathcal{Z}_0 \times \mathcal{Y}),$$
$$\operatorname{Gr} Y_1 := \{(z_0, z_1, y) \mid y \in Y_1(z_0, z_1)\} \in \mathcal{B}(\mathcal{Z}_0 \times Z_1 \times \mathcal{Y}).$$

Lemma A.6. *For every measurable $C \subseteq \mathcal{Z}_0$, $Y_C : \mathcal{Z}_0 \times \mathcal{Z}_1 \rightrightarrows \mathcal{Y}$ is nonempty- and compact-valued, and has a measurable graph.*

Proof. Since $Y_C(z_0, z_1) \in \{Y_0(z_0), Y_1(z_0, z_1)\}$ for all $(z_0, z_1) \in \mathcal{Z}_0 \times \mathcal{Z}_1$, Y_C is nonempty- and compact-valued. The graph of Y_C can be decomposed as

$$G_0 \cup G_1,$$

where

$$G_0 := \{(z_0, z_1, y) \mid z_0 \notin C \text{ and } y \in Y_0(z_0)\}$$

and

$$G_1 := \{(z_0, z_1, y) \mid z_0 \in C \text{ and } y \in Y_1(z_0, z_1)\}.$$

Since C and the graphs of Y_0 and Y_1 are measurable, these sets are also measurable. □

Definition A.3. Let $\Delta_0 : \mathcal{Z}_0 \to \mathbb{R}_{\geqslant 0}$ be a bounded nonnegative measurable function such that

$$\Delta_0(z_0) = \operatorname{diam} g(\{z_0\} \times Y_0(z_0)) \quad \mathsf{P}_0\text{-a.e.}$$

Let $\Delta_1 : \mathcal{Z}_0 \times \mathcal{Z}_1 \to \mathbb{R}_{\geqslant 0}$ be a bounded nonnegative measurable function such that

$$\Delta_1(z_0, z_1) = \operatorname{diam} g(\{z_0\} \times Y_1(z_0, z_1)) \quad \mathsf{P}_{01}\text{-a.e.}$$

Such functions exist by Lemma A.2(iii).

The following lemma is immediate from Definition A.3 and the restriction $Y_1(z_0, z_1) \subseteq Y_0(z_0)$.

Lemma A.7. $\overline{\Delta}_1, v : \mathcal{Z}_0 \to \mathbb{R}$ *defined by*

$$\overline{\Delta}_1(z_0) := \mathsf{E}_{01}\left[\Delta_1(\mathbf{z}_0, \mathbf{z}_1) \mid \mathbf{z}_0 = z_0\right]$$

and

$$v(z_0) := \Delta_0(z_0) - \overline{\Delta}_1(z_0)$$

are bounded, nonnegative, and measurable.

Define the loss of a randomised probing rule by

$$L((q_j, C_j)_{j=1}^{\infty}) := \sum_{j=1}^{\infty} q_j \operatorname{diam}_d \Theta_\tau(Y_{C_j}, \mathsf{P}_{01}).$$

Lemma A.8 (Loss decomposition)**.** *Under Assumption 1,*

$$\begin{aligned} L((q_j, C_j)_{j=1}^{\infty}) &= \mathsf{E}_{01}\left[\Delta_1(\mathbf{z}_0, \mathbf{z}_1)\phi(\mathbf{z}_0) + \Delta_0(\mathbf{z}_0)(1 - \phi(\mathbf{z}_0))\right] \\ &= -\mathsf{E}_0\left[v(\mathbf{z}_0)\phi(\mathbf{z}_0)\right] + \mathsf{E}_0\left[\Delta_0(\mathbf{z}_0)\right]. \end{aligned}$$

Proof. Assumption 1 implies

$$L((q_j, C_j)_{j=1}^{\infty}) = \sum_{j=1}^{\infty} q_j \mathsf{E}_{01}\left[\Delta_{C_j}(\mathbf{z}_0, \mathbf{z}_1)\right] = \mathsf{E}_{01}\left[\sum_{j=1}^{\infty} q_j \Delta_{C_j}(\mathbf{z}_0, \mathbf{z}_1)\right].$$

The integrand can be written

$$\begin{aligned} \sum_{j=1}^{\infty} q_j \Delta_{C_j}(z_0, z_1) &= \sum_{j=1}^{\infty} q_j \Big(\Delta_1(z_0, z_1) 1\{z_0 \in C_j\} + \Delta_0(z_0) 1\{z_0 \notin C_j\}\Big) \\ &= \Delta_1(z_0, z_1) \sum_{j=1}^{\infty} q_j 1\{z_0 \in C_j\} + \Delta_0(z_0)\left(1 - \sum_{j=1}^{\infty} q_j 1\{z_0 \in C_j\}\right). \end{aligned}$$

Substituting in $\phi(z_0) = \sum_{j=1}^{\infty} q_j 1\{z_0 \in C_j\}$,

$$\begin{aligned} L((q_j, C_j)_{j=1}^{\infty}) &= \mathsf{E}_{01}\left[\Delta_1\phi + \Delta_0 \cdot (1 - \phi)\right] \\ &= \mathsf{E}_0\left[\overline{\Delta}_1\phi + \Delta_0 \cdot (1 - \phi)\right] \\ &= -\mathsf{E}_0\left[v\phi\right] + \mathsf{E}_0\left[\Delta_0\right], \end{aligned}$$

where the second equality uses the law of iterated expectations and the third uses $v := \Delta_0 - \overline{\Delta}_1$. □

Lemma A.9. *Let $f_v : \mathcal{Z}_0 \to \mathbb{R}_{\geqslant 0}$ and $f_c : \mathcal{Z}_0 \to \mathbb{R}_{>0}$ be bounded measurable functions, and let $b > 0$. The problem*

$$\sup_{\phi \in \Phi} \mathsf{E}_0\left[f_v(\mathbf{z}_0)\phi(\mathbf{z}_0)\right] \quad \text{s.t.} \quad \mathsf{E}_0\left[f_c(\mathbf{z}_0)\phi(\mathbf{z}_0)\right] \leqslant b \tag{15}$$

is solved by a feasible rule ϕ^* if and only if, for some $\lambda^* \geqslant 0$ and some measurable $\gamma^* : \mathcal{Z}_0 \to [0,1]$,

$$\phi^*(z_0) = \mathbf{1}\left\{\frac{f_v(z_0)}{f_c(z_0)} > \lambda^*\right\} + \gamma^*(z_0)\mathbf{1}\left\{\frac{f_v(z_0)}{f_c(z_0)} = \lambda^*\right\} \quad \mathsf{P}_0\text{-}a.e. \tag{16}$$

and

$$\lambda^*\left(b - \mathsf{E}_0\left[f_c(\mathbf{z}_0)\phi^*(\mathbf{z}_0)\right]\right) = 0.$$

Proof. If $\mathsf{E}_0[f_v] = 0$ then every feasible ϕ is optimal and satisfies the threshold formulation and complementary slackness condition with $\lambda^* = 0$ and $\gamma^* = \phi^*$. If $b \geqslant \mathsf{E}_0[f_c\mathbf{1}_{\{f_v>0\}}]$ then the optimal rules are those that take value 1 a.e. on $\{z_0 \mid f_v(z_0) > 0\}$. Such rules satisfy the threshold formulation and complementary slackness condition with $\lambda^* = 0$ and $\gamma^* = \phi^*$. Suppose, then, that

$$0 < b < \mathsf{E}_0[f_c\mathbf{1}_{\{f_v>0\}}] \quad \text{and} \quad \mathsf{E}_0[f_v] > 0.$$

Define probability measures P_c, P_v by

$$\mathsf{P}_c(A) := \frac{\mathsf{E}_0[f_c\mathbf{1}_A]}{\mathsf{E}_0[f_c]}, \qquad \mathsf{P}_v(A) := \frac{\mathsf{E}_0[f_v\mathbf{1}_A]}{\mathsf{E}_0[f_v]}.$$

Denote the corresponding expectations by E_c and E_v. Then the budget-allocation problem (15) can be written as

$$\sup_{\phi\in\Phi} \mathsf{E}_v\left[\phi\right] \quad \text{s.t.} \quad \mathsf{E}_c\left[\phi\right] \leqslant \frac{b}{\mathsf{E}_0\left[f_c\right]}.$$

This is the most-powerful-test problem solved by the Neyman-Pearson Lemma (Lehmann and Romano, 2022, Theorem 3.2.1), which says that a rule ϕ^* is optimal if and only if there exists a constant λ^* and a measurable function γ^* such that

$$\phi^*(z_0) = \begin{cases} 1, & f_v(z_0) > \lambda^* f_c(z_0), \\ \gamma^*(z_0), & f_v(z_0) = \lambda^* f_c(z_0), \\ 0, & f_v(z_0) < \lambda^* f_c(z_0), \end{cases} \qquad \mathsf{P}_0\text{-a.e.},$$

and

$$\mathsf{E}_c\left[\phi^*\right] = \frac{b}{\mathsf{E}_0\left[f_c\right]},$$

or equivalently,

$$\mathsf{E}_0[f_c\phi^*] = b. \qquad \square$$

Proof of Theorem 1. By Lemma A.8, the loss can be written as

$$L(\phi) = \mathsf{E}_0\left[\Delta_0\right] - \mathsf{E}_0\left[v\phi\right].$$

The solution set is unchanged if we subtract the constant term $\mathsf{E}_0\left[\Delta_0\right]$ from the objective. The problem can then be written as

$$\sup_{\phi\in\Phi} \mathsf{E}_0\left[v\phi\right] \quad \text{s.t.} \quad \mathsf{E}_0\left[c\phi\right] \leqslant b.$$

Since v is bounded, nonnegative, and measurable by Lemma A.7 in the appendix, and c is bounded, positive, and measurable by assumption, we can apply Lemma A.9. □

Corollary A.1 (Two-rule randomisation)**.** *The boundary randomisation function γ^* in Theorem 1 can be chosen to be constant. Accordingly, there exists an optimal rule that can be implemented by randomising between at most two nonrandomised rules,*

$$\phi^* = (1-q)\mathbf{1}_C + q\mathbf{1}_{C'},$$

for some $q \in [0,1]$, where

$$C := \left\{ z_0 \in \mathcal{Z}_0 \;\middle|\; \frac{v(z_0)}{c(z_0)} > \lambda^* \right\}, \qquad C' := \left\{ z_0 \in \mathcal{Z}_0 \;\middle|\; \frac{v(z_0)}{c(z_0)} \geqslant \lambda^* \right\}.$$

Proof. This follows directly from the proof of the Neyman-Pearson Lemma (Lehmann and Romano, 2022, Theorem 3.2.1). □

Corollary A.2 (Nonrandomisation under nonatomicity)**.** *Suppose that the restriction of P_0 to the boundary set*

$$\left\{ z_0 \in \mathcal{Z}_0 \;\middle|\; \frac{v(z_0)}{c(z_0)} = \lambda^* \right\}$$

is nonatomic. Then the optimal rule (4) can be chosen to be nonrandomised; that is, $\gamma^* : \mathcal{Z}_0 \to [0,1]$ can be chosen to take values in $\{0,1\}$ P_0-a.e.

Proof. Though the result was stated in the notation of Theorem 1, we prove it in the more general notation of Lemma A.9. We know that problem (15) is solved by a threshold rule $\phi^* : \mathcal{Z}_0 \to [0,1]$ of the form (16),

$$\phi^*(z_0) := \mathbf{1}\left\{ \frac{f_v(z_0)}{f_c(z_0)} > \lambda^* \right\} + \gamma^*(z_0)\mathbf{1}\left\{ \frac{f_v(z_0)}{f_c(z_0)} = \lambda^* \right\} \quad \mathsf{P}_0\text{-a.e.} \tag{17}$$

We want to prove that there exists a threshold rule $\phi^\dagger$ that satisfies

$$\mathsf{E}_0\left[f_c \phi^\dagger \right] = \mathsf{E}_0\left[f_c \phi^* \right] \tag{18}$$

and

$$\mathsf{E}_0[f_v \phi^\dagger] = \mathsf{E}_0[f_v \phi^*], \tag{19}$$

and differs from ϕ^* only in that a $\{0,1\}$-valued function $\gamma^\dagger$ replaces γ^* in (17). Denote the boundary set

$$A := \left\{ \frac{f_v(z_0)}{f_c(z_0)} = \lambda^* \right\}.$$

Since ϕ^* and $\phi^\dagger$ agree on A^{C}, it is sufficient to show that the two above equalities (18) and (19) hold on A. If $\mathsf{P}_0(A) = 0$ then ϕ^* is already nonrandomised and we are done. Therefore, assume that $\mathsf{P}_0(A) > 0$. Let P_c and E_c be as defined in the proof of Lemma A.9.

By hypothesis, the restriction of P_0 to A is nonatomic, and since P_c is absolutely continuous with respect to P_0, the restriction of P_c to A is also nonatomic. Let

$$\delta := \mathsf{E}_c\left[\gamma^* \mathbf{1}_A\right].$$

By the intermediate value property of nonatomic measures (Aliprantis and Border, 2006, Theorem 10.52), there exists a measurable set $A' \subseteq A$ such that

$$\mathsf{P}_c(A') = \delta.$$

Setting $\gamma^\dagger := \mathbf{1}_{A'}$,

$$\begin{aligned}\mathsf{E}_0\left[f_c \phi^\dagger \mathbf{1}_A\right] = \mathsf{E}_0\left[f_c \gamma^\dagger \mathbf{1}_A\right] = \mathsf{E}_0\left[f_c \mathbf{1}_{A'}\right] = \mathsf{E}_0[f_c]\ \mathsf{P}_c\left(A'\right) = \mathsf{E}_0[f_c]\ \delta \\ = \mathsf{E}_0[f_c]\ \mathsf{E}_c\left[\gamma^* \mathbf{1}_A\right] = \mathsf{E}_0\left[f_c \gamma^* \mathbf{1}_A\right] = \mathsf{E}_0\left[f_c \phi^* \mathbf{1}_A\right].\end{aligned}$$

Moreover, $f_v = \lambda^* f_c$ on A, so

$$\mathsf{E}_0\left[f_v \phi^\dagger \mathbf{1}_A\right] = \lambda^* \mathsf{E}_0\left[f_c \phi^\dagger \mathbf{1}_A\right] = \lambda^* \mathsf{E}_0\left[f_c \phi^* \mathbf{1}_A\right] = \mathsf{E}_0\left[f_v \phi^* \mathbf{1}_A\right]. \qquad \square$$

Corollary A.3 (Nonrandomised approximation)**.** *There exists a nonrandomised rule $\phi^0 \in \Phi_b$ such that*

$$0 \leqslant L(\phi^0) - L^*(b) \leqslant \sup_{z_0 \in \mathcal{Z}_0} \mathsf{P}_0(\{z_0\}) v(z_0).$$

Proof. Let ϕ^* be the optimal rule given by Theorem 1. If $\lambda^* = 0$ then take $\phi^0 := \mathbf{1}\{v > 0\}$. Suppose, then, that $\lambda^* > 0$. Using the notation in the proof of Corollary A.2, if the restriction of P_c to A is nonatomic, then this corollary gives a nonrandomised rule that attains $L^*(b)$ exactly. The same is true if this restriction has atoms, but the nonatomic part has mass at least δ. Otherwise, set $\gamma^*(z_0) = 1$ on the nonatomic part and note that the remaining mass of P_c on A is made of countably many atomic contributions. If $\delta = \mathsf{P}_c(A)$, the optimal rule is nonrandomised and we are done. If $\delta < \mathsf{P}_c(A)$, then we can obtain an optimal γ^* by assigning $\gamma^*(z_0) = 1$ to finitely many atoms, $\gamma^*(z_0) \in (0,1)$ to at most one atom z_0^*, and $\gamma^*(z_0) = 0$ to all other atoms. If no randomised atom is needed we are done. Otherwise, let ϕ^0 differ from ϕ^* only in assigning $\phi^0(z_0^*) = 0$. Then ϕ^0 is feasible and

$$\begin{aligned}L(\phi^0) - L^*(b) &= \mathsf{E}_0\left[v(\phi^* - \phi^0)\right] \\ &= \mathsf{P}_0(\{z_0^*\}) v(z_0^*) \phi^*(z_0^*) \\ &\leqslant \sup_{z_0 \in \mathcal{Z}_0} \mathsf{P}_0(\{z_0\}) v(z_0).\end{aligned}$$

$\square$

Corollary A.4 (Coarsely targeted probing)**.** *Fix a measurable mapping $\kappa : \mathcal{Z}_0 \to \mathcal{K} \subseteq \mathbb{R}^{d_\mathcal{K}}$. Let*

$$\Phi_\kappa := \{\phi \in \Phi \mid \phi = f \circ \kappa\ \mathsf{P}_0\text{-a.e. for some measurable } f : \mathcal{K} \to [0,1]\}.$$

That is, Φ_κ is the set of probing rules that are almost-surely constant on the level sets of κ. Define

$$v_\kappa(t) := \mathsf{E}_0[v(\mathbf{z}_0) \mid \kappa(\mathbf{z}_0) = t], \qquad c_\kappa(t) := \mathsf{E}_0[c(\mathbf{z}_0) \mid \kappa(\mathbf{z}_0) = t].$$

Then the problem

$$\inf_{\phi \in \Phi_b \cap \Phi_\kappa} L(\phi) \tag{20}$$

is solved by a pointwise threshold rule of the form

$$\phi^*(z_0) := \mathbf{1}\left\{\frac{v_\kappa(\kappa(z_0))}{c_\kappa(\kappa(z_0))} > \lambda^*\right\} + \gamma^*(z_0)\mathbf{1}\left\{\frac{v_\kappa(\kappa(z_0))}{c_\kappa(\kappa(z_0))} = \lambda^*\right\} \quad \mathsf{P}_0\text{-}a.e.,$$

where $\gamma^* : \mathcal{Z}_0 \to [0,1]$ is almost-surely constant on the level sets of κ.

Proof. Fix $\phi \in \Phi_b \cap \Phi_\kappa$. By definition of Φ_κ, there exists a measurable function $f : \mathcal{K} \to [0,1]$ satisfying

$$\phi(z_0) = (f \circ \kappa)(z_0) \qquad \mathsf{P}_0\text{-a.e.}$$

By the Doob-Dynkin lemma (Kallenberg, 2021, Lemma 1.14), there exist measurable functions $v_\kappa, c_\kappa : \mathcal{K} \to \mathbb{R}$ such that

$$v_\kappa(\kappa(\mathbf{z}_0)) = \mathsf{E}_0[v(\mathbf{z}_0) \mid \sigma(\kappa)], \qquad c_\kappa(\kappa(\mathbf{z}_0)) = \mathsf{E}_0[c(\mathbf{z}_0) \mid \sigma(\kappa)] \qquad \mathsf{P}_0\text{-a.s.},$$

where $\sigma(\kappa)$ is the σ-algebra generated by κ. By Lemma A.8 and the law of iterated expectations,

$$\begin{aligned}
L(\phi) - \mathsf{E}_0\left[\Delta_0(\mathbf{z}_0)\right] &= -\mathsf{E}_0\left[v(\mathbf{z}_0)\phi(\mathbf{z}_0)\right] \\
&= -\mathsf{E}_0[\mathsf{E}_0[v(\mathbf{z}_0)f(\kappa(\mathbf{z}_0)) \mid \sigma(\kappa)]] \\
&= -\mathsf{E}_0[\mathsf{E}_0[v(\mathbf{z}_0) \mid \sigma(\kappa)]f(\kappa(\mathbf{z}_0))] \\
&= -\mathsf{E}_0[v_\kappa(\kappa(\mathbf{z}_0))f(\kappa(\mathbf{z}_0))] \\
&= -\mathsf{E}_{\kappa_\# \mathsf{P}_0}[v_\kappa(\mathbf{t})f(\mathbf{t})].
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathsf{E}_0\left[c(\mathbf{z}_0)\phi(\mathbf{z}_0)\right] &= \mathsf{E}_0\left[\mathsf{E}_0[c(\mathbf{z}_0)f(\kappa(\mathbf{z}_0)) \mid \sigma(\kappa)]\right] \\
&= \mathsf{E}_0\left[\mathsf{E}_0[c(\mathbf{z}_0) \mid \sigma(\kappa)]f(\kappa(\mathbf{z}_0))\right] \\
&= \mathsf{E}_0\left[c_\kappa(\kappa(\mathbf{z}_0))f(\kappa(\mathbf{z}_0))\right] \\
&= \mathsf{E}_{\kappa_\# \mathsf{P}_0}\left[c_\kappa(\mathbf{t})f(\mathbf{t})\right].
\end{aligned}$$

Therefore, the coarse-targeting problem (20) can be written as

$$\sup_{f \in [0,1]^{\mathcal{K}}} \mathsf{E}_{\kappa_\# \mathsf{P}_0}\left[v_\kappa(\mathbf{t})f(\mathbf{t})\right] \quad \text{s.t.} \quad \mathsf{E}_{\kappa_\# \mathsf{P}_0}\left[c_\kappa(\mathbf{t})f(\mathbf{t})\right] \leqslant b,$$

where the supremum is taken over measurable f. Now apply Lemma A.9. □

Proof of Proposition 2. Expanding the budget set cannot increase the value function, so L^* is weakly decreasing. Fix $t \in [0,1]$ and $b_1, b_2 \in [0, b^U]$. If ϕ_1, ϕ_2 are the optimal rules at budgets b_1 and b_2, respectively, then $t\phi_1 + (1-t)\phi_2$ is feasible at $tb_1 + (1-t)b_2$ and attains loss

$$tL^*(b_1) + (1-t)L^*(b_2).$$

Therefore, L^* is convex. The untargeted rule $\phi \equiv b/b^U$ is feasible and attains diameter reduction $\frac{b}{b^U}(L(0) - L(1))$ so

$$L(0) - L^*(b) \geqslant \frac{b}{b^U}(L(0) - L(1)).$$

If v/c is constant P_0-a.e., then Theorem 1 says that untargeted probing is optimal, so this holds with equality. Conversely, if this holds with equality then untargeted probing is optimal. If $b \in (0, b^U)$, then the untargeted rule takes values in $(0,1)$ everywhere. But Theorem 1 says that any optimal rule takes value 1 a.e. on $\{v/c > \lambda^*\}$ and 0 a.e. on $\{v/c < \lambda^*\}$. Thus untargeted probing can be optimal only if both sets have probability zero. It follows that $v/c = \lambda^*$ P_0-a.e. Since $\mathsf{E}_0[v] = L(0) - L(1) > 0$ and $v = \lambda^* c$ a.e., $\lambda^* > 0$. □

Let Q_c be the quantile function of v/c under the cost-weighted probability distribution P_c defined in the proof of Lemma A.9. Define the corresponding Lorenz curve by

$$\mathrm{Lor}(\alpha) := \frac{\int_0^\alpha Q_c(s)\,\mathrm{d}s}{\int_0^1 Q_c(s)\,\mathrm{d}s}.$$

The dual Lorenz curve (Koshevoy and Mosler, 1996),

$$1 - \mathrm{Lor}(1-\alpha),$$

reverses the order of the standard Lorenz curve, ordering v/c from highest to lowest. The following corollary shows that the maximum fraction of the full-probing gain $L(0) - L(1)$ that can be attained at budget $b \in [0, b^U]$ is given by the value of the dual Lorenz curve at $\alpha = b/b^U$. As a consequence, if one cost-weighted distribution of v/c is more unequal than another in the sense of Lorenz dominance, a weakly larger fraction of its full-probing gain will be attainable at every fraction of full-probing cost.

Corollary A.5 (Lorenz representation)**.**

$$\frac{L(0) - L^*(b)}{L(0) - L(1)} = 1 - \mathrm{Lor}\left(1 - \frac{b}{b^U}\right).$$

In particular, suppose that $j = 1, 2$ index alternative distributions of $(v_j(\mathbf{z}_0), c_j(\mathbf{z}_0))$. Then

$$\mathrm{Lor}_1(\alpha) \leqslant \mathrm{Lor}_2(\alpha), \qquad \alpha \in [0,1]$$

if and only if

$$\frac{L_1(0) - L_1^*(\alpha b_1^U)}{L_1(0) - L_1(1)} \geqslant \frac{L_2(0) - L_2^*(\alpha b_2^U)}{L_2(0) - L_2(1)}, \qquad \alpha \in [0,1].$$

Proof. Consider the measurable correspondence $S : \mathcal{Z}_0 \rightrightarrows \mathbb{R}^2$ which maps z_0 to the line segment between $(0,0)$ and $(1, v(z_0)/c(z_0))$. Lemma A.5 says that Φ is the set of all measurable mappings $\phi : \mathcal{Z}_0 \to [0,1]$, so every measurable selection from S can be written $(\phi, \phi v/c)$, or equivalently, $(1-\phi, (1-\phi)v/c)$. Each such selection is a random variable, and the collection of their expectations is

$$\left\{ \left(\mathsf{E}_c[1-\phi], \mathsf{E}_c\left[(1-\phi)\frac{v}{c}\right] \right) \,\middle|\, \phi \in \Phi \right\}. \tag{21}$$

By Molchanov and Molinari (2018, Definition 3.1, Example 3.20 and subsequent discussion), this selection expectation is the lift zonoid of the random variable $v(\mathbf{z}_0)/c(\mathbf{z}_0)$ under P_c, and thus has lower boundary given by the scaled Lorenz curve

$$\left\{ \left(\alpha, \int_0^\alpha Q_c(s)\,\mathrm{d}s \right) \,\middle|\, \alpha \in [0,1] \right\}.$$

But

$$\begin{aligned} \left(\mathsf{E}_c[1-\phi], \mathsf{E}_c\left[(1-\phi)\frac{v}{c}\right] \right) &= \left(\frac{\mathsf{E}_0[c(1-\phi)]}{b^U}, \frac{\mathsf{E}_0[v(1-\phi)]}{b^U} \right) \\ &= \left(1 - \frac{\mathsf{E}_0[c\phi]}{b^U}, \frac{L(\phi) - L(1)}{b^U} \right). \end{aligned}$$

Because $v \geqslant 0$, at each $b \leqslant b^U$ there is an optimal rule that exhausts the budget: $L(\phi^*) = L^*(b)$ and $\mathsf{E}_0[c\phi^*] = b$. Thus, the lower boundary of the selection expectation (21) is given by

$$\left\{ \left(1 - \frac{b}{b^U}, \frac{L^*(b) - L(1)}{b^U} \right) \,\middle|\, b \in \left[0, b^U\right] \right\}.$$

Then

$$\frac{L^*(b) - L(1)}{b^U} = \int_0^{1-b/b^U} Q_c(s)\,\mathrm{d}s$$

for all $b \in [0, b^U]$. In particular, choosing $b = 0$ gives

$$\frac{L(0) - L(1)}{b^U} = \int_0^1 Q_c(s)\,\mathrm{d}s.$$

Therefore,

$$\frac{L^*(b) - L(1)}{L(0) - L(1)} = \mathrm{Lor}\left(1 - \frac{b}{b^U}\right).$$

The desired identity follows.

□

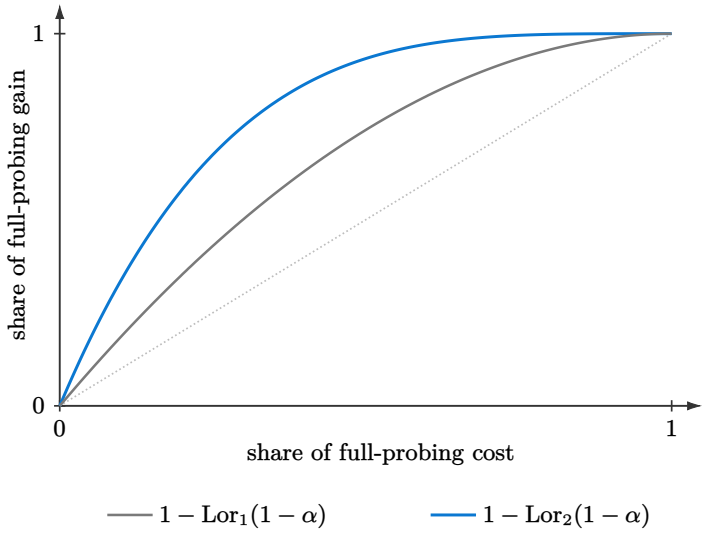


Figure 6: Visualisation of Corollary A.5. Plotting the maximal attainable share of full-probing gain, $\mathsf{E}_0[v\phi^*]/\mathsf{E}_0[v]$, against the share of full-probing cost, $\mathsf{E}_0[c\phi^*]/\mathsf{E}_0[c] = b/b^U$, gives the dual Lorenz curve of v/c under $z_0 \sim \mathsf{P}_c$. In particular, if v_1/c_1 is more equally distributed than v_2/c_2, in the sense that $\mathrm{Lor}_1 \geqslant \mathrm{Lor}_2$, then v_2/c_2 attains a weakly larger share of full-probing gain at every share of full-probing cost.

A.5 Distributional Targets

Proof of Proposition 3. Kantorovich-Rubinstein duality (Villani, 2009, Remark 6.5) implies that for any two probability distributions $\mu, \nu \in \mathcal{P}(\mathcal{G})$,

$$W_1(\mu,\nu) = \sup_{f \in \mathrm{Lip}_1(\mathcal{G})} \left| \int_{\mathcal{G}} f \,\mathrm{d}\mu - \int_{\mathcal{G}} f \,\mathrm{d}\nu \right|,$$

where $\mathrm{Lip}_1(\mathcal{G})$ is the space of 1-Lipschitz mappings from $\mathcal{G}$ into $\mathbb{R}$. Restricting to those distributions on $\mathcal{G}$ that are the pushforward under g of a distribution in $\mathcal{R}(Y,\mathsf{P})$,

$$\begin{aligned}
\mathrm{diam}_{W_1}\, \Theta_{\mathrm{dis}}(Y,\mathsf{P}) &= \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \sup_{f \in \mathrm{Lip}_1(\mathcal{G})} \left| \int_{\mathcal{G}} f \,\mathrm{d}g_{\#}\mathsf{R} - \int_{\mathcal{G}} f \,\mathrm{d}g_{\#}\mathsf{R}' \right| \\
&= \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \sup_{f \in \mathrm{Lip}_1(\mathcal{G})} \left| \mathsf{E}_{\mathsf{R}}[(f\circ g)(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[(f\circ g)(\mathbf{z},\mathbf{y})] \right| \\
&= \sup_{f \in \mathrm{Lip}_1(\mathcal{G})} \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \left| \mathsf{E}_{\mathsf{R}}[(f\circ g)(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[(f\circ g)(\mathbf{z},\mathbf{y})] \right| \\
&= \sup_{f \in \mathrm{Lip}_1(\mathcal{G})} \mathrm{diam}\left\{ \mathsf{E}_{\mathsf{R}}[(f\circ g)(\mathbf{z},\mathbf{y})] \mid \mathsf{R} \in \mathcal{R}(Y,\mathsf{P}) \right\}. \qquad (22)
\end{aligned}$$

Since f is 1-Lipschitz and real-valued, and $g : \mathcal{Z} \times \mathcal{Y} \to \mathbb{R}$ is bounded and measurable, $f \circ g$ is bounded, measurable, and real-valued. Applying Proposition 1 to the parameter

$\mathsf{E}_{\mathsf{R}}[(f \circ g)(\mathbf{z}, \mathbf{y})]$ gives

$$\operatorname{diam} \{\mathsf{E}_{\mathsf{R}}[(f \circ g)(\mathbf{z}, \mathbf{y})] \mid \mathsf{R} \in \mathcal{R}(Y, \mathsf{P})\} = \mathsf{E}_{\mathsf{P}} [\Delta_f(\mathbf{z})], \tag{23}$$

where

$$\Delta_f(z) := \operatorname{diam}(f \circ g)(\{z\} \times Y(z))$$

up to P-a.e. equivalence. Since f is 1-Lipschitz,

$$\Delta_f(z) \leqslant \Delta(z) \qquad \mathsf{P}\text{-a.e.}$$

This implies

$$\sup_{f \in \mathrm{Lip}_1(\mathcal{G})} \mathsf{E}_{\mathsf{P}}[\Delta_f(\mathbf{z})] \leqslant \mathsf{E}_{\mathsf{P}}[\Delta(\mathbf{z})].$$

In fact, this holds with equality, since the identity function is 1-Lipschitz. Combining this equality with equalities (22) and (23),

$$\operatorname{diam}_{W_1} \Theta_{\mathrm{dis}}(Y, \mathsf{P}) = \sup_{f \in \mathrm{Lip}_1(\mathcal{G})} \mathsf{E}_{\mathsf{P}} [\Delta_f(\mathbf{z})] = \mathsf{E}_{\mathsf{P}} [\Delta(\mathbf{z})] .$$

For all $\mathsf{R}, \mathsf{R}' \in \mathcal{R}(Y, \mathsf{P})$,

$$\|F_{\mathsf{R}} - F_{\mathsf{R}'}\|_{L^1(\mathcal{G})} = \|Q_{\mathsf{R}} - Q_{\mathsf{R}'}\|_{L^1([0,1])} = W_1(g_{\#}\mathsf{R}, g_{\#}\mathsf{R}'),$$

by Panaretos and Zemel (2019, Section 1.2.3). Taking the supremum over $\mathsf{R}, \mathsf{R}' \in \mathcal{R}(Y, \mathsf{P})$ gives the L^1 diameter decompositions for the CDF and quantile functions. □

We can alternatively equip the space of Borel probability measures $\mathcal{P}(\mathcal{G})$ with the total variation (TV) distance,

$$d_{\mathrm{TV}}(\mu, \nu) := \sup_{B \in \mathcal{B}(\mathcal{G})} |\mu(B) - \nu(B)|.$$

The TV distance between two distributions is the maximum difference between the probabilities they assign to an event (Gibbs and Su, 2002, p424). It has a coupling characterisation analogous to the definition of Wasserstein-1 distance, but with the discrete metric $d_{\mathrm{discrete}}(t, s) := \mathbf{1}\{t \neq s\}$ replacing absolute distance.

Corollary A.6 (TV diameter decomposition)**.** *Suppose that $\mathcal{G}$ can be partitioned into $A, \mathcal{G} \setminus A \in \mathcal{B}(\mathcal{G})$ such that for P-a.e. z, every non-singleton $g(\{z\} \times Y(z))$ intersects both. Then*

$$\operatorname{diam}_{d_{\mathrm{TV}}} \Theta_{\mathrm{dis}}(Y, \mathsf{P}) = \mathsf{P} (\Delta(\mathbf{z}) > 0) .$$

Proof.

$$\begin{aligned}
\operatorname{diam}_{d_{\mathrm{TV}}} \Theta_{\mathrm{dis}}(Y, \mathsf{P}) &= \sup_{\mathsf{R}, \mathsf{R}' \in \mathcal{R}(Y, \mathsf{P})} \sup_{B \in \mathcal{B}(\mathcal{G})} \left| \mathsf{R}(g(\mathbf{z}, \mathbf{y}) \in B) - \mathsf{R}'(g(\mathbf{z}, \mathbf{y}) \in B) \right| \\
&= \sup_{B \in \mathcal{B}(\mathcal{G})} \sup_{\mathsf{R}, \mathsf{R}' \in \mathcal{R}(Y, \mathsf{P})} \left| \mathsf{E}_{\mathsf{R}} [(\mathbf{1}_B \circ g)(\mathbf{z}, \mathbf{y})] - \mathsf{E}_{\mathsf{R}'} [(\mathbf{1}_B \circ g)(\mathbf{z}, \mathbf{y})] \right| \\
&= \sup_{B \in \mathcal{B}(\mathcal{G})} \mathsf{E}_{\mathsf{P}}[\operatorname{diam} (\mathbf{1}_B \circ g)(\{\mathbf{z}\} \times Y(\mathbf{z}))] \\
&\leqslant \mathsf{P} (\Delta(\mathbf{z}) > 0) ,
\end{aligned}$$

where the third equality applies Proposition 1. By choosing $B = A$, the final inequality holds with equality. □

The Kolmogorov distance on $\mathcal{P}(\mathcal{G})$ is

$$d_K(\mu, \nu) := \sup_{t \in \mathbb{R}} |F_\mu(t) - F_\nu(t)|.$$

It is the $L^\infty(\mathcal{G})$ distance between the corresponding CDFs (Gibbs and Su, 2002).

Corollary A.7 (Kolmogorov diameter decomposition)**.** *Suppose that there exists $t_0 \in \mathcal{G}$ such that for P-a.e. z, every non-singleton $g(\{z\} \times Y(z))$ intersects both $(-\infty, t_0]$ and (t_0, ∞). Then*

$$\operatorname{diam}_{d_K} \Theta_{\text{dis}}(Y, \mathsf{P}) = \mathsf{P}\left(\Delta(\mathbf{z}) > 0\right).$$

Proof. The proof parallels that of Corollary A.6, using $F_{\mathsf{R}}(t) := \mathsf{E}_{\mathsf{R}}\left[(\mathbf{1}_{(-\infty,t]} \circ g)(\mathbf{z}, \mathbf{y})\right]$. □

It follows from Corollaries A.6 and A.7 that if the relevant partitioning condition is satisfied both before probing and after implementing any nonrandomised probing rule, then Assumption 1 is satisfied with observation-level contributions $\mathbf{1}\{\Delta_0(z_0) > 0\}$ and $\mathbf{1}\{\Delta_C(z_0, z_1) > 0\}$ replacing $\Delta_0(z_0)$ and $\Delta_C(z_0, z_1)$, respectively. The optimal probing rule is then given by Theorem 1 with

$$v(z_0) := \mathbf{1}\{\Delta_0(z_0) > 0\} - \mathsf{P}_{01}(\Delta_1(\mathbf{z}_0, \mathbf{z}_1) > 0 \mid \mathbf{z}_0 = z_0).$$

This rule differs from the W_1-diameter rule in that it assigns probing value based only on the probability of reducing $g(\{z_0\} \times Y_0(z_0))$ to a singleton.

Remark A.2 (Failure of additive decomposition under alternative probability metrics). Let $g(z, y) := y$, and consider a dataset consisting of exactly two observations with disjoint outcome sets $Y_1 := [0, 1]$ and $Y_2 := [2, 3]$. Probing reveals the corresponding outcome exactly. Figure 7 shows the CDF envelopes of the identified set before probing (left), after probing Y_1 (centre), and after probing both observations (right). Probing either observation alone leaves the Kolmogorov diameter unchanged, whereas probing both reduces it to zero. Thus, the diameter reduction from probing an observation depends on whether the other observation has been probed, precluding an additive observation-level decomposition. More generally, if $p \in [1, \infty]$, the L^p diameter of the CDF identified set is $2^{1/p-1}$ before probing, $1/2$ after probing either observation, and zero after probing both. For the quantile function, the corresponding diameters are 1, $2^{-1/p}$, and zero. By the quantile representation of Wasserstein-p distances, these are also the W_p diameters of the distribution identified set. Thus, none of these identified-set diameters admits an additive observation-level decomposition when $p > 1$.

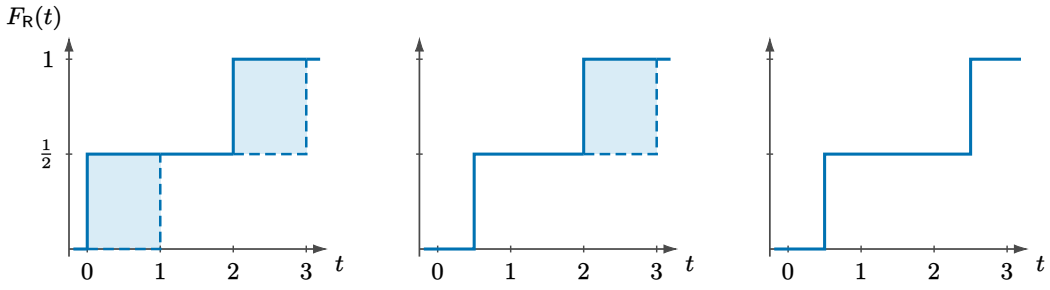


Figure 7: Failure of additive decomposition.

A.6 Estimated Probing Values

Proof of Proposition 4.

(i) Lemma A.8 gives

$$\begin{aligned} L(\widehat{\phi}) - L(\phi^*) &= \mathsf{E}_0\left[v(\phi^* - \widehat{\phi})\right] \\ &= \mathsf{E}_0\left[(v - \widehat{v})(\phi^* - \widehat{\phi})\right] + \mathsf{E}_0\left[\widehat{v}(\phi^* - \widehat{\phi})\right]. \end{aligned} \tag{24}$$

The second term is nonpositive because $\widehat{\phi}$ maximises the estimated objective over the budget set. Thus,

$$\begin{aligned} L(\widehat{\phi}) - L(\phi^*) &\leqslant \mathsf{E}_0\left[(v - \widehat{v})(\phi^* - \widehat{\phi})\right] \\ &\leqslant \mathsf{E}_0\left[|v - \widehat{v}|\,|\phi^* - \widehat{\phi}|\right]. \end{aligned}$$

(ii) By Lemma A.9, $\widehat{\phi} \in \Phi_b$ is a threshold plug-in rule of the form

$$\widehat{\phi} = \mathbf{1}\left\{\frac{\widehat{v}}{c} > \widehat{\lambda}\right\} + \widehat{\gamma}\mathbf{1}\left\{\frac{\widehat{v}}{c} = \widehat{\lambda}\right\} \qquad \mathsf{P}_0\text{-a.e.}$$

for some $\widehat{\lambda} \geqslant 0$ and measurable $\widehat{\gamma} : \mathcal{Z}_0 \to [0,1]$. Fix a weakly increasing function $h : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$. It suffices to consider h for which the expectation in the statement of (ii) is finite. Then,

$$\begin{aligned} &\mathsf{E}_0\left[ch\left(\frac{\widehat{v}}{c}\right)(\phi^* - \widehat{\phi})\right] \\ &= h(\widehat{\lambda})\mathsf{E}_0\left[c(\phi^* - \widehat{\phi})\right] + \mathsf{E}_0\left[c\left(h\left(\frac{\widehat{v}}{c}\right) - h(\widehat{\lambda})\right)(\phi^* - \widehat{\phi})\right]. \end{aligned}$$

The first term is nonpositive because h is nonnegative and $\widehat{\phi}$ costs at least as much as ϕ^*. The second term is nonpositive because $\widehat{\phi} = 1 \geqslant \phi^*$ when $h(\widehat{v}/c) > h(\widehat{\lambda})$ and $\widehat{\phi} = 0 \leqslant \phi^*$ when $h(\widehat{v}/c) < h(\widehat{\lambda})$. Thus,

$$\mathsf{E}_0\left[ch\left(\frac{\widehat{v}}{c}\right)(\phi^* - \widehat{\phi})\right] \leqslant 0.$$

Subtracting this expression from the right-hand side of (24),

$$\begin{aligned} & L(\widehat{\phi}) - L(\phi^*) \\ & \leqslant \mathsf{E}_0\left[c\left(\frac{v}{c} - h\left(\frac{\widehat{v}}{c}\right)\right)\left(\phi^* - \widehat{\phi}\right)\right] \\ & \leqslant \mathsf{E}_0\left[c\left|\frac{v}{c} - h\left(\frac{\widehat{v}}{c}\right)\right|\left|\phi^* - \widehat{\phi}\right|\right]. \end{aligned}$$

Since h was chosen arbitrarily, this is true for all weakly increasing $h : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$.

(iii) follows immediately from (ii). □

A.7 Multiple Linear Constraints

Proof of Proposition 5. By Lemma A.8, the probing problem (5) can be written as

$$\sup_{\phi \in \Phi} \mathsf{E}_0[v\phi] \qquad \text{s.t.} \qquad \mathsf{E}_0[c_k\phi] \leqslant b_k, \quad k = 1, \ldots, K. \tag{25}$$

Suppose that $\phi^* \in \Phi_{b_1,\ldots,b_K}$ satisfies conditions (6) and (7). If all multipliers vanish, condition (6) gives $\phi^* = 1$ on $\{v > 0\}$, which solves (25). Otherwise, relabel the constraints so that $\lambda_k^* > 0$ for $k = 1, \ldots, J$, where $1 \leqslant J \leqslant K$. Then the generalisation of the Neyman-Pearson Lemma to multiple null hypotheses (Lehmann and Romano, 2022, Theorem 3.6.1) implies that ϕ^* solves (25) among rules satisfying the first J constraints. Since every rule in $\Phi_{b_1,\ldots,b_K}$ satisfies these first J constraints and $\phi^* \in \Phi_{b_1,\ldots,b_K}$, ϕ^* is optimal among rules in $\Phi_{b_1,\ldots,b_K}$. □

B Diameter as a Proxy for Downstream Regret

The value of data probing may alternatively be measured by how much it reduces downstream decision regret. However, if the eventual downstream decision is not known when the data are collected—that is, the data may subsequently be used in a variety of ways by a variety of decision-makers—the identified-set diameter is a natural conservative criterion. In particular, it can bound the worst-case regret over a collection of downstream decisions. If this collection is sufficiently rich, the bound is sharp.

Let $(\mathcal{T}, d)$ be an ambient parameter space equipped with the metric d. An identified set is a nonempty, bounded $\Theta \subseteq \mathcal{T}$. A downstream decision problem is a pair $(\mathcal{A}_j, \ell_j)$, where $\mathcal{A}_j$ is a nonempty action set and $\ell_j : \mathcal{A}_j \times \mathcal{T} \to \mathbb{R}$ a loss function. For each $\theta \in \mathcal{T}$, the oracle value of the downstream problem $(\mathcal{A}_j, \ell_j)$ is

$$\ell_j^*(\theta) := \inf_{a \in \mathcal{A}_j} \ell_j(a, \theta)$$

and is assumed to be finite. The minimax-regret (MMR) value of the downstream problem over an identified set $\Theta \subseteq \mathcal{T}$ is

$$V_j(\Theta) := \inf_{a \in \mathcal{A}_j} \sup_{\theta \in \Theta} \left\{\ell_j(a, \theta) - \ell_j^*(\theta)\right\}.$$

Collect all potential downstream problems in the set

$$\{(\mathcal{A}_j, \ell_j) \mid j \in \mathcal{J}\}.$$

Then the worst-case MMR value over potential downstream problems is

$$\overline{V}(\Theta) := \sup_{j \in \mathcal{J}} V_j(\Theta).$$

Let $\mathcal{C}$ denote some collection of subsets of $\mathcal{T}$. Consider the researcher's problem

$$\inf_{\Theta \in \mathcal{C}} \overline{V}(\Theta).$$

The following results specify conditions under which $\operatorname{diam} \Theta$ provides an upper bound for $\overline{V}(\Theta)$, so that minimising the diameter of the identified set is a conservative proxy for minimising the worst-case MMR value. They also provide more restrictive conditions under which this bound is tight, so that minimising the diameter of the identified set also minimises the worst-case MMR.

Assumption B.1 (Uniform equicontinuity of downstream losses)**.** The family $\{\ell_j(a, \cdot) \mid a \in \mathcal{A}_j \text{ and } j \in \mathcal{J}\}$ is uniformly equicontinuous on $\mathcal{T}$. That is, there exists a weakly increasing modulus $\omega : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$ with $\lim_{t \to 0} \omega(t) = \omega(0) = 0$ such that for all actions $a \in \mathcal{A}_j$, all decisions $j \in \mathcal{J}$, and all parameter values $\theta, \theta' \in \mathcal{T}$,

$$|\ell_j(a, \theta) - \ell_j(a, \theta')| \leqslant \omega(d(\theta, \theta')).$$

Proposition B.1 (Diameter bound for downstream regret)**.** *Under Assumption B.1, the worst-case downstream MMR is bounded above by a weakly increasing transformation of diameter,*

$$\overline{V}(\Theta) \leqslant \omega(\operatorname{diam} \Theta).$$

In particular, if $\{(\mathcal{A}_j, \ell_j) \mid j \in \mathcal{J}\}$ is sufficiently rich that for every possible identified set $\Theta \in \mathcal{C}$ and every $\varepsilon > 0$, there exists $j \in \mathcal{J}$ satisfying

$$V_j(\Theta) \geqslant \omega(\operatorname{diam} \Theta) - \varepsilon,$$

then

$$\overline{V}(\Theta) = \omega(\operatorname{diam} \Theta).$$

Proof. Note first that $\ell_j^*(\cdot)$ inherits the modulus $\omega(\cdot)$:

$$\begin{aligned} \ell_j(a, \theta) &\leqslant \ell_j(a, \theta') + \omega(d(\theta, \theta')) \\ \ell_j^*(\theta) &\leqslant \ell_j^*(\theta') + \omega(d(\theta, \theta')). \end{aligned}$$

Fix $j \in \mathcal{J}$ and $\delta > 0$. Choose $\theta_0 \in \Theta$ such that

$$\ell_j^*(\theta_0) \leqslant \inf_{\theta \in \Theta} \ell_j^*(\theta) + \frac{\delta}{2}.$$

Choose a_0 such that

$$\ell_j(a_0, \theta_0) \leqslant \ell_j^*(\theta_0) + \frac{\delta}{2}.$$

For any $\theta \in \Theta$,

$$\begin{aligned}
\ell_j(a_0, \theta) &\leqslant \ell_j(a_0, \theta_0) + \omega(d(\theta, \theta_0)) \\
&\leqslant \ell_j^*(\theta_0) + \frac{\delta}{2} + \omega(d(\theta, \theta_0)) \\
&\leqslant \ell_j^*(\theta) + \delta + \omega(d(\theta, \theta_0)) \\
\ell_j(a_0, \theta) - \ell_j^*(\theta) &\leqslant \omega(d(\theta, \theta_0)) + \delta.
\end{aligned}$$

By monotonicity of the modulus ω,

$$\begin{aligned}
\sup_{\theta \in \Theta} \left\{ \ell_j(a_0, \theta) - \ell_j^*(\theta) \right\} &\leqslant \omega \left(\sup_{\theta \in \Theta} d(\theta, \theta_0) \right) + \delta \\
&\leqslant \omega(\operatorname{diam} \Theta) + \delta.
\end{aligned}$$

Taking $\delta \to 0$,

$$V_j(\Theta) \leqslant \omega(\operatorname{diam} \Theta),$$

and so

$$\overline{V}(\Theta) \leqslant \omega(\operatorname{diam} \Theta).$$

□

The following corollary gives a lower-level condition for $\mathcal{J}$ that satisfies the richness condition required for equality in Proposition B.1.

Corollary B.1 (Sharpness of the diameter-regret bound)**.** *Under Assumption B.1, if ω is continuous and subadditive,*[10] *that is,*

$$\omega(s + t) \leqslant \omega(s) + \omega(t),$$

and $\{(\mathcal{A}_j, \ell_j) \mid j \in \mathcal{J}\}$ contains all two-action no-randomisation downstream problems of the form

$$\mathcal{A}_j := \{0, 1\}, \qquad \ell_j(a, \theta) := \begin{cases} \omega(d(\theta, \theta_0)), & a = 0, \\ \omega(d(\theta, \theta_1)), & a = 1, \end{cases}$$

for $\theta_0, \theta_1 \in \mathcal{T}$, then

$$\overline{V}(\Theta) = \omega(\operatorname{diam} \Theta). \tag{26}$$

[10] Hölder continuous functions, for example, have continuous subadditive moduli, $\omega(t) := Ct^\alpha$, $\alpha \in (0, 1]$.

Proof. First we show that the described two-action problem has a loss that satisfies the uniform equicontinuity requirement. Fix $a \in \{0,1\}$ and $\theta, \theta' \in \mathcal{T}$. Suppose, without loss of generality, that $d(\theta, \theta_a) \geqslant d(\theta', \theta_a)$. The triangle inequality implies

$$d(\theta, \theta_a) - d(\theta_a, \theta') \leqslant d(\theta, \theta').$$

Subadditivity and monotonicity of ω then imply

$$\omega(d(\theta, \theta_a)) - \omega(d(\theta_a, \theta')) \leqslant \omega(d(\theta, \theta_a) - d(\theta_a, \theta')) \leqslant \omega(d(\theta, \theta')).$$

By symmetry in θ and θ',

$$|\ell_j(a, \theta) - \ell_j(a, \theta')| = |\omega(d(\theta, \theta_a)) - \omega(d(\theta_a, \theta'))| \leqslant \omega(d(\theta, \theta')).$$

Now we prove the equality (26). The first part of Proposition B.1 gives

$$\overline{V}(\Theta) \leqslant \omega(\operatorname{diam}\Theta).$$

We prove the reverse inequality. If $\operatorname{diam}\Theta = 0$ then this follows immediately from the nonnegativity of regret. Suppose, then, that $\operatorname{diam}\Theta > 0$. Fix $\delta \in (0, \operatorname{diam}\Theta)$ and choose $\theta_0, \theta_1 \in \Theta$ such that

$$d(\theta_0, \theta_1) > \operatorname{diam}\Theta - \delta.$$

The oracle value at $\theta \in \Theta$ is

$$\ell_j^*(\theta) = \min\{\omega(d(\theta, \theta_0)), \omega(d(\theta, \theta_1))\},$$

so the regret of an action $a \in \{0,1\}$ at θ_{1-a} is

$$\omega(d(\theta_a, \theta_{1-a})) - \omega(d(\theta_a, \theta_a)) = \omega(d(\theta_0, \theta_1)) \geqslant \omega(\operatorname{diam}\Theta - \delta).$$

Then

$$\overline{V}(\Theta) \geqslant V_j(\Theta) \geqslant \omega(\operatorname{diam}\Theta - \delta).$$

By continuity of ω, taking $\delta \to 0$,

$$\overline{V}(\Theta) \geqslant \omega(\operatorname{diam}\Theta).$$

Together with Proposition B.1, this proves (26). □

The following proposition gives a case in which the upper bound $\omega(\operatorname{diam}\Theta)$ is not sharp, but there exists a tighter bound that is sharp.

Proposition B.2 (Binary treatment-choice regret identity)**.** *Suppose that the downstream decision is a binary treatment choice problem. The action space is the fraction of individuals treated, $a \in [0,1]$. The welfare loss associated with treatment fraction a is*

$$\ell_{t,f}(a, \theta) := a(t - f(\theta)),$$

where $t \in \mathbb{R}$ is a threshold and $f : \mathcal{T} \to \mathbb{R}$ maps the parameter to a welfare contrast. Let $\mathrm{UC}_\omega(\mathcal{T}, \mathbb{R})$ denote the space of uniformly continuous functions mapping $\mathcal{T}$ into $\mathbb{R}$ with common modulus ω. Suppose that the collection of possible welfare-contrast functions f ranges over a sufficiently rich subset of $\mathrm{UC}_\omega(\mathcal{T}, \mathbb{R})$ that for all $\Theta \subseteq \mathcal{T}$ and all $\varepsilon > 0$ there exists a possible f satisfying[11]

$$\operatorname{diam} f(\Theta) \geqslant \omega(\operatorname{diam} \Theta) - \varepsilon.$$

Suppose that at each fixed f, t ranges over an interval containing $f(\mathcal{T})$. Then

$$\overline{V}(\Theta) = \frac{1}{4}\,\omega(\operatorname{diam} \Theta).$$

In particular, if $\mathcal{T} \subseteq \mathbb{R}$, d is absolute distance, and f is known to be the identity function $f(\theta) := \theta$, then

$$\overline{V}(\Theta) = \frac{1}{4}\,\operatorname{diam} \Theta.$$

Proof. The regret of action a is

$$(a - \mathbf{1}\{t \leqslant f(\theta)\})\,(t - f(\theta)) = a\,(t - f(\theta))_+ + (1 - a)\,(f(\theta) - t)_+\,,$$

where $x_+ := \max\{x, 0\}$. Define $f^U(\Theta) := \sup_{\theta \in \Theta} f(\theta)$ and $f^L(\Theta) := \inf_{\theta \in \Theta} f(\theta)$. Suppose $t > f^U(\Theta)$. Then $a = 0$ is uniformly loss-minimising over $\theta \in \Theta$, so $V_{t,f}(\Theta) = 0$. Similarly, if $t < f^L(\Theta)$ then $V_{t,f}(\Theta) = 0$. Suppose, then, that $t \in [f^L(\Theta), f^U(\Theta)]$. By assumption, there exists a possible t for which this is true. The worst-case regret of action a over $\theta \in \Theta$ is then

$$\max\left\{a(t - f^L(\Theta)), (1 - a)\left(f^U(\Theta) - t\right)\right\}. \tag{27}$$

If $f^L(\Theta) = f^U(\Theta)$, then the maximum (27) evaluates to 0, so $V_{t,f}(\Theta) = 0 = \frac{1}{4}\operatorname{diam} f(\Theta) \leqslant \frac{1}{4}\omega(\operatorname{diam} \Theta)$. Now suppose $f^L(\Theta) \neq f^U(\Theta)$. The optimal action equalises the entries of the maximum, giving

$$a^*_{t,f}(\Theta) := \frac{f^U(\Theta) - t}{f^U(\Theta) - f^L(\Theta)},$$

so that the MMR value is

$$V_{t,f}(\Theta) := \frac{(f^U(\Theta) - t)(t - f^L(\Theta))}{f^U(\Theta) - f^L(\Theta)} \leqslant \frac{(\frac{1}{2}\operatorname{diam} f(\Theta))^2}{\operatorname{diam} f(\Theta)} = \frac{1}{4}\operatorname{diam} f(\Theta).$$

Since t can take any value in the convex hull of $f(\mathcal{T})$, it can, in particular, take the value $\frac{1}{2}(f^L(\Theta) + f^U(\Theta))$ so that the above inequality holds with equality for some t. Thus,

$$\overline{V}(\Theta) \leqslant \frac{1}{4}\,\omega(\operatorname{diam} \Theta),$$

[11] If ω is continuous and subadditive, some sufficient subsets are: the full space $\mathrm{UC}_\omega(\mathcal{T}, \mathbb{R})$, which is the space of all C-Lipschitz functions if $\omega(t) := Ct$; a subset containing all functions of the form $f(\theta) := \omega(d(\theta, \theta_0))$ for all $\theta_0 \in \mathcal{T}$; if $\mathcal{T} \subseteq \mathbb{R}$ is equipped with the standard absolute distance and $\omega(t) := Ct$, any set of C-Lipschitz functions that contains in particular the mapping $f(\theta) := C\theta$.

where we use that all possible functions f are in $\mathrm{UC}_\omega(\mathcal{T}, \mathbb{R})$, so $\operatorname{diam} f(\Theta) \leqslant \omega(\operatorname{diam}\Theta)$. The richness condition means that for all $\varepsilon > 0$ there exists a possible f such that $\operatorname{diam} f(\Theta) \geqslant \omega(\operatorname{diam}\Theta) - \varepsilon$. Taking $\varepsilon \to 0$,

$$\overline{V}(\Theta) = \frac{1}{4}\,\omega(\operatorname{diam}\Theta).$$

The special case in which $\mathcal{T} \subseteq \mathbb{R}$ and $f(\theta) := \theta$ is known follows by noting that then $\operatorname{diam} f(\Theta) = \operatorname{diam}\Theta$. □

The following proposition gives a case in which diameter also provides a lower bound on minimax loss, and under additional assumptions, minimax loss is a weakly increasing transformation of diameter.

Proposition B.3 (Diameter bounds for point estimation)**.** *Let $\Theta \subseteq \mathcal{A} \subseteq \mathcal{T}$ with Θ nonempty and compact, and let $h : \mathbb{R}_{\geqslant 0} \to \mathbb{R}_{\geqslant 0}$ be a weakly increasing function that vanishes at 0. Let the downstream loss function $\ell(a, \theta) := h(d(a, \theta))$ be known. Then*

$$V(\Theta) := \inf_{a \in \mathcal{A}} \sup_{\theta \in \Theta} \ell(a, \theta)$$

satisfies

$$h\left(\frac{1}{2}\operatorname{diam}\Theta\right) \leqslant V(\Theta) \leqslant h(\operatorname{diam}\Theta). \tag{28}$$

In particular, if $\mathcal{T} \subseteq \mathbb{R}$, Θ is a compact interval, and d is the standard absolute distance,

$$V(\Theta) = h\left(\frac{1}{2}\operatorname{diam}\Theta\right).$$

Proof. The upper bound is immediate from $\ell = h \circ d$. For the lower bound,

$$d(\theta, \theta') \leqslant d(\theta, a) + d(a, \theta') \leqslant 2 \sup_{\theta \in \Theta} d(a, \theta)$$

by the triangle inequality, so

$$\sup_{\theta \in \Theta} d(a, \theta) \geqslant \frac{1}{2}\operatorname{diam}\Theta.$$

Monotonicity of h and compactness of Θ then give the lower bound in (28). For the special case of an interval identified set on the real line, the lower bound in (28) is attained by choosing $a^* := \frac{1}{2}(\inf\Theta + \sup\Theta)$. □

C Penalised and Ambiguity-Robust Probing

Penalising probing cost rather than constraining it preserves the pointwise threshold rule, except that boundary randomisation becomes optional. Moreover, the penalised formulation can incorporate ambiguity—non-probabilistic uncertainty—over probing effectiveness through a minimax regret (MMR) criterion. The MMR rule compares optimistic and pessimistic value-cost ratios with a threshold: probe when both exceed the threshold, do not

probe when both fall below it, and randomise when the threshold lies between them. If probing rules are restricted to be nonrandomised, the MMR rule retains the threshold form of the no-ambiguity rule, with marginal value replaced by its midpoint over the ambiguity interval.

Proposition C.1 (Penalised probing rule)**.** *Fix $\lambda \geqslant 0$. Under Assumption 1, the problem*

$$\inf_{\phi \in \Phi} L(\phi) + \lambda \, \mathsf{E}_0[c(\mathbf{z}_0)\phi(\mathbf{z}_0)]$$

is solved by a rule ϕ^* if and only if

$$\phi^*(z_0) = \mathbf{1}\left\{\frac{v(z_0)}{c(z_0)} > \lambda\right\} + \gamma(z_0)\mathbf{1}\left\{\frac{v(z_0)}{c(z_0)} = \lambda\right\} \quad \mathsf{P}_0\text{-}a.e.,$$

for some measurable $\gamma : \mathcal{Z}_0 \to [0,1]$.

Proof. Applying Lemma A.8 and subtracting a constant term from the objective gives the equivalent optimisation problem

$$\sup_{\phi \in \Phi} \left\{\mathsf{E}_0\left[v\phi\right] - \lambda \, \mathsf{E}_0[c\phi]\right\},$$

which can be written as

$$\sup_{\phi \in \Phi} \mathsf{E}_0\left[(v - \lambda c)\phi\right].$$

Any solution must maximise the integrand pointwise P_0-a.e., and so must satisfy

$$\phi^*(z_0) = \begin{cases} 1, & v(z_0) > \lambda \, c(z_0), \\ 0, & v(z_0) < \lambda \, c(z_0), \\ \gamma(z_0), & v(z_0) = \lambda \, c(z_0), \end{cases} \qquad \mathsf{P}_0\text{-a.e.},$$

for some measurable function $\gamma : \mathcal{Z}_0 \to [0,1]$. Conversely, any rule satisfying this condition maximises the integrand pointwise P_0-a.e., and thus maximises the expectation. □

Suppose the researcher is unwilling to specify a single expected post-probing image diameter $\overline{\Delta}_1 : \mathcal{Z}_0 \to \mathbb{R}_{\geqslant 0}$, or equivalently a single marginal value function $v : \mathcal{Z}_0 \to \mathbb{R}_{\geqslant 0}$. Let $\mathcal{V}$ denote a nonempty family of measurable candidate marginal-value functions, each of which is nonnegative and bounded above by Δ_0. For each $v \in \mathcal{V}$, denote the candidate-specific loss function,

$$L(\phi; v) := \mathsf{E}_0[\Delta_0] - \mathsf{E}_0[v\phi].$$

Assumption C.1 (Endpoint-rectangularity of $\mathcal{V}$)**.** There exist bounded measurable functions

$$v^L, v^U : \mathcal{Z}_0 \to \mathbb{R}_{\geqslant 0}$$

such that for every $v \in \mathcal{V}$,

$$v^L(z_0) \leqslant v(z_0) \leqslant v^U(z_0) \qquad \mathsf{P}_0\text{-a.e.},$$

and for every measurable $A \subseteq \mathcal{Z}_0$, there exists a $v^A \in \mathcal{V}$ satisfying

$$v^A(z_0) = \begin{cases} v^U(z_0), & z_0 \in A, \\ v^L(z_0), & z_0 \notin A, \end{cases} \qquad \mathsf{P}_0\text{-a.e.}$$

Regret. Fix $\lambda \geqslant 0$. Define the regret of probing rule ϕ under candidate marginal value function $v \in \mathcal{V}$ to be

$$\mathrm{Reg}(\phi; v) := L(\phi; v) + \lambda \, \mathsf{E}_0[c\phi] - \inf_{\phi' \in \Phi} \left\{ L(\phi'; v) + \lambda \, \mathsf{E}_0[c\phi'] \right\}.$$

The minimax regret probing problem is

$$\inf_{\phi \in \Phi} \sup_{v \in \mathcal{V}} \mathrm{Reg}(\phi; v).$$

Remark C.1. The endpoint rectangularity of $\mathcal{V}$ implies that the maximum-regret criterion $\sup_v \mathrm{Reg}(\phi; v)$ inherits the additive decomposition of $L(\phi; v)$ across observations.

Proposition C.2 (Minimax-regret probing rule)**.** *Under Assumptions 1 and C.1, the minimax regret problem*

$$\inf_{\phi \in \Phi} \sup_{v \in \mathcal{V}} \mathrm{Reg}(\phi; v) \tag{29}$$

is solved by a rule ϕ^* if and only if

$$\phi^*(z_0) = \begin{cases} 1, & \lambda c(z_0) < v^L(z_0), \\ 0, & \lambda c(z_0) > v^U(z_0), \\ \dfrac{v^U(z_0) - \lambda c(z_0)}{v^U(z_0) - v^L(z_0)}, & v^L(z_0) \leqslant \lambda c(z_0) \leqslant v^U(z_0) \text{ and } v^L(z_0) < v^U(z_0), \\ \gamma(z_0), & \lambda c(z_0) = v^L(z_0) = v^U(z_0), \end{cases}$$

P_0*-a.e., for some measurable* $\gamma : \mathcal{Z}_0 \to [0, 1]$.

Proof. By definition of $L(\phi; v)$,

$$L(\phi; v) + \lambda \, \mathsf{E}_0[c\phi] = \mathsf{E}_0\left[(\lambda c - v)\phi\right] + \mathsf{E}_0\left[\Delta_0\right],$$

and so

$$\begin{aligned} \mathrm{Reg}(\phi; v) &= L(\phi; v) + \lambda \, \mathsf{E}_0[c\phi] - \inf_{\phi' \in \Phi} \left\{ L(\phi'; v) + \lambda \, \mathsf{E}_0[c\phi'] \right\} \\ &= \mathsf{E}_0\left[(\lambda c - v)\phi\right] - \inf_{\phi' \in \Phi} \left\{ \mathsf{E}_0\left[(\lambda c - v)\phi'\right] \right\} \\ &= \mathsf{E}_0\big[(\lambda c - v)\left(\phi - \mathbf{1}\left\{\lambda c - v \leqslant 0\right\}\right)\big] \\ &= \mathsf{E}_0\big[(\lambda c - v)_+ \, \phi + (v - \lambda c)_+ \, (1 - \phi)\big]. \end{aligned}$$

Pass to the full-P_0 set on which v is bounded by v^L and v^U. The integrand is a convex function of $v(z_0)$, so it is maximised at one of the endpoints $v^L(z_0)$, $v^U(z_0)$. That is,

$$\begin{aligned}&\sup_{v(z_0)\in[v^L(z_0),v^U(z_0)]} \{(\lambda c(z_0) - v(z_0))_+ \,\phi + (v(z_0) - \lambda c(z_0))_+ \,(1-\phi)\}\\ &= \max\{(\lambda c(z_0) - v^L(z_0))_+ \,\phi(z_0), (v^U(z_0) - \lambda c(z_0))_+ \,(1-\phi(z_0))\}.\end{aligned}$$

Assumption C.1 implies that there is a single $v \in \mathcal{V}$ that equals v^L wherever this maximum is $(\lambda c(z_0) - v^L(z_0))_+ \,\phi(z_0)$ and v^U otherwise. This v attains the pointwise supremum P_0-a.e. It follows that maximum regret is

$$\begin{aligned}\sup_{v\in\mathcal{V}} \mathrm{Reg}(\phi; v) &= \sup_{v\in\mathcal{V}} \mathsf{E}_0\big[(\lambda c - v)_+ \,\phi + (v - \lambda c)_+ \,(1-\phi)\big]\\ &= \mathsf{E}_0\big[\max\{(\lambda c - v^L)_+ \,\phi, (v^U - \lambda c)_+ \,(1-\phi)\}\big].\end{aligned}$$

A rule ϕ^* minimises this expectation if and only if it minimises the integrand P_0-a.e. This is the case if and only if it satisfies

(i) $\phi^*(z_0) = 1$ when $\lambda c(z_0) < v^L(z_0)$;

(ii) $\phi^*(z_0) = 0$ when $\lambda c(z_0) > v^U(z_0)$;

(iii) and when $v^L(z_0) \leqslant \lambda c(z_0) \leqslant v^U(z_0)$ and $v^L(z_0) < v^U(z_0)$,

$$\begin{aligned}\big(\lambda c(z_0) - v^L(z_0)\big)\ \phi^*(z_0) &= \big(v^U(z_0) - \lambda c(z_0)\big)\ (1-\phi^*(z_0))\\ \big(v^U(z_0) - \lambda c(z_0) + \lambda c(z_0) - v^L(z_0)\big)\ \phi^*(z_0) &= v^U(z_0) - \lambda c(z_0)\\ \phi^*(z_0) &= \frac{v^U(z_0) - \lambda c(z_0)}{v^U(z_0) - v^L(z_0)}.\end{aligned}$$

When both inequalities hold with equality, ϕ^* can choose any value in $[0,1]$. □

Remark C.2 (Hedging versus boundary randomisation). When λ lies between $v^L(z_0)/c(z_0)$ and $v^U(z_0)/c(z_0)$, the maximum regret of probing occurs when $v(z_0) = v^L(z_0)$ and the maximum regret of not probing occurs when $v(z_0) = v^U(z_0)$. The minimax regret rule hedges against these two states by randomising between the optimal action under each. This differs from boundary randomisation in Theorem 1, which serves only to satisfy the expected budget constraint.

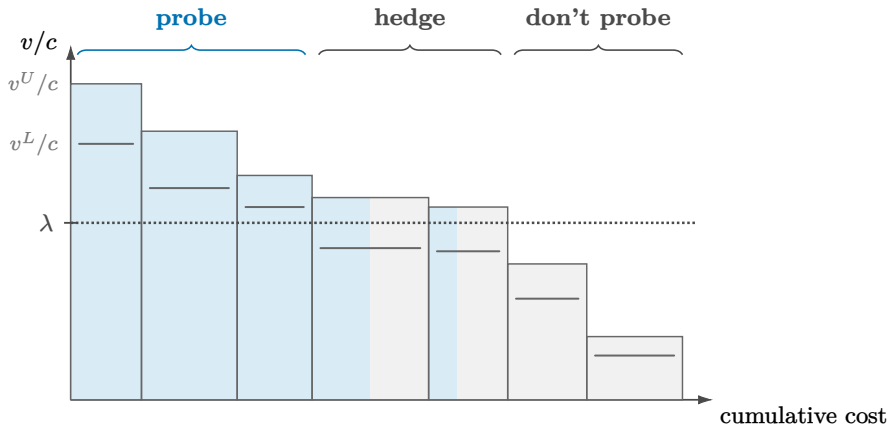


Figure 8: Visualisation of Proposition C.2. The optimal rule probes where $v^L/c > \lambda$; does not probe where $v^U/c < \lambda$; and randomises probing where $v^L/c \leqslant \lambda \leqslant v^U/c$ and $v^L < v^U$, with probabilities given by the ratio of $v^U/c - \lambda$ to $v^U/c - v^L/c$. Where $v^L = v^U = \lambda c$, any probing probability is optimal.

Corollary C.1 (Nonrandomised minimax-regret rule)**.** *Suppose probing rules are restricted to being nonrandomised. Then the minimax regret problem* (29) *is solved by* ϕ^* *if and only if*

$$\phi^*(z_0) = \mathbf{1}\left\{\frac{v^{\text{mid}}(z_0)}{c(z_0)} > \lambda\right\} + \gamma(z_0)\mathbf{1}\left\{\frac{v^{\text{mid}}(z_0)}{c(z_0)} = \lambda\right\} \quad \mathsf{P}_0\text{-}a.e.,$$

for some measurable $\gamma : \mathcal{Z}_0 \to \{0,1\}$ *and*

$$v^{\text{mid}}(z_0) := \frac{v^U(z_0) + v^L(z_0)}{2}.$$

Proof. The proof differs from that of Proposition C.2 only in the choice set we have for minimising the final integrand,

$$\max\Big\{\big(\lambda c(z_0) - v^L(z_0)\big)_+ \phi(z_0), \big(v^U(z_0) - \lambda c(z_0)\big)_+ (1 - \phi(z_0))\Big\}.$$

Since $\phi(z_0) \in \{0,1\}$, the integrand is minimised by setting $\phi(z_0) = 1$ if

$$v^U(z_0) - \lambda c(z_0) > \lambda c(z_0) - v^L(z_0),$$

or equivalently,

$$\frac{v^U(z_0) + v^L(z_0)}{2} > \lambda\, c(z_0).$$

If instead this holds with equality, any choice of $\phi(z_0)$ is optimal; if the reverse strict inequality holds, set $\phi(z_0) = 0$. □

D Multivariate Random Variables

The baseline model assumes that g is real-valued. In some applications, however, g may take values in $\mathbb{R}^J$. For example, the researcher may wish to sharpen identification jointly for several survey items. I consider two approaches. First, for vector expectations, targeting a linear functional of the expectation reduces the problem to the scalar case. Second, I give geometric conditions under which the identified-set diameter induced by an ℓ^p norm decomposes into expected image diameter.

The simplest way to accommodate vector expectations is to target a fixed linear functional of the vector expectation,

$$u^{\mathsf{T}}\mathsf{E}_{\mathsf{R}}\left[g(\mathbf{z},\mathbf{y})\right] = \mathsf{E}_{\mathsf{R}}\left[u^{\mathsf{T}}g(\mathbf{z},\mathbf{y})\right].$$

Proposition 1 and Theorem 1 can then be applied directly to the real-valued function

$$(z,y) \mapsto u^{\mathsf{T}}g(z,y).$$

This function could represent an additive welfare criterion, an index formed by summing responses to several survey items, or a within-person treatment effect. Fixing $u \in \mathbb{R}^J$ fixes a direction and measures identification uncertainty by the width of the identified set in that direction.

Norm-induced diameters avoid fixing a direction of interest in advance. However, because different outcome-set images $g(\{z\} \times Y(z))$ may attain their diameters in different directions, identified-set diameter need not decompose into expected image diameter. To ensure that it does, we require restrictions aligning the direction of diameter attainment across outcome-set images. Let $g : \mathcal{Z} \times \mathcal{Y} \to \mathbb{R}^J$ be bounded and measurable, and let

$$\Theta_{\mathrm{vec}}(Y,\mathsf{P}) := \{\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] \mid \mathsf{R} \in \mathcal{R}(Y,\mathsf{P})\}.$$

Equip $\mathbb{R}^J$ with a norm $\|\cdot\|$, and define the measurable image diameter

$$\Delta_{\|\cdot\|}(z) = \mathrm{diam}_{\|\cdot\|}\, g(\{z\} \times Y(z)) \qquad \mathsf{P}\text{-a.e.}$$

Such a measurable function exists by Lemma A.2(iii).

Lemma D.1.

$$\mathrm{diam}_{\|\cdot\|}\, \Theta_{\mathrm{vec}}(Y,\mathsf{P}) \leqslant \mathsf{E}_{\mathsf{P}}\left[\Delta_{\|\cdot\|}(\mathbf{z})\right]. \tag{30}$$

Proof. Fix $\mathsf{R}, \mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})$. By the law of iterated expectations and Jensen's inequality,

$$\begin{aligned}\|\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[g(\mathbf{z},\mathbf{y})]\| &= \|\mathsf{E}_{\mathsf{P}}\left[\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y}) \mid \mathbf{z}]\right] - \mathsf{E}_{\mathsf{P}}\left[\mathsf{E}_{\mathsf{R}'}[g(\mathbf{z},\mathbf{y}) \mid \mathbf{z}]\right]\| \\ &\leqslant \mathsf{E}_{\mathsf{P}}\left\|\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y}) \mid \mathbf{z}] - \mathsf{E}_{\mathsf{R}'}[g(\mathbf{z},\mathbf{y}) \mid \mathbf{z}]\right\|.\end{aligned}$$

For P-almost every z, conditional on $\mathbf{z} = z$, $g(\mathbf{z},\mathbf{y})$ is supported on $g(\{z\} \times Y(z))$ under both R and R'. It follows that both conditional expectations are points in the closed convex hull of $g(\{\mathbf{z}\} \times Y(\mathbf{z}))$, which has diameter $\Delta_{\|\cdot\|}(\mathbf{z})$. Therefore,

$$\|\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[g(\mathbf{z},\mathbf{y})]\| \leqslant \mathsf{E}_{\mathsf{P}}\left[\Delta_{\|\cdot\|}(\mathbf{z})\right].$$

Taking the supremum over $\mathsf{R}, \mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})$ gives the result. □

Remark D.1. Inequality (30) may hold strictly. For example, suppose that $g(z, y) := y \in \mathbb{R}^2$ and P is the uniform distribution on $\{z_a, z_b, z_c\}$. Let $Y(z_a)$ be the horizontal line segment from the origin to $(3, 0)$, and $Y(z_b) = Y(z_c)$ the shorter vertical line segment from the origin to $(0, 2)$. The expected ℓ^2 diameter of these outcome sets is $7/3$. The identified set for the expectation is the rectangle $[0, 1] \times [0, 4/3]$, which has ℓ^2 diameter $5/3 < 7/3$. Under the ℓ^∞ norm the analogous inequality is $4/3 < 7/3$. Note that equality holds under the ℓ^1 norm: Proposition D.2 generalises this observation.

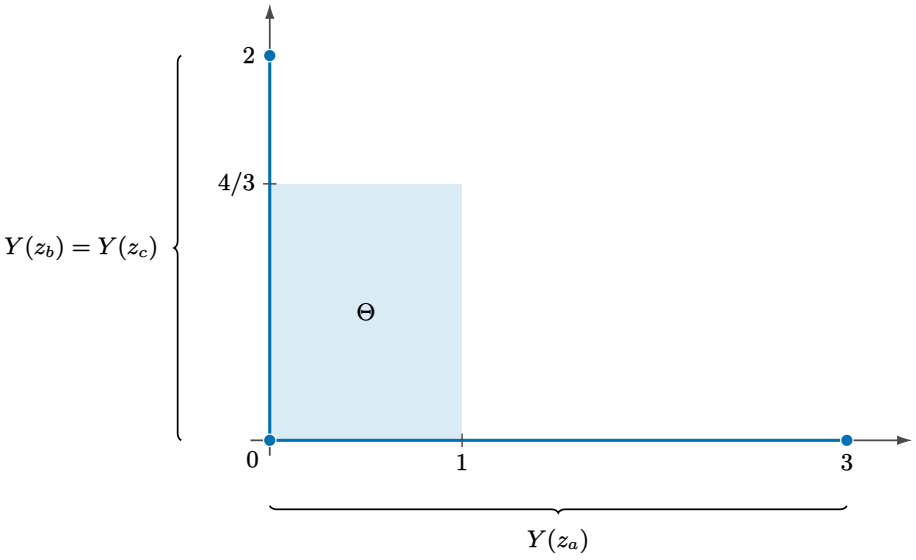


Figure 9: Visualisation of Remark D.1.

Proposition D.1 (ℓ^p diameter decomposition)**.**

(i) Fix $p \in [1, \infty)$ and $u \in \mathbb{R}^J$ with $\|u\|_p = 1$. Suppose that for P-a.e. z, there exist $x, x' \in \operatorname{cl} g(\{z\} \times Y(z))$ such that

$$x' - x = \Delta_{\|\cdot\|_p}(z) u. \tag{31}$$

Then

$$\operatorname{diam}_{\|\cdot\|_p} \Theta_{\mathrm{vec}}(Y, \mathsf{P}) = \mathsf{E}_{\mathsf{P}} \left[\Delta_{\|\cdot\|_p}(\mathbf{z}) \right].$$

(ii) If $p = \infty$, the same conclusion holds under the weaker condition that for some fixed $j^* \in \{1, \ldots, J\}$ and every $j \in \{1, \ldots, J\}$,

$$\operatorname{diam} g_j(\{z\} \times Y(z)) \leqslant \operatorname{diam} g_{j^*}(\{z\} \times Y(z)) \qquad \mathsf{P}\text{-a.e.} \tag{32}$$

Proof.

(i) Let $q \in [1,\infty]$ be such that $1/p + 1/q = 1$. Since the ℓ^q norm is dual to ℓ^p, there exists $\xi \in \mathbb{R}^J$ with

$$\|\xi\|_q = 1, \qquad \xi^{\mathsf{T}} u = \|u\|_p = 1,$$

by a corollary of the Hahn-Banach Theorem (Megginson, 1998, Corollary 1.9.8). Then, for P-a.e. z,

$$\Delta_{\|\cdot\|_p}(z) = \Delta_{\|\cdot\|_p}(z)\|u\|_p = \Delta_{\|\cdot\|_p}(z)\xi^{\mathsf{T}} u = \xi^{\mathsf{T}}(x' - x) \leqslant \operatorname{diam} \xi^{\mathsf{T}} g(\{z\} \times Y(z)).$$

Applying Proposition 1 to $(z,y) \mapsto \xi^{\mathsf{T}} g(z,y)$,

$$\begin{aligned}
\mathsf{E}_{\mathsf{P}}\left[\Delta_{\|\cdot\|_p}(\mathbf{z})\right] &\leqslant \mathsf{E}_{\mathsf{P}}\left[\operatorname{diam} \xi^{\mathsf{T}} g(\{\mathbf{z}\} \times Y(\mathbf{z}))\right] \\
&= \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \left|\xi^{\mathsf{T}} \left(\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[g(\mathbf{z},\mathbf{y})]\right)\right| \\
&\leqslant \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \|\xi\|_q \left\|\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[g(\mathbf{z},\mathbf{y})]\right\|_p \\
&= \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \left\|\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[g(\mathbf{z},\mathbf{y})]\right\|_p \\
&= \operatorname{diam}_{\|\cdot\|_p} \Theta_{\mathrm{vec}}(Y,\mathsf{P}),
\end{aligned}$$

where the second inequality follows from the definition of a dual norm (Aliprantis and Border, 2006, Definition 6.7). The claim now follows from Lemma D.1.

(ii) Under condition (32),

$$\begin{aligned}
\mathsf{E}_{\mathsf{P}}\left[\Delta_{\|\cdot\|_\infty}(\mathbf{z})\right] &= \mathsf{E}_{\mathsf{P}}\left[\operatorname{diam} g_{j^*}(\{\mathbf{z}\} \times Y(\mathbf{z}))\right] \\
&= \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \left|\mathsf{E}_{\mathsf{R}}[g_{j^*}(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[g_{j^*}(\mathbf{z},\mathbf{y})]\right| \\
&\leqslant \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \left\|\mathsf{E}_{\mathsf{R}}[g(\mathbf{z},\mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[g(\mathbf{z},\mathbf{y})]\right\|_\infty \\
&= \operatorname{diam}_{\|\cdot\|_\infty} \Theta_{\mathrm{vec}}(Y,\mathsf{P}),
\end{aligned}$$

where the second equality uses Proposition 1. The reverse inequality follows from Lemma D.1.

□

Example D.1 (Bounded measurement error). Suppose that $p \in [1,\infty)$ and consider the bounded measurement error model,

$$y = y^{\mathrm{obs}} + \varepsilon, \qquad \|\varepsilon\|_p \leqslant r(z), \qquad r(z) \geqslant 0.$$

If $g(z,y) := y$, then the outcome-set images are ℓ^p-balls. Under the ℓ^p norm, these satisfy condition (31) for any u with $\|u\|_p = 1$.

For $w_1, \dots, w_J > 0$, let $\|\cdot\|_{1,w}$ denote the $(w_1, \dots, w_J)$-weighted ℓ^1 norm,

$$\|x\|_{1,w} := \sum_{j=1}^{J} w_j |x_j|.$$

Proposition D.2 (Weighted ℓ^1 diameter decomposition)**.** *Fix $\sigma \in \{-1, 1\}^J$. Define the reoriented mapping $h : \mathcal{Z} \times \mathcal{Y} \to \mathbb{R}^J$ by*

$$h(z, y) := \sigma \odot g(z, y),$$

where $\odot$ denotes the componentwise product. Let $h^L, h^U : \mathcal{Z} \to \mathbb{R}^J$ be defined by

$$h_j^L(z) := \inf_{y \in Y(z)} h_j(z, y), \qquad h_j^U(z) := \sup_{y \in Y(z)} h_j(z, y).$$

Suppose that

$$\{h^L(z), h^U(z)\} \subseteq \mathrm{cl}\left(h(\{z\} \times Y(z))\right) \qquad \mathsf{P}\text{-}a.e. \tag{33}$$

Then

$$\mathrm{diam}_{\|\cdot\|_{1,w}} \Theta_{\mathrm{vec}}(Y, \mathsf{P}) = \mathsf{E}_{\mathsf{P}}\left[\Delta_{\|\cdot\|_{1,w}}(\mathbf{z})\right].$$

Proof. The mapping $x \mapsto \sigma \odot x$ is an isometry for $\|\cdot\|_{1,w}$ since

$$\|\sigma \odot x - \sigma \odot x'\|_{1,w} = \sum_{j=1}^{J} w_j |\sigma_j (x_j - x_j')| = \sum_{j=1}^{J} w_j |x_j - x_j'| = \|x - x'\|_{1,w}.$$

Thus, by the opposite-corner condition (33),

$$\begin{aligned}
\Delta_{\|\cdot\|_{1,w}}(z) &= \sum_{j=1}^{J} w_j (h_j^U(z) - h_j^L(z)) \\
&= w^{\mathsf{T}}(h^U(z) - h^L(z)) \\
&= \mathrm{diam}\, w^{\mathsf{T}} h(\{z\} \times Y(z)).
\end{aligned}$$

Applying Proposition 1 to $(z, y) \mapsto w^{\mathsf{T}} h(z, y)$,

$$\begin{aligned}
\mathsf{E}_{\mathsf{P}}\left[\Delta_{\|\cdot\|_{1,w}}(\mathbf{z})\right] &= \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \left| w^{\mathsf{T}} \left(\mathsf{E}_{\mathsf{R}}[h(\mathbf{z}, \mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[h(\mathbf{z}, \mathbf{y})]\right) \right| \\
&\leqslant \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \sum_{j=1}^{J} w_j \left| \mathsf{E}_{\mathsf{R}}[h_j(\mathbf{z}, \mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[h_j(\mathbf{z}, \mathbf{y})] \right| \\
&= \sup_{\mathsf{R},\mathsf{R}' \in \mathcal{R}(Y,\mathsf{P})} \sum_{j=1}^{J} w_j \left| \mathsf{E}_{\mathsf{R}}[g_j(\mathbf{z}, \mathbf{y})] - \mathsf{E}_{\mathsf{R}'}[g_j(\mathbf{z}, \mathbf{y})] \right| \\
&= \mathrm{diam}_{\|\cdot\|_{1,w}} \Theta_{\mathrm{vec}}(Y, \mathsf{P}),
\end{aligned}$$

where the penultimate equality uses that $x \mapsto \sigma \odot x$ is an isometry. The reverse inequality follows from Lemma D.1. □

Example D.2. In $\mathbb{R}^2$, condition (33) means that one of the following is true:

(i) the closure of almost every outcome-set image contains its north-east and south-west corners; or

(ii) the closure of almost every outcome-set image contains its north-west and south-east corners.

Condition (i) is satisfied if the closure of almost every outcome-set image is an axis-aligned rectangle, a single point, a line segment with nonnegative slope, or any other region that can be constructed from two ordered points $(a, b) \leqslant (c, d)$ and some part of the axis-aligned rectangle between them.

Example D.3 (Vectors of CDF values). Suppose $\mathcal{Y} \subseteq \mathbb{R}$ and let $t_1 < \cdots < t_J \in \mathbb{R}$. If

$$g_j(z, y) := \mathbf{1}\{y \leqslant t_j\},$$

then, denoting CDF values $F_{\mathsf{R}}(t_j) := \mathsf{R}(\mathbf{y} \leqslant t_j)$,

$$\mathsf{E}_{\mathsf{R}}\left[g(\mathbf{z}, \mathbf{y})\right] = \left(F_{\mathsf{R}}(t_j)\right)_{j=1}^{J}.$$

Every possible realisation of $g(\mathbf{z}, \mathbf{y})$ has the form $(0, \ldots, 0, 1, \ldots, 1)$. These vectors are totally ordered with respect to the coordinatewise order, so every outcome-set image contains its coordinatewise supremum and infimum.

Example D.4 (Two interval-valued outcomes). Suppose that $g(z, y) := y = (y_1, y_2)$, $\mathbf{z} := \left(\mathbf{y}_1^L, \mathbf{y}_1^U, \mathbf{y}_2^L, \mathbf{y}_2^U\right) \in \mathbb{R}^4$, $\mathbf{y}_1^L \leqslant \mathbf{y}_1 \leqslant \mathbf{y}_1^U$, and $\mathbf{y}_2^L \leqslant \mathbf{y}_2 \leqslant \mathbf{y}_2^U$. If there are no further restrictions, then the outcome set is the axis-aligned rectangle

$$Y(z) := \left[y_1^L, y_1^U\right] \times \left[y_2^L, y_2^U\right],$$

which satisfies the ℓ^1 condition (33).

Remark D.2. If the relevant sufficient condition (31), (32), or (33) holds for almost every outcome-set image both before and after probing, with the same u, j^*, or σ respectively, then Assumption 1 holds for the parameter-metric pair $(\mathsf{E}_{\mathsf{R}}[g(\mathbf{z}, \mathbf{y})], \|\cdot\|)$, and Theorem 1 can be applied to obtain an optimal probing rule.

Remark D.3 (Multivariate distributions). Under the hypotheses of either Proposition D.1 or Proposition D.2, the identified-set diameter decomposition also holds for distributions under the Wasserstein-1 distance generated by the corresponding norm on $\mathbb{R}^J$. The proof largely parallels that of Proposition 3: the only substantive change is that the identity function step must be replaced with an alternative $f : \mathbb{R}^J \to \mathbb{R}$ that is 1-Lipschitz with respect to that norm, and satisfies

$$\operatorname{diam}(f \circ g)(\{z\} \times Y(z)) \geqslant \operatorname{diam}_{\|\cdot\|} g(\{z\} \times Y(z)), \qquad \mathsf{P}\text{-a.e.}$$

Under the hypotheses of Proposition D.1(i), this is $f(x) := \xi^{\mathsf{T}} x$, with ξ as constructed in the proof of that proposition; under the hypotheses of part (ii), this is the projection $f(x) := x_{j^*}$; and under the hypotheses of Proposition D.2, this is $f(x) := w^{\mathsf{T}}(\sigma \odot x)$. Unlike in the univariate case, the W_1 result does not automatically extend to L^1-diameter results for multivariate CDFs.

E Multiple Alternative Probe Types

The baseline model includes only one probe type. This section extends the model to several mutually exclusive probe types. Let the available probe types be indexed by

$$a \in \mathcal{A} := \{0, 1, \ldots, A\},$$

where $A \geqslant 1$ and $a = 0$ denotes no probing. Call $a \in \mathcal{A}$ an *action*. Retain the bounded measurable function $g : \mathcal{Z}_0 \times \mathcal{Y} \to \mathbb{R}$ of Section 3. Each action $a \geqslant 1$ returns a refinement $z_a \in \mathcal{Z}_a$ and an outcome set

$$Y_a(z_0, z_a) \subseteq Y_0(z_0).$$

For each $a \geqslant 1$, impose Standing Assumption A.4 with $\mathcal{Z}_1$ and Y_1 replaced by $\mathcal{Z}_a$ and Y_a, respectively. Define corresponding measurable image diameters

$$\Delta_a(z_0, z_a) := \operatorname{diam} g(\{z_0\} \times Y_a(z_0, z_a)), \qquad \Delta_0(z_0) = \operatorname{diam} g(\{z_0\} \times Y_0(z_0)).$$

As in the baseline model, pass to measurable a.e. equivalent versions when necessary. Let $\mathsf{P}_{\mathcal{A}}$ be the joint distribution of $(\mathbf{z}_0, \mathbf{z}_1, \ldots, \mathbf{z}_A)$, with $\mathbf{z}_0$-marginal P_0 and $(\mathbf{z}_0, \mathbf{z}_a)$-marginal P_{0a}; denote the corresponding expectations by $\mathsf{E}_{\mathcal{A}}$ and E_{0a}. The researcher need not know the full distribution $\mathsf{P}_{\mathcal{A}}$: the decision-relevant quantities are

$$\overline{\Delta}_a(z_0) := \mathsf{E}_{0a}\left[\Delta_a(\mathbf{z}_0, \mathbf{z}_a) \mid \mathbf{z}_0 = z_0\right], \qquad v_a(z_0) := \Delta_0(z_0) - \overline{\Delta}_a(z_0), \qquad a \geqslant 1.$$

Each v_a is bounded, nonnegative and measurable. Set $v_0 := 0$. A probing rule ϕ is a measurable mapping from $\mathcal{Z}_0$ into the subprobability simplex

$$\left\{ x \in [0,1]^A \;\middle|\; \sum_{a=1}^{A} x_a \leqslant 1 \right\}.$$

Here, ϕ_a is the probability assigned to action $a \geqslant 1$, and the residual probability $\phi_0(z_0) := 1 - \sum_{a \geqslant 1} \phi_a(z_0)$ is assigned to no probing. Denote the space of all such probing rules by Φ. The multiple-action counterpart to the baseline model's loss (3) is

$$L(\phi) := \mathsf{E}_0[\Delta_0] - \mathsf{E}_0\left[\sum_{a=1}^{A} v_a \phi_a\right].$$

Action $a \geqslant 0$ has bounded measurable pointwise cost $c_a(z_0)$. Assume $c_a(z_0) > 0$ for $a \geqslant 1$ and set $c_0 \equiv 0$. For $b > 0$, Φ_b collects the probing rules satisfying the expected budget constraint

$$\mathsf{E}_0\left[\sum_{a=1}^{A} c_a \phi_a\right] \leqslant b.$$

The following result is a multiple-action analogue of Theorem 1.

Proposition E.1 (Optimal probing with multiple probe types)**.** *The problem*

$$\inf_{\phi\in\Phi_b} L(\phi)$$

is solved by $\phi^*\in\Phi_b$ if and only if, for some $\lambda^*\geqslant 0$,

$$\phi_a^*(z_0)>0 \implies a\in\arg\max_{a'\in\mathcal{A}}\left\{v_{a'}(z_0)-\lambda^* c_{a'}(z_0)\right\}, \tag{34}$$

for every $a\in\mathcal{A}$ and P_0-a.e. z_0, and

$$\lambda^*\left(b-\mathsf{E}_0\left[\sum_{a=1}^{A} c_a\phi_a^*\right]\right)=0. \tag{35}$$

Moreover, ϕ^* can be chosen so as to randomise between at most two nonrandomised rules.

Proof. First, we prove the existence of a rule satisfying (34) and (35) that mixes between at most two nonrandomised rules. For $\lambda\geqslant 0$, define

$$f(\lambda,z_0):=\max_{a\in\mathcal{A}}\{v_a(z_0)-\lambda c_a(z_0)\}$$

and

$$h(\lambda):=\begin{cases}\lambda b+\mathsf{E}_0\left[f(\lambda,\mathbf{z}_0)\right], & \lambda\geqslant 0,\\ +\infty, & \lambda<0.\end{cases}$$

Since $c_0\equiv v_0\equiv 0$, we have $h(\lambda)\geqslant\lambda b$, so the lower contour sets satisfy

$$\{\lambda\in\mathbb{R}\mid h(\lambda)\leqslant t\}\subseteq\left[0,\frac{t}{b}\right],\qquad t\geqslant 0.$$

For $\lambda,\lambda'\geqslant 0$,

$$\begin{aligned}|f(\lambda,z_0)-f(\lambda',z_0)| &= \left|\max_{a\in\mathcal{A}}\{v_a(z_0)-\lambda c_a(z_0)\}-\max_{a'\in\mathcal{A}}\left\{v_{a'}(z_0)-\lambda' c_{a'}(z_0)\right\}\right|\\ &\leqslant \max_{a\in\mathcal{A}}\left|\left\{(\lambda'-\lambda)c_a(z_0)\right\}\right|\\ &= |\lambda-\lambda'|\max_{a\in\mathcal{A}} c_a(z_0),\end{aligned} \tag{36}$$

so that

$$\begin{aligned}|h(\lambda)-h(\lambda')| &\leqslant |\lambda-\lambda'|b+\mathsf{E}_0|f(\lambda,\mathbf{z}_0)-f(\lambda',\mathbf{z}_0)|\\ &\leqslant |\lambda-\lambda'|\left(b+\mathsf{E}_0\left[\max_{a\in\mathcal{A}} c_a(\mathbf{z}_0)\right]\right).\end{aligned}$$

Thus, h is (Lipschitz) continuous, and in particular lower semicontinuous, on $[0,+\infty)$. Together with $h(\lambda)=+\infty$ for $\lambda<0$, this implies that h is lower semicontinuous on $\mathbb{R}$. It

follows from Rockafellar and Wets (1998, Theorem 1.9) that h attains a minimum at some $\lambda^* \geqslant 0$. Since $\mathcal{A}$ is finite,

$$z_0 \rightrightarrows \arg\max_{a \in \mathcal{A}} \{v_a(z_0) - \lambda^* c_a(z_0)\}$$

is nonempty-valued. Let $a^L, a^U : \mathcal{Z}_0 \to \{0, \ldots, A\}$ select the lowest-cost and highest-cost elements of this solution correspondence, respectively; break cost ties by choosing the lowest-index action. They are measurable selections since each of their level sets is defined by finitely many comparisons between measurable functions: first $v_a - \lambda^* c_a$, then c_a, and finally between indices. Let ϕ^L and ϕ^U be the corresponding deterministic probing rules, and write

$$b^L := \mathsf{E}_0\left[c_{a^L(\mathbf{z}_0)}(\mathbf{z}_0)\right], \qquad b^U := \mathsf{E}_0\left[c_{a^U(\mathbf{z}_0)}(\mathbf{z}_0)\right].$$

Fix z_0 and define $f_{z_0} : \mathbb{R} \to \mathbb{R}$ by

$$f_{z_0}(\lambda) := f(\lambda, z_0) = \max_{a \in \mathcal{A}} \{v_a(z_0) - \lambda c_a(z_0)\}.$$

For each fixed pair (a, z_0), $\lambda \mapsto v_a(z_0) - \lambda c_a(z_0)$ is affine. Danskin's Theorem (Bertsekas, 2009, Proposition A.3.2(b)) therefore implies that f_{z_0} has subdifferential

$$\begin{aligned}\partial f_{z_0}(\lambda) &= \operatorname{conv}\left\{-c_a(z_0) \;\middle|\; a \in \arg\max_{a' \in \mathcal{A}} \{v_{a'}(z_0) - \lambda c_{a'}(z_0)\}\right\} \\ &= \left[\partial_- f_{z_0}(\lambda), \partial_+ f_{z_0}(\lambda)\right],\end{aligned}$$

where the endpoints of the interval are the left and right derivatives of f_{z_0} at λ. It follows that

$$\partial_- f_{z_0}(\lambda^*) = -c_{a^U(z_0)}(z_0), \qquad \partial_+ f_{z_0}(\lambda^*) = -c_{a^L(z_0)}(z_0).$$

By the Lipschitz bound (36), for all $k \in \mathbb{N}$, the quotient is dominated by an integrable function:

$$\left|\frac{f\left(\lambda^* + \frac{1}{k}, z_0\right) - f\left(\lambda^*, z_0\right)}{\frac{1}{k}}\right| \leqslant \max_{a \in \mathcal{A}} c_a(z_0).$$

Hence, by the Dominated Convergence Theorem (Aliprantis and Border, 2006, Theorem 11.21),

$$\begin{aligned}\mathsf{E}_0\left[\partial_+ f_{\mathbf{z}_0}(\lambda^*)\right] &= \mathsf{E}_0\left[\lim_{k \to \infty} \frac{f\left(\lambda^* + \frac{1}{k}, \mathbf{z}_0\right) - f\left(\lambda^*, \mathbf{z}_0\right)}{\frac{1}{k}}\right] \\ &= \lim_{k \to \infty} \mathsf{E}_0\left[\frac{f\left(\lambda^* + \frac{1}{k}, \mathbf{z}_0\right) - f\left(\lambda^*, \mathbf{z}_0\right)}{\frac{1}{k}}\right] \\ &= \lim_{k \to \infty} \frac{\mathsf{E}_0[f\left(\lambda^* + \frac{1}{k}, \mathbf{z}_0\right)] - \mathsf{E}_0[f\left(\lambda^*, \mathbf{z}_0\right)]}{\frac{1}{k}}.\end{aligned}$$

The last expression is the right derivative of $\lambda \mapsto \mathsf{E}_0\left[f_{\mathbf{z}_0}(\lambda)\right]$ at λ^*, so

$$\partial_+ h(\lambda^*) = b + \mathsf{E}_0\left[\partial_+ f_{\mathbf{z}_0}(\lambda^*)\right] = b - \mathsf{E}_0[c_{a^L(\mathbf{z}_0)}(\mathbf{z}_0)] = b - b^L. \tag{37}$$

An analogous argument gives

$$\partial_- h(\lambda^*) = b - b^U$$

for $\lambda^* > 0$.

If indeed $\lambda^* > 0$, then it is an interior minimiser of a convex function h so

$$\partial_- h(\lambda^*) \leqslant 0 \leqslant \partial_+ h(\lambda^*),$$

and thus

$$b^L \leqslant b \leqslant b^U.$$

Set

$$t := \begin{cases} \dfrac{b - b^L}{b^U - b^L}, & b^L \neq b^U, \\ 0, & b^L = b^U, \end{cases}$$

and

$$\phi^* := (1-t)\phi^L + t\phi^U.$$

This is a measurable probing rule that mixes between the two nonrandomised rules ϕ^L and ϕ^U. It is supported on

$$\{a^L(z_0), a^U(z_0)\} \subseteq \arg\max_{a \in \mathcal{A}} \{v_a(z_0) - \lambda^* c_a(z_0)\},$$

thereby satisfying the support condition (34). When $t = 0$ it has cost $b^L = b$; when $t > 0$ it has cost

$$(1-t)b^L + tb^U = b^L + t(b^U - b^L) = b^L + b - b^L = b.$$

Therefore, it also satisfies complementary slackness (35).

Suppose instead that $\lambda^* = 0$. Since h attains a minimum at 0, $\partial_+ h(\lambda^*) \geqslant 0$ so expression (37) implies that $b^L \leqslant b$. Hence ϕ^L is feasible. Since $\lambda^* = 0$, complementary slackness is immediate and ϕ^L deterministically chooses an element of $\arg\max_a v_a(z_0)$ at each z_0, so ϕ^L satisfies the support condition (34).

Now, we prove necessity and sufficiency of (34) and (35) for optimality. For any $\phi \in \Phi_b$,

$$\begin{aligned} \mathsf{E}_0\left[\sum_{a=0}^{A} v_a \phi_a\right] &= \lambda^* \mathsf{E}_0\left[\sum_{a=0}^{A} c_a \phi_a\right] + \mathsf{E}_0\left[\sum_{a=0}^{A} (v_a - \lambda^* c_a)\phi_a\right] \\ &\leqslant \lambda^* b + \mathsf{E}_0\left[\max_{a \in \mathcal{A}} \{v_a - \lambda^* c_a\}\right]. \end{aligned}$$

If ϕ^* satisfies (34) and (35), then this inequality holds with equality for ϕ^*. Thus, for every $\phi \in \Phi_b$,

$$\mathsf{E}_0\left[\sum_{a=0}^{A} v_a \phi_a\right] \leqslant \mathsf{E}_0\left[\sum_{a=0}^{A} v_a \phi_a^*\right],$$

so ϕ^* is optimal. Moreover, in the first part of this proof we constructed a rule that satisfies (34) and (35). Let ϕ^* now denote this rule, which we now know is optimal. If $\phi^\dagger \in \Phi_b$ is any other optimal rule, then $L(\phi^\dagger) = L(\phi^*)$, so

$$\begin{aligned}
&\mathsf{E}_0\left[\sum_{a=0}^{A} v_a \phi_a^\dagger\right] = \lambda^* \mathsf{E}_0\left[\sum_{a=0}^{A} c_a \phi_a^\dagger\right] + \mathsf{E}_0\left[\sum_{a=0}^{A} (v_a - \lambda^* c_a)\phi_a^\dagger\right] \\
&\leqslant \lambda^* b + \mathsf{E}_0\left[\max_{a\in\mathcal{A}} \{v_a - \lambda^* c_a\}\right] = \mathsf{E}_0\left[\sum_{a=0}^{A} v_a \phi_a^*\right] = \mathsf{E}_0\left[\sum_{a=0}^{A} v_a \phi_a^\dagger\right].
\end{aligned}$$

Therefore, the middle inequality must hold with equality, implying that (34) and (35) must be satisfied by $\phi^\dagger$ at λ^*. □

Remark E.1 (Pointwise-dominance reduction). Suppose that, at P_0-almost every z_0, there is an action $a \geqslant 1$ that is weakly dominant among the positive actions $a' \geqslant 1$: a has weakly lower cost $c_a(z_0)$ and weakly greater marginal value $v_a(z_0)$ than every other $a' \geqslant 1$. Then the problem reduces to the binary probing problem where the probe at almost every z_0 is a dominant positive action at that z_0, breaking ties using any measurable convention.

Remark E.2 (Probing possibility frontiers). At each fixed z_0, the set of cost-value pairs attainable by randomising among actions is

$$\operatorname{conv}\{(c_a(z_0), v_a(z_0)) \mid a = 0, 1, \ldots, A\}.$$

Proposition E.1 says that any optimal rule maximises $(c, v) \mapsto v - \lambda^* c$ over this set, for P_0-a.e. z_0. This optimum is attained on the upper frontier of the convex hull (see Figure 10).

The finite-sample plug-in problem is

$$\max_{\phi\in\Phi} \frac{1}{n}\sum_{i=1}^{n}\sum_{a=1}^{A} v_{ai}\phi_{ai} \qquad \text{s.t.} \qquad \frac{1}{n}\sum_{i=1}^{n}\sum_{a=1}^{A} c_{ai}\phi_{ai} \leqslant b.$$

The following algorithm is a particular case of the MCKP-Greedy algorithm (Kellerer, Pferschy, and Pisinger, 2004, Figure 11.2), and has runtime $O(nA\log(nA+1))$.

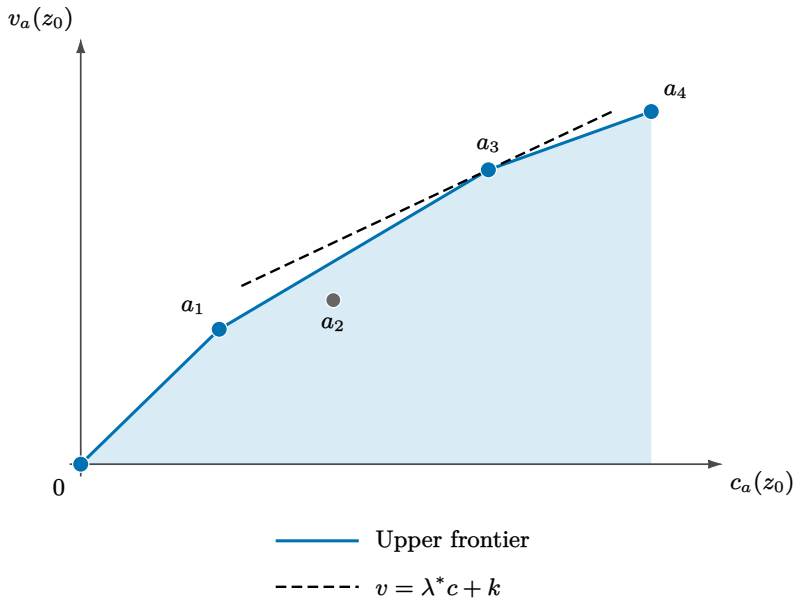


Figure 10: Probing possibility frontier at z_0. Action a_2 is dominated by a mixture over a_1 and a_3. Every point on the frontier can be attained by mixing over at most two actions. In this case, $v - \lambda^* c$ is maximised at a_3, so the optimal rule deterministically selects a_3 at z_0.

Algorithm 1 Finite-sample implementation with multiple actions

1: For each unit $i = 1, \ldots, n$, construct the probing possibility frontier described in Remark E.2, retaining one action at each vertex. Index these actions by increasing cost: $a_{i,0} := 0$, $a_{i,1}$, …, a_{i,J_i}. Note that the j indices are unit-specific.

2: For each frontier segment (i, j), compute its incremental value-cost ratio,

$$e_{ij} := \frac{v_{a_{i,j},i} - v_{a_{i,j-1},i}}{c_{a_{i,j},i} - c_{a_{i,j-1},i}}.$$

3: Initialise with no probing at each unit and residual budget $\overline{b} \leftarrow b$.

4: Select increments in decreasing order of e_{ij}. Whenever increment (i, j) is selected, replace action $a_{i,j-1}$ at unit i with the next frontier action, $a_{i,j}$; and update the residual budget $\overline{b} \leftarrow \overline{b} - \frac{1}{n}(c_{a_{i,j},i} - c_{a_{i,j-1},i})$.

5: Continue until no positive increments remain or $\overline{b}$ is smaller than the cost of the next increment, $\frac{1}{n}(c_{a_{i,j},i} - c_{a_{i,j-1},i})$. If necessary, randomise between consecutive frontier actions at the boundary, $a_{i,j-1}$ and $a_{i,j}$, to exactly exhaust the budget.

F Empirical Implementation Details and Extensions

Job insecurity. The pilot sample size of 3,708 is larger than that in Manski and Straub (2000) because I only exclude nonrespondents to loss_i whereas they also exclude nonrespondents to find_i, to a subsequent job-quit probability question, and to certain demographic questions. For the Black-White contrast, pilot group proportions $\widehat{\mathsf{P}}_0(B)$ and $\widehat{\mathsf{P}}_0(W)$ are used to construct the probing rule; the corresponding evaluation group proportions are used to construct the estimated identified set.

Stock-return expectations. In the survey, the \$1,000 threshold was one of several thresholds respondents were asked to consider. I judge the probe successful if a response was provided for the \$1,000 threshold, or if the answer to the \$1,000 threshold question was implied by a reported zero percent chance of the investment's value exceeding some threshold less than \$1,000. The plug-in probing rule is conditioned on respondents' age, education level, sex, race, employment status, marital status, home-ownership status, and individual income; and for each of the five expectations modules that appeared in the survey before pnr_i—the weather, crime, health insurance, employment, and income modules—an indicator for at least one nonresponse in that module. The ridge penalty is selected by minimising mean squared prediction error in five-fold cross-validation within the wave-12 pilot sample. The model is then estimated using the full pilot sample and held fixed for evaluation on waves 13 and 14.

Expected and realised probe shares. Rules are estimated using expected probe shares in the pilot sample. Randomisation and differences between pilot and evaluation samples can produce a different realised probe share. Figure 11 compares planned shares with mean realised shares.

Best linear predictors. Manski and Straub (2000) also report a mean best linear predictor (BLP) of job insecurity. I retain the specification given in their Table 5 except for omitting survey year indicators (since these differ between pilot and evaluation samples): regressors are the binary indicators for age, schooling, sex, race, self-employed, and working for a family business. I focus here on the coefficient of the Black respondent indicator—the base category is White. In addition to excluding nonrespondents to the opening loss_i question, I also exclude individuals for whom any of the demographic regressors are unavailable. In particular, this implies exclusion of all Wave 16 respondents. With these restrictions, the pilot sample consists of 3,636 respondents and the evaluation sample consists of 2,141 respondents. Let $x_i = x(z_{0i}) \in \mathbb{R}^J$ contain the regressors and an intercept term. Let

$$\widehat{Q}^{\text{eval}} := \frac{1}{n} \sum_{i=1}^{n} x_i x_i^{\mathsf{T}} \tag{38}$$



Figure 11: Planned and mean realised probe share under targeted probing.

be nonsingular, where $i = 1, \ldots, n$ index the evaluation sample. The Black respondent coefficient can be written as

$$\beta_B^{\text{eval}} := \frac{1}{n} \sum_{i=1}^{n} v_i^{\text{eval}} \cdot \mathsf{ins}_i, \qquad v_i^{\text{eval}} := e_B^{\mathsf{T}} \left(\widehat{Q}^{\text{eval}} \right)^{-1} x_i,$$

where e_B is a standard basis vector. A valid response to find_i reveals ins_i and thus $v_i^{\text{eval}} \cdot \mathsf{ins}_i$. The targeted rule probes in decreasing order of

$$\Delta_{0i} := \left| v_i^{\text{pilot}} \right| \cdot \mathsf{loss}_i, \qquad v_i^{\text{pilot}} := e_B^{\mathsf{T}} \left(\widehat{Q}^{\text{pilot}} \right)^{-1} x_i,$$

where $\widehat{Q}^{\text{pilot}}$ is the pilot-sample counterpart to (38). The rule is evaluated by the diameter of the estimated identified set for β_B^{eval}.

Table 4 compares this BLP targeted rule with the two targeted rules discussed in Section 5.1. All three rules use the same complete-case samples described above. The contrast rule targets the unadjusted Black-White mean contrast. It initially probes in decreasing order of $\left| w_i^{\text{pilot}} \right| \cdot \mathsf{loss}_i$, where w_i^{pilot} is given in (10) and the subsequent discussion; after probing all Black and White respondents, it probes the remaining respondents in order of loss_i. The overall-mean rule probes all respondents in order of loss_i. The BLP rule performs slightly better than the contrast rule at each tabulated probe share and both substantially outperform the overall-mean rule at 10% and 25% shares.

	Full-probing gain share (%)		
Probe share (%)	Overall-mean rule	Contrast rule	BLP rule
10	42.3	65.2	66.9
25	74.5	85.4	86.0
50	92.6	94.2	95.0
75	98.1	97.4	98.6

Table 4: Full-probing gain shares for the Black respondent BLP coefficient under alternative probing rules.

Sampling uncertainty. The empirical implementation in the main text evaluates the probing rules by the reduction in estimated identified-set diameter. The performance of the rules can also be evaluated by the diameter of a Wald-type 95% confidence interval for the identified set of the mean (Stoye, 2009, Section 2),

$$\mathcal{C}_n := \left[\widehat{\theta}^L - \frac{c\widehat{\sigma}^L}{\sqrt{n}}, \widehat{\theta}^U + \frac{c\widehat{\sigma}^U}{\sqrt{n}}\right],$$

where n is the size of the evaluation sample, c is the 0.975 quantile of the standard normal distribution, $\widehat{\theta}^L$ and $\widehat{\theta}^U$ are the endpoints of the estimated identified set $\widehat{\Theta}_n$, and $\widehat{\sigma}^L/\sqrt{n}$ and $\widehat{\sigma}^U/\sqrt{n}$ are the corresponding estimated standard errors.[12] Where applicable, the confidence interval is clipped to the parameter space. The plug-in rule targets the diameter of $\widehat{\Theta}_n := [\widehat{\theta}^L, \widehat{\theta}^U]$ rather than that of $\mathcal{C}_n$. The difference between the two diameters vanishes as $n \to \infty$.

The shaded region in Figure 12 is the averaged estimated identified set for mean job insecurity that appeared in Figure 3; the dashed lines give the corresponding averaged confidence-interval limits. The endpoints of both sets are averaged across probe allocation realisations—for confidence intervals this is done numerically, by drawing 2,000 realisations at each budget grid point. Figure 13 reproduces the top panel of Figure 4, but with confidence intervals for the identified sets replacing the estimated identified sets. The averaged 95% confidence limits exclude zero after 93.2%, 55.5%, and 43.9% of positive-loss_i respondents are probed under untargeted probing, overall-mean targeted probing and contrast targeted probing, respectively.

[12]For conditional means, standard errors are computed within the corresponding evaluation group. For conditional mean contrasts, standard errors account for estimation of evaluation group proportions.

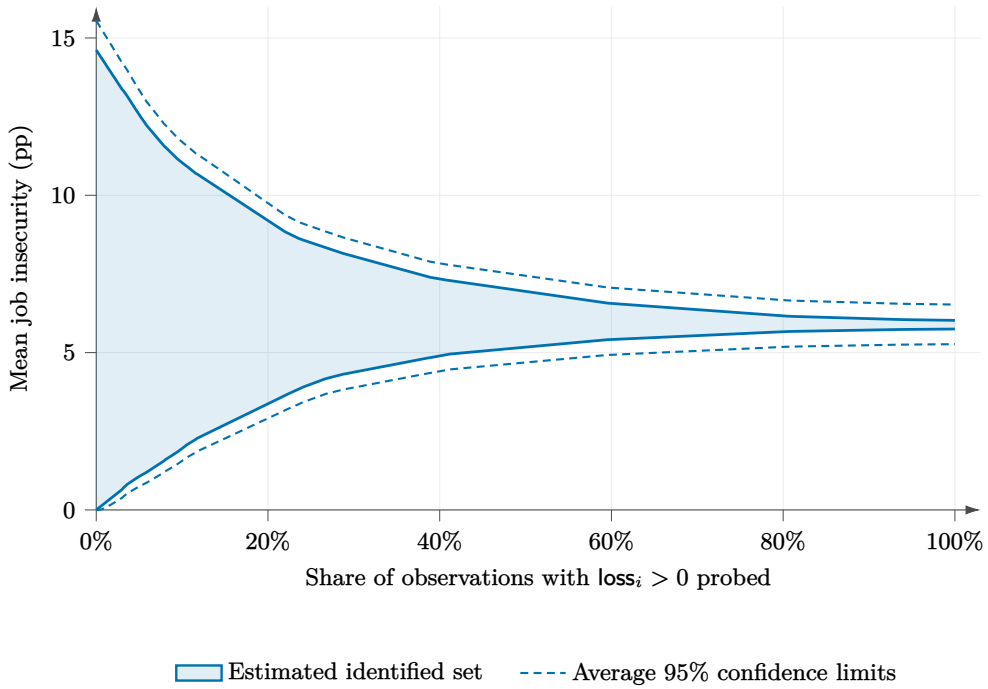


Figure 12: Estimated identified set and averaged 95% confidence intervals for mean job insecurity under targeted probing.

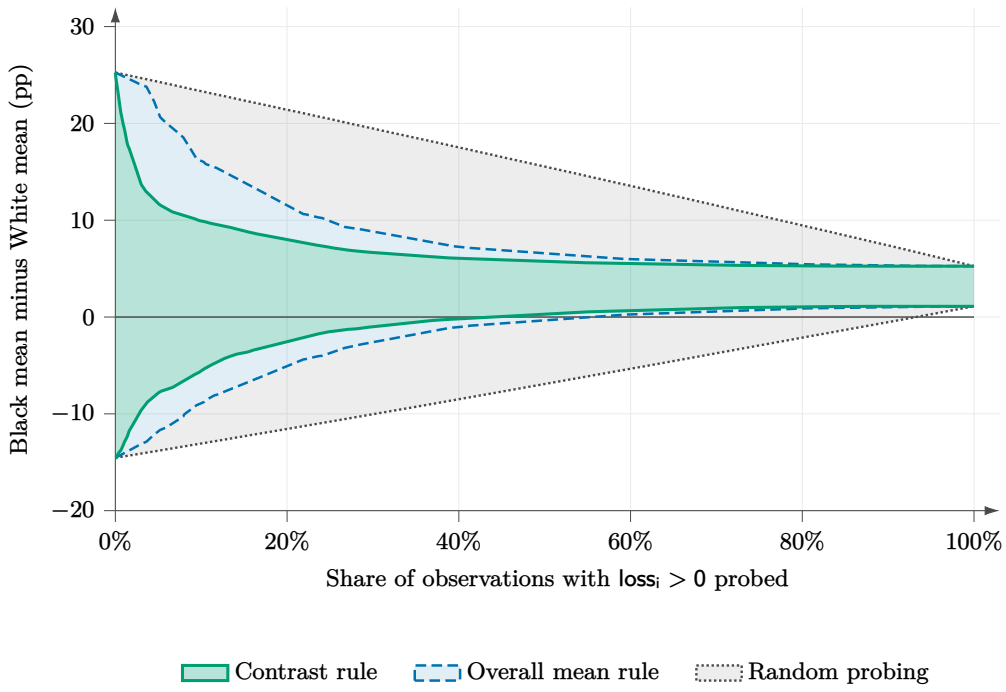


Figure 13: Averaged 95% confidence intervals for the Black-White mean job-insecurity contrast under alternative probing rules.